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FACULTY OF SOCIAL SCIENCES
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**Present Bias in Quasi-Hyperbolic
Discounting: A Meta-Analysis**

Bachelor's thesis

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Declaration of Authorship

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Abstract

Present bias is a vital component in modeling behavioral phenomena, including self-control, procrastination, and dynamic inconsistency, yet empirical estimates of the quasi-hyperbolic parameter β vary greatly. This thesis provides a quantitative synthesis of 1,050 estimates from 130 primary studies, analyzing selective reporting and heterogeneity in the literature. The unweighted mean estimate of β is 0.919, suggesting modest present bias. Nonlinear publication-bias correction methods, however, revise the implied mean to around 0.95–0.99, implying a much milder present bias. To investigate the drivers of heterogeneity, we employ Bayesian and frequentist model averaging across a large set of study characteristics. The results suggest methodological choices as the main determinants of reported estimates: controlling for nonlinear utility, equalizing transaction costs, and using a varying front-end delay are all associated with systematically higher β estimates, while delivery of the immediate reward on a different day lowers β . Evidence consistent with selective reporting in favor of smaller β estimates is concentrated in field and real-effort studies, while switch-point estimates display an opposite, likely mechanical, pattern. After controlling for these factors, the gap between monetary and real-effort estimates is substantially reduced.

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Abstrakt

Současné zkreslení představuje klíčovou složku modelování behaviorálních jevů, jako jsou sebekontrola, prokrastinace a dynamická nekonzistence, přesto se empirické odhady kvazihyperbolického parametru β výrazně liší. Tato práce poskytuje kvantitativní syntézu 1 050 odhadů ze 130 primárních studií a analyzuje publikační zkreslení a heterogenitu v literatuře. Nevážený průměrný odhad parametru β činí 0,919, což naznačuje mírné současné zkreslení. Nelineární metody korekce publikačního zkreslení však revidují implikovaný průměr přibližně na 0,95–0,99, což implikuje podstatně slabší současné zkreslení. Pro identifikaci zdrojů heterogenity využíváme bayesovské i frekventistické přístupy k průměrování modelů napříč rozsáhlou sadou charakteristik studií. Výsledky ukazují, že hlavními determinanty vykazovaných odhadů jsou metodologické volby: kontrola nelineární užitkové funkce, vyrovnání transakčních nákladů a variabilní počáteční zpoždění jsou systematicky spojeny s vyššími odhady β , zatímco doručení okamžité odměny v jiný den vede k jejich snížení. Důkazy konzistentní se selektivním vykazováním se koncentrují především v terénních studiích a studiích založených na skutečném úsilí, zatímco odhady založené na indifferenčních bodech vykazují opačný, pravděpodobně mechanický vzorec. Po zohlednění těchto faktorů se rozdíl mezi peněžními odhady a odhady založenými na úsilí výrazně zmenšuje.

Klasifikace JEL

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Klíčová slova

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Acronyms

BMA	Bayesian Model Averaging
CRRA	Constant Relative Risk Aversion
CTB	Convex Time Budget
DMPL	Double Multiple-Price List
DPB	Degree of Present Bias
EDU	Exponentially Discounted Utility
EK	Endogenous Kink
FAT	Funnel Asymmetry Test
FED	Front-End Delay
FMA	Frequentist Model Averaging
MAIVE	Meta-Analysis Instrumental Variable Estimator
MMA	Mallows Model Averaging
MPL	Multiple Price List
PEESE	Precision-Effect Estimate with Standard Error
PET	Precision Effect Test
PIP	Posterior Inclusion Probability
PMP	Posterior Model Probability
QHD	Quasi-Hyperbolic Discounting
RTMA	Right-Truncated Meta-Analysis
WAAP	Weighted Average of Adequately Powered estimates

Chapter 1

Introduction

Intertemporal choices—decisions involving tradeoffs between the present and the future—are ubiquitous in life. One might spend money now rather than save or invest it for the future, or decide to work on a task tonight instead of postponing it to another day. Theoretical and experimental economics, as well as general anecdotal evidence, suggest that individuals, when making intertemporal choices, exhibit a so-called present bias, a tendency to overvalue immediate utility compared to future outcomes. Present bias induces a phenomenon referred to as time-inconsistency: a situation in which individuals' preferences change over time, leading them to abandon previously made plans in favor of more immediate utility. In this capacity, present bias helps explain, for example, impulse spending, procrastination, or difficulties in quitting smoking or maintaining a healthy diet.

The effect of present bias and time-inconsistent preferences has, therefore, been extensively studied in numerous areas, such as personal finance and savings (Beshears *et al.* 2015; Bonan *et al.* 2022), health (Fang & Wang 2015; Mørkbak *et al.* 2017), including dietary decisions (Courtemanche *et al.* 2015; Zhang *et al.* 2024) and vaccination uptake (Andreoni *et al.* 2023; Attema *et al.* 2025), substance use (Lemenze & Murray 2013; Chaloupka *et al.* 2019), labor market decisions (Kaur *et al.* 2015; Abebe *et al.* 2021; Backes-Gellner *et al.* 2021), adoption of new technologies (Creed & Santos 2023), conservation behavior (Boonmanunt *et al.* 2020), or commitment devices (Ashraf *et al.* 2006; John 2020).

The Quasi-Hyperbolic Discounting (QHD) model, introduced by Phelps & Pollak (1968) and popularized by Laibson (1997), quantifies present bias as the parameter β alongside the long-term discount factor δ with the aim of explain-

ing an individual’s time preferences. This formulation has proved influential in economics, as it provided a simple framework accommodating a wide range of empirically observed behavior (Frederick *et al.* 2002).

Many studies have set out to experimentally measure the average β across individuals, eliciting present bias from intertemporal-choice tasks. The findings are, nevertheless, far from conclusive. The estimates of β vary not only across domains (such as money or real effort) and subject pools, but also across elicitation tasks, estimation methods, and practical design choices. Moreover, several factors may contaminate the identification of present bias: nonlinear forms of the utility function, transaction costs, distrust in receiving payments, or imprecise timing of immediate outcomes have been shown to confound the time preferences (Frederick *et al.* 2002; Harrison & Lau 2005; Balakrishnan *et al.* 2020). Additionally, a literature built around an intuitively appealing behavioral mechanism is naturally exposed to selective reporting: estimates consistent with strong present bias may be more likely to get highlighted or published than estimates implying future bias or no bias at all (Stanley 2005; Ioannidis *et al.* 2017).

Two meta-analyses already show that the present bias literature deserves much scrutiny. Imai *et al.* (2021) synthesize estimates of β from studies employing the Convex Time Budget (CTB) protocol, and report modest present bias with heterogeneous results across monetary and non-monetary domains. In their recent work, Cheung *et al.* (2025) broaden the scope to a wider set of quasi-hyperbolic discounting studies and likewise find substantial heterogeneity. This thesis builds on these contributions in a number of ways. First, we collect 1,050 estimates of β from 130 primary studies, including both published and unpublished work, creating the largest dataset to date. Second, we examine selective reporting using a broader battery of linear, nonlinear, and endogeneity-robust methods, along with subgroup-analyses, which enables us to uncover possible sources of selection including p -hacking and other research biases. Third, we analyze heterogeneity in reported estimates using Bayesian and frequentist model-averaging methods, jointly evaluating publication-bias patterns and study characteristics, which allows us to validate the findings of the literature and of the selective-reporting analysis while controlling for covariates. From these findings, we derive a best-practice estimate and validate several findings of the literature.

The raw literature suggests moderate present bias, with an unweighted mean β estimate of 0.919; most nonlinear publication-bias methods, how-

ever, move the implied mean to around 0.95–0.99, indicating only mild present bias after correction. The heterogeneity analysis suggests that methodological choices have a considerable impact: control for nonlinear utility, equalization of transaction costs, and front-end delay variation are all associated with higher β estimates, whereas delivery of the immediate payment on a different day is associated with lower β estimates. The estimate–standard error relationship varies greatly across subgroups, with stronger evidence consistent with selective reporting in field and real-effort studies and an opposite pattern for switch-point estimates, which is more plausibly mechanical than selective. After controlling for these factors, the gap between monetary and real-effort estimates is substantially reduced.

The thesis proceeds as follows. Chapter 2 reviews the theory of quasi-hyperbolic discounting, measurement of present bias, and prior meta-analyses. Chapter 3 describes the data collection strategy and presents preliminary descriptive evidence. Chapter 4 examines selective reporting using funnel-plot asymmetry tests, nonlinear corrections, endogeneity-robust methods, caliper tests, and subgroup analyses. Chapter 5 studies heterogeneity through model averaging. Chapter 6 discusses the implied best-practice estimate, and Chapter 7 concludes.

Chapter 2

Theoretical Framework

2.1 Quasi-Hyperbolic Discounting

The long-established model of intertemporal choices widely used in economics is the Exponentially Discounted Utility (EDU) model proposed by Samuelson (1937), which assumes the discount factor δ as the sole parameter governing an individual's time preferences. When making plans in the present, an individual then attaches the weight δ^t to the utility received from consumption t periods into the future. While the model offered an elegant and simple framework to economists, even Samuelson himself expressed doubts as to its descriptive ability. Indeed, later empirical research at the end of the 20th century reported several inadequacies of the EDU model, particularly in explaining real-world behavior. Importantly, observed patterns exhibited discount rates that were not constant, but rather declining in time (Frederick *et al.* 2002). This is known as hyperbolic discounting, and implies that individuals tend to be more impatient for the near future than for more distant outcomes. Consequently, their decisions may display time-inconsistency, a situation in which individuals' preferences flip, and immediate utility is suddenly preferred to long-term plans, possibly leading to procrastination, impulsivity or addiction (Lipman & Attema 2024).

From the family of hyperbolic time preference models, the one most prevalent in economics is the Quasi-Hyperbolic Discounting (QHD) model. First formulated by Phelps & Pollak (1968) and later generalized by Laibson (1997), it captures dynamically inconsistent choices by extending the EDU model by the parameter β , designed to measure present bias, a tendency to overvalue the utility gained from present consumption (Frederick *et al.* 2002). In the QHD

model, an individual at present ($t = 0$) values a consumption stream $(x_0, \dots, x_T)$ according to

$$U(x_0, \dots, x_T) = u(x_0) + \beta \sum_{t=1}^T \delta^t u(x_t), \quad (2.1)$$

where $\delta > 0$ is the standard discount factor and $\beta > 0$ captures present bias. The parameter δ then reflects long-run impatience between future periods, whereas β represents the additional discount applied to all future utilities relative to the present. The lower the β , the stronger the present bias. In the case when $\beta = 1$, the equation reduces to the standard EDU model, and implies time consistency. The situation when $\beta > 1$ is referred to as future bias, and signifies that the individual places more weight on future rather than present utility (Imai *et al.* 2021; Cheung *et al.* 2025).

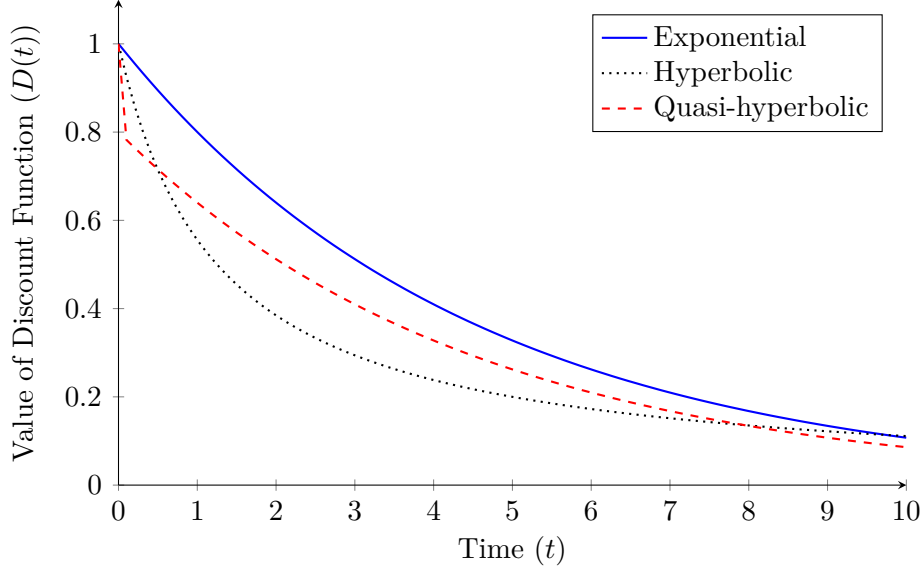


Figure 2.1: Discounting functions over time

Notes: This figure illustrates the value of the discount function $D(t)$ over time for three specifications of intertemporal preferences: Exponential ($D(t) = \delta^t$), Hyperbolic ($D(t) = \frac{1}{1+\rho t}$), and Quasi-hyperbolic ($D(t) = \{1, \beta\delta, \beta\delta^2, \dots, \beta\delta^t\}$). The illustrative values for the parameters are $\delta = 0.8$, $\beta = 0.8$, $\rho = 0.8$.

Figure 2.1 shows the value of the discount function $D(t)$ for exponential preferences ($D(t) = \delta^t$), a form of hyperbolic preferences as proposed by Mazur (1984), where $D(t) = \frac{1}{1+\rho t}$, ρ being the discount rate, and quasi-hyperbolic preferences ($D(t) = \{1, \beta\delta, \beta\delta^2, \dots, \beta\delta^t\}$). The defining element of the quasi-hyperbolic model is the ‘jump’ at $t = 0$ induced by present bias, after which it behaves identically to the exponential model (Andersen *et al.* 2014b).

2.2 Measuring Present Bias

Various methods are available for eliciting and estimating time preference parameters, each bearing different strengths and weaknesses. The basic premise in most of them is nevertheless similar in nature: ask subjects to compare or allocate a budget between a smaller reward x_t at time t and a larger delayed reward x_{t+k} at time $t + k$. The parameters are then elicited by asking a set of questions, changing the values x_t , x_{t+k} , base period t , and delay k . Note that what is referred to here as ‘reward’ is not necessarily monetary, it can also relate to time spent working on a laborious real-effort tasks (Augenblick & Rabin 2019), food consumption (Cheung *et al.* 2022), health outcomes (Fredslund *et al.* 2018), or environmental goods (Meyer 2013).

It is generally believed that time preferences vary across domains, that is, individuals discount differently, for example, monetary and health outcomes. The low correlation between time preference parameters across domains is referred to as domain independence (Loewenstein *et al.* 2003; Matousek *et al.* 2022). Augenblick *et al.* (2015) conduct a study investigating choices for real-effort tasks and monetary payments and find very limited time inconsistency for monetary rewards, whereas significant evidence of present bias for real effort. Similarly, Cerrone *et al.* (2024), in their experiment in which participants choose the number of tasks at different points in time for various wages (Augenblick & Rabin 2019), find considerable present bias for real effort, but very little evidence of present bias for money. Cheung *et al.* (2025), in their meta-analysis, find substantial differences in reported estimates of β between monetary and non-monetary domains, with non-monetary domains exhibiting stronger present bias. Imai *et al.* (2021) report comparable findings, though the difference between domains for money and real effort becomes less apparent after correcting for selective reporting.

On theoretical grounds, the methodological choice of real or hypothetical incentives appears vital to the credibility of elicited time-preference parameters: real stakes induce authentic decision-making, whereas selecting between two hypothetical rewards misrepresents the participants’ true preferences. The consensus in the literature, however, suggests the opposite—no substantial difference between real and hypothetical incentives (Brañas-Garza *et al.* 2023; Locey *et al.* 2011; Johnson & Bickel 2002; Lipman & Attema 2024). Both Cheung *et al.* (2025), in their meta-analysis of both QHD parameters, and Matousek *et al.* (2022), in their meta-analysis of individual discount rates, find no effect

of the genuineness of rewards.

The most straightforward method to elicit time preference parameters is the Multiple Price List (MPL) task (also called a choice task), in which subjects are presented with a set of binary outcomes at different times, and are asked to choose between them. Participants usually face a sequence of decisions between a smaller sooner and a larger later reward; researchers then vary the dates and sizes of rewards to infer points at which subjects switch between the sooner and later reward. Choice lists are the simplest method, and are therefore easy to administer and implement to larger study designs (Andreoni *et al.* 2015); for examples, see Albrecht *et al.* (2011); Ackert *et al.* (2020); Harrison *et al.* (2025) or Haushofer *et al.* (2013). However, traditional MPL experiments have been shown to harbor some undesirable side effects, most notably the anchoring effect, in which subjects' first choice psychologically influences subsequent decisions (Frederick *et al.* 2002). There are, however, modifications that remedy this issue, such as using the iterative version of MPL (Harrison *et al.* 2005), in which participants face increasingly finer intervals based on previous choices, or by randomizing the ordering of questions and the default options (Ashton 2015).

In a Matching task, subjects are asked to identify a value that would make them indifferent between two intertemporal outcomes. Researchers commonly ask: *how much money would you require in addition to the sooner reward if you had to wait k months (days, weeks, years, ...) to get the money?* This allows for a more precise estimate, since choices do not depend on predefined options. Still, there seem to be large discrepancies in the elicited time preference parameters depending on whether subjects matched on amount (i.e. asking how much additional money is required to compensate for the delay) or time (i.e. asking how long one would be willing to wait to receive a larger reward), and on other methodological decisions (Frederick *et al.* 2002). For examples of matching-task QHD studies, see Fredslund *et al.* (2018) or Madsen & Kjær (2024).

Augenblick & Rabin (2019) introduce a real-effort task, which in this thesis will be referred to as a work-for-wage task. Subjects are asked to identify a number of (unpleasant) tasks to complete (or the amount of time they are willing to spend working) at present and future dates for various wages. While Augenblick & Rabin (2019) elicit only time preferences relating to real effort, Cerrone & Lades (2017), by varying the timing of the payments as well as of the tasks themselves, augment the study design so that it also allows for

elicitation of monetary time preferences. Studies employing a work-for-wage task, among others, also elicit the individuals' perceived present bias β^* by comparing their predictions with actual immediate work choices. An individual is deemed sophisticated when they are aware of their present bias, $\beta^* = \beta$, and naive when they are unaware of or underestimate their self-control difficulties, $\beta^* > \beta$. Work-for-wage studies are not the only ones eliciting this parameter, but their longitudinal design creates a convenient environment for it. Wu (2021) conducted a meta-analysis of 47 estimates of present bias β and its perceived counterpart β^* from 10 studies, and found people understand only 36% of their present bias, rejecting the hypothesis of sophistication.

The method most frequently used for eliciting the parameters of QHD is the Convex Time Budget (CTB) protocol developed by Andreoni & Sprenger (2012); for diverse examples, see Aycinena & Rentschler (2018); Bartoš *et al.* (2021); Banerji *et al.* (2018); Cerrone & Lades (2017); Kölle & Wenner (2023) or Yang & Carlsson (2016). Subjects are asked to allocate a fixed budget between the two dates t and $t + k$, having the option to choose any linear combination (x_t, x_{t+k}) along the budget line. By varying the dates, budget and the interest rate between the two periods (the slope of the budget line), researchers can precisely identify the parameters of the utility function, of which a commonly used form is the Constant Relative Risk Aversion (CRRA) utility function:

$$U(x_t, x_{t+k}) = \frac{1}{\alpha}(x_t + \omega_t)^\alpha + \beta^{1\{t=0\}}\delta^k \frac{1}{\alpha}(x_{t+k} + \omega_{t+k})^\alpha,$$

where α is the utility curvature parameter, ω is a background consumption parameter and β and δ are the time preference parameters.

Imai *et al.* (2021), Andreoni & Sprenger (2012), and Augenblick *et al.* (2015) provide great summaries of how researchers typically estimate the time-preference parameters using regression methods in CTB studies and in quasi-hyperbolic discounting studies in general; here, we provide a brief illustration. From the above utility function subject to the intertemporal budget constraint

$$(1 + r)x_t + x_{t+k} = I,$$

where $1 + r$ is the gross interest and I is the available budget, one can derive the intertemporal Euler equation

$$\frac{x_t + \omega_t}{x_{t+k} + \omega_{t+k}} = \left(\beta^{1\{t=0\}}\delta^k(1 + r)\right)^{\frac{1}{\alpha-1}}$$

that allows for estimation of the parameters α , β and δ . Because observed allocations are often bounded by corner solutions (i.e., individuals may allocate the entire budget to one period), the dependent variable is censored. Therefore, the equation is typically estimated using a two-limit Tobit model of the form:

$$\log \left(\frac{x_t + \omega_t}{x_{t+k} + \omega_{t+k}} \right) = \frac{\log \beta}{\alpha - 1} \cdot \mathbf{1}\{t = 0\} + \frac{\log \delta}{\alpha - 1} \cdot k + \frac{1}{\alpha - 1} \cdot \log(1 + r),$$

recovering the time-preference parameters via a nonlinear combination of the estimated coefficients. Nonlinear least squares (see, for example, Balakrishnan *et al.* (2020); Riis-Vestergaard *et al.* (2018)) and Maximum likelihood estimation (for a detailed description, see Andersen *et al.* (2014b; 2008); for examples, see Krause (2024); Meier & Sprenger (2015)) are also frequently used (Cheung 2016). In some studies, the parameters are calculated directly from inferred switching points by solving a set of indifference equations for δ and β based on the subjects' responses to the elicitation task, such as in Attema *et al.* (2025) or Schleich *et al.* (2019). To illustrate, consider two choices faced by a participant: one involving only future outcomes, (x_{A1}, x_{B1}) , and another involving a present-future trade-off, (x_{A2}, x_{B2}) . Suppose the participant is shown to be indifferent between the options in both choices. The parameters can then be calculated solving:

$$\begin{aligned} u(x_{A1}) &= \delta u(x_{B1}), \\ u(x_{A2}) &= \beta \delta u(x_{B2}). \end{aligned}$$

In the above example, and in most quasi-hyperbolic discounting studies, β and δ are elicited using a varying Front-End Delay (FED), i.e. by shifting the timing of the sooner reward. The discount factor is then estimated using the variation in the future-only choices and present bias based on the choices involving the present. A small subset of the analyzed studies, however, measure time-preference parameters without FED variation; for examples, see Bonan *et al.* (2022); Courtemanche *et al.* (2015) or Tanaka *et al.* (2010).

Potential confounding factors that hinder proper measurement of time preferences are numerous. One important issue frequently discussed in the literature is accounting for nonlinear utility. The default approach to estimation assumes a linear form of the utility function in the magnitude of the outcome. If this assumption does not hold, the estimation of the time preference parameters becomes biased (Frederick *et al.* 2002). Cheung (2016) provides a great

succinct survey of the issue and the methods to control for nonlinear utility; here, we provide a brief summary. From Equation 2.1 it is evident that the value of the discounted utility function $U(x_0, \dots, x_T)$ depends on both the time preference parameters and also the instantaneous utility function $u(\cdot)$. To then provide unbiased estimates of the former, one has to also understand the form of the latter. Falsely assuming a linear utility function would lead to overstating the effect of intertemporal discounting.

There exist two main approaches to tackling the problem: equating utility under risk and over time and inferring utility curvature directly from intertemporal choices. To illustrate the former, consider risk preferences under expected utility, which follow:

$$E[v(x)] = \sum_s p(s)v(x_s),$$

where $p(s)$ is the probability of state s and $v(x_s) = (1/\alpha)x_s^\alpha$ is the Bernoulli utility, whose curvature reflects the individual's risk aversion. The utility function is elicited using a complementary task, such as the choice task based on Holt & Laury (2002). Similar to a multiple-price list, participants face a set of choices between a safer and a riskier gamble; the probability of the higher outcome in each of the gambles progressively increases with each choice. Researchers then search for the switching point, where the participant changes to the risky option, and the utility curvature parameter α is identified based on this elicited risk attitude (Andreoni *et al.* 2015). It is then assumed that $v(\cdot)$ and the instantaneous utility function $u(\cdot)$ are identical. The time preference parameters are then estimated conditional on the curvature of $u(\cdot)$.

This two-fold elicitation of time and risk preferences is employed in the Double Multiple-Price List (DMPL) study design. Aptly named, the procedure consists of two MPLs: the first is a standard intertemporal MPL to elicit, potentially confounded, time preferences, and the second uses the risk-preference choice task to elicit information on the utility function that allows for a proper estimation of the discounting parameters (Andreoni *et al.* 2015). For examples of the DMPL, see Bradford *et al.* (2017); Andersen *et al.* (2014a) or Sawada & Kuroishi (2024); for an alternative risk-preference task, see Fuerst & Singh (2018).

The alternative approach circumvents the need for the strong assumption of equating utility under risk and over time but relies on not restricting subjects' intertemporal choices to the binary, instead allowing them to choose a convex combination of the two outcomes x_t and x_{t+k} . This is exactly the design of

the CTB protocol (Andreoni & Sprenger 2012), which then elicits the time preference parameters along with the utility-curvature parameter α by varying the dates, payments, and the exchange rate between them. Andreoni *et al.* (2015) compare the performance of the CTB and DMPL designs and find that CTB outperforms the DMPL in predicting intertemporal choice.

Two other widely-documented issues that hinder proper identification of time-preference parameters pertain to practical nuances in implementation. Firstly, the differences in transaction costs between the sooner and later payment may confound intertemporal preferences with the effort cost of receiving the delayed payment. For example, if the reward has to be received in cash at the place where the experimental session took place, the additional trip the individual has to make in the case of the delayed payment is also reflected in the individual's decisions. Transaction costs can be equalized in a variety of ways, including front-end delays on *all* sooner outcomes (see Andreoni & Sprenger (2012); Denant-Boemont *et al.* (2017)); here, however, we arrive at another concern: the uniform delay effectively shifts all outcomes into the future and thus 'immediate' rewards may be perceived as future rather than present by the quasi-hyperbolic discounter. Balakrishnan *et al.* (2020) conduct an experiment in Nairobi, in which they control for transaction costs by using the Kenyan M-Pesa mobile money payment system, which allows real-time payments to participants. They compare truly immediate payments (at the end of the experiment session) and 'immediate' payments made by the end of the day. They find substantial present bias for participants in the immediate-payment group, but little evidence of present bias when the earliest payment is at the end of the day. Other methods ensuring equal transaction costs include forcing the participants to receive some amount of the reward at both the sooner and later date (Aycinena *et al.* 2019; Cheung *et al.* 2022; Kuhn *et al.* 2017) or by using real-time bank or digital transfer (Balakrishnan *et al.* 2020; Lindner & Rose 2017; Liu *et al.* 2014; Harrison *et al.* 2018).

Secondly, the level of uncertainty in receiving the delayed payment may also confound the elicited time preferences. If the participants do not fully trust the researchers, they may view the delayed options with a certain risk factor, which would further diminish the perceived utility from any option not received immediately. The common method to tackle the trust issue is a simple guarantee or certificate from an official source, such as a notarized guarantee from a university professor (Coller *et al.* 2012) or from a bank (Meier & Sprenger 2015), or by providing contact information of the researchers directly responsible for

payments (Delaney & Lades 2017; Liu *et al.* 2014; Andreoni & Sprenger 2012; Sun & Potters 2022).

2.3 Prior Meta-Analyses

Two prior meta-analyses have been conducted on the topic of present bias. The quintessential experimental-economics meta-analysis by Imai *et al.* (2021) provides a synthesis of 220 estimates from 28 studies exclusively using the CTB protocol. Overall, they find statistically significant present bias, with a meta-analytic average of β between 0.95 and 0.97. Results for monetary rewards indicate no present bias on average (β close to one), while estimates for real effort report a lower average β of 0.88. They do, however, find evidence of selective reporting favoring smaller β estimates (stronger present bias) in real-effort studies. The authors employ linear and two non-linear publication correction methods (the Andrews & Kasy (2019) selection model and the stem-based method by Furukawa (2021)) and report corrected estimates for real effort from 0.90–0.93 to 0.99.

Subsequently, the authors analyze the drivers of heterogeneity in present bias estimates utilizing Bayesian Model Averaging (BMA). They find systematically lower estimates of β (stronger present bias) for studies which deliver immediate rewards (at $t = 0$) by the end of the experiment, as opposed to delivery by the end of the day, corroborating the findings of Balakrishnan *et al.* (2020). Furthermore, they find stronger present bias for lab studies compared to studies conducted in the field and no substantial differences between econometric strategies.

Recently, Cheung *et al.* (2025) conducted a meta-analysis of both parameters of quasi-hyperbolic discounting using 108 estimates from 86 studies. They find a significant present bias overall, with a meta-analytic estimate of β of 0.919 for money and 0.750 for non-monetary rewards. They detect selective reporting in favor of smaller β estimates, and use linear and nonlinear methods (Endogenous kink by Bom & Rachinger (2019), WAAP by Ioannidis *et al.* (2017), and MAIVE by Irsova *et al.* (2025)) to compute corrected estimates of around 0.938 for money and 0.750 for non-monetary rewards. For real effort, a subset of the non-monetary rewards, they find an implied β of 0.815.

Regarding heterogeneity, the authors also use BMA and corroborate much of the earlier empirical findings: correcting for nonlinear utility is an important driver of heterogeneity, with higher β estimates (weaker present bias) reported

by studies that adjust for this confound, and studies eliciting using hypothetical incentives do not report significantly different estimates compared to studies that use real rewards. Moreover, they find higher β estimates (weaker present bias) associated with the CTB protocol, and lower β estimates for Asian samples.

While both prior meta-analyses are relatively recent and employ up-to-date methods to address publication bias and model uncertainty, they seem to provide markedly different results, at times diverging substantially. These discrepancies may partly stem from the limited samples both studies restrict themselves to: Imai *et al.* (2021) focus their attention only on the CTB literature, analyzing 28 studies, while Cheung *et al.* (2025) synthesize 86 studies (not limited to CTB), however mainly recording only one headline estimate per study.

The first main contribution of this thesis is, therefore, substantially broader empirical coverage. We collect 1,050 estimates of the present-bias parameter β from 130 primary studies. Including all comparable estimates, rather than only those preferred by authors, increases statistical power, mitigates the loss of information from single-estimate aggregation, and enables a much more thorough analysis of within-study variation, subgroup-specific patterns, and the role of selective reporting.

A second contribution, relative to both previous works, is a much more comprehensive and meticulous treatment of selective reporting. We apply a battery of linear funnel-asymmetry tests, several nonlinear correction methods, endogeneity-robust approaches, caliper tests and subgroup-specific analyses of selection—broader than either prior work. This allows us to gauge the extent of selection and estimate the corrected effect and, more importantly, they also make it possible to identify the potential sources of selection. We also shed some light on other possible mechanisms affecting the estimate–standard error relationship, which standard publication-bias tests may confuse with selective reporting.

Third, to our knowledge, this is the first meta-analysis of present bias to apply RTMA (Mathur 2024), a method explicitly designed to tackle p -hacking and within-study selection. Although both p -hacking-robust methods, RTMA and MAIVE, have their identifying assumptions violated in our data and therefore cannot reliably detect within-study selection in this literature, the documented limitations are themselves an informative result for future meta-analyses on this topic.

Fourth, alongside the standard Bayesian model averaging used in both prior works, we apply Mallows frequentist model averaging with cluster-robust standard errors, implemented from the publicly available replication tools for meta-analysis. This provides a complementary check on inference under model uncertainty that, to our knowledge, has not been used in the present-bias literature.

Fifth, building on the BMA results, we derive a best-practice estimate of $\beta = 0.97$ conditional on a methodologically defensible set of study-design choices. Imai *et al.* (2021) provide subgroup-specific estimates corrected for selective reporting, but do not construct a fitted best-practice estimate from their model-averaging results. Cheung *et al.* (2025) do not pursue this exercise either. Our estimate provides a single, design-conditional benchmark for calibration in applied work.

Finally, this thesis makes a substantive contribution by qualifying several conclusions of the existing literature. The raw mean of the collected estimates is 0.919, which, at face value, suggests moderate present bias, but the nonlinear bias-correction methods consistently move the corrected estimate into the range of roughly 0.95–0.99, implying that the literature likely overstates the average size of present bias. Moreover, once study-characteristics are controlled for in the model-averaging framework, the remaining heterogeneity appears to be driven less by the canonical monetary–real effort distinction and more by methodological choices such as control for nonlinear utility, transaction-cost equalization, varying front-end delay, and payment timing. We then estimate the effect implied by the best practice in the literature to show that part of what earlier work interpreted as behavioral heterogeneity may in fact reflect selective reporting and differences in study design.

Chapter 3

Data

To provide a quantitative synthesis of studies measuring present bias, we create a dataset with 1,050 estimates of the quasi-hyperbolic discounting model parameter β from 130 primary studies. This section outlines the data collection process, from initial literature search and inclusion criteria to constructing the dataset. Lastly, it provides an introductory analysis using descriptive statistics.

3.1 Literature Search

A meta-analysis begins with an exhaustive search for eligible studies. Before conducting our own search however, we had the opportunity to build on the datasets of previous meta-analyses. The dataset of Imai *et al.* (2021) is publicly available in their online replication package. The authors of Cheung *et al.* (2025) graciously provided their data upon request. The latter's latest addition to their dataset was carried out in August 2021; we therefore had space for including newly published studies, as well as any past studies the authors may have missed. Furthermore, Cheung *et al.* (2025) recorded only the main estimate per study (with exceptions for cross-method or cross-domain comparisons; see their online appendix for details). Our preference was instead to collect all reported estimates, regardless of the authors' stated priority. We therefore carefully screened all the included studies for additional results. We also searched for any updated or published version of working papers, so that all results were up-to-date. After this procedure, and after merging the two datasets, we were equipped with a solid foundation to begin our own search.

A comprehensive picture of the study collection procedure is provided by the PRISMA flow diagram in Figure A.1 (Appendix A). The search was con-

ducted using Google Scholar, since it allows for full-text screening of keywords. The query was iteratively refined to maximize recall of studies reporting estimates of present bias. The final Google Scholar query used is:

("present bias" OR "quasi-hyperbolic discounting" OR "beta-delta model" OR " β - δ model" OR "hyperbolic discount*" OR "non-constant discount*") (self-control OR "present bias" OR "quasi-hyperbolic discounting") (experiment OR estimation OR elicitation OR measur* OR comput* OR test*).

The search was terminated on December 8, 2025. We examined the title and abstract of the first 500 results (Irsova *et al.* 2024), and the following inclusion criteria were then applied. Studies that were solely theoretical or certain to not include present bias estimates were excluded. Next, a full-text screening was conducted. Studies not reporting empirical estimates or a measure of their precision (standard errors, or way to compute them) were then omitted from further analysis, as well as any duplicates. Note that in some cases, estimates of parameters labeled as ‘present bias’ from other time preference models were reported. We include only present bias estimates that are in the form of the quasi-hyperbolic discounting model described in Chapter 2. We therefore exclude, for instance, ‘present bias’ parameters from fixed-cost models (Benhabib *et al.* 2010) or from other non-QHD functional forms.

Following the guidance of Stanley (2001), we omit no studies based on their perceived quality or data. The whole philosophy of conducting a meta-analysis after all is to dissect these differences. Hence, graduate and dissertations theses, as well as working or unpublished papers, are included in the analysis.

After having a complete set of studies reporting present bias estimates, we decided to exclude a number of studies based on a conceptual criterion. We included only studies in which the time preference parameters were estimated based on individuals’ decisions in experimental or survey settings designed for elicitation. This conceptual restriction ensures comparability across the measurement framework. We therefore excluded a number of studies that estimate the parameters from observational data in institutional settings. An example of such a study is Abaluck *et al.* (2018), who use data on prescription drug use from Medicare Part D enrollees to estimate various parameters, including time preferences. The final list of the 130 included primary studies is presented in Table 3.1, and the complete list of excluded studies can be found in Table A.1 in Appendix A.

Table 3.1: List of included studies

Abebe <i>et al.</i> (2021)	Creed & Santos (2023)	Lührmann <i>et al.</i> (2018)
Ackert <i>et al.</i> (2020)	Dang (2023)	Lu & Zhuang (2014)
Akesaka (2019)	Danzer & Zeidler (2024)	Lusher (2016)
Akin & Yavaş (2025)	Delaney & Lades (2017)	Mørkbak <i>et al.</i> (2017)
Albrecht <i>et al.</i> (2011)	Denant-Boemont <i>et al.</i> (2017)	Madsen & Kjær (2024)
Al Lily <i>et al.</i> (2025)	Eberth <i>et al.</i> (2020)	McClure <i>et al.</i> (2007)
Ambrus <i>et al.</i> (2015)	Eil (2012)	Meier & Sprenger (2007)
Andersen <i>et al.</i> (2014b)	Engle-Warnick <i>et al.</i> (2021)	Meier & Sprenger (2015)
Andersen <i>et al.</i> (2014a)	Enke <i>et al.</i> (2025)	Meissner <i>et al.</i> (2022)
Andreoni & Sprenger (2012)	Erdem & Can (2013)	Meyer (2013)
Andreoni <i>et al.</i> (2015)	Fedyk (2024)	Meyer (2016)
Andreoni <i>et al.</i> (2023)	Fredslund <i>et al.</i> (2018)	Meyer (2015)
Ashton (2015)	Freeman & Laughren (2024)	Meyer (2018)
Attema <i>et al.</i> (2025)	Fuerst & Singh (2018)	Mograbi (2022)
Augenblick & Rabin (2019)	Glimcher <i>et al.</i> (2007)	Nguyen (2011)
Augenblick <i>et al.</i> (2015)	Goda <i>et al.</i> (2019)	Nguyen (2016)
Aycinena & Rentschler (2018)	Goda <i>et al.</i> (2015)	Nguyen <i>et al.</i> (2016)
Aycinena <i>et al.</i> (2019)	Goossens <i>et al.</i> (2022)	Nuscheler & Roeder (2016)
Backes-Gellner <i>et al.</i> (2021)	Guan <i>et al.</i> (2022)	Olivola & Wang (2016)
Balakrishnan <i>et al.</i> (2020)	Harrison <i>et al.</i> (2025)	Ong <i>et al.</i> (2025)
Banerji <i>et al.</i> (2018)	Harrison <i>et al.</i> (2018)	Patiño & Gómez-García (2019)
Bao <i>et al.</i> (2018)	Harrison <i>et al.</i> (2022)	Patterson & Shank (2020)
Barcellos & Carvalho (2014)	Haushofer <i>et al.</i> (2013)	Rahardi & Dartanto (2021)
Bartoš <i>et al.</i> (2021)	Holloway <i>et al.</i> (2025)	Reddinger (2025)
Belzil & Sidibe (2016)	Horn & Kiss (2020)	Riis-Vestergaard <i>et al.</i> (2018)
Bonan <i>et al.</i> (2022)	Hvide & Lee (2016)	Rodriguez-Lara & Ponti (2017)
Boonmanunt <i>et al.</i> (2020)	Ida (2014)	Santos <i>et al.</i> (2022)
Bousquet (2017)	Imai & Camerer (2018)	Sawada & Kuroishi (2024)
Bradford <i>et al.</i> (2017)	Imas <i>et al.</i> (2022)	Schleich <i>et al.</i> (2019)
Brown <i>et al.</i> (2009)	Jones & Mahajan (2015)	Schurer <i>et al.</i> (2024)
Burks <i>et al.</i> (2012)	Kiessling <i>et al.</i> (2022)	Sial <i>et al.</i> (2024)
Can & Erdem (2013)	Kim (2023)	Sun & Potters (2022)
Carvalho <i>et al.</i> (2016b)	Kim & Nguyen (2024)	Sutter <i>et al.</i> (2013)
Carvalho <i>et al.</i> (2016a)	Kölle & Wenner (2023)	Tanaka & Yamano (2015)
Cavagnaro <i>et al.</i> (2016)	Koch & Nafziger (2019)	Tanaka <i>et al.</i> (2010)
Cerrone <i>et al.</i> (2024)	Kosse & Pfeiffer (2013)	Toubia <i>et al.</i> (2013)
Cerrone & Lades (2017)	Krause (2024)	Wang <i>et al.</i> (2016)
Chaloupka <i>et al.</i> (2019)	Kuhn <i>et al.</i> (2017)	Werthschulte & Löschel (2021)
Chan (2017)	Lemenze & Murray (2013)	Yamamoto <i>et al.</i> (2025)
Chaudhry & Hussain (2023)	Lerner <i>et al.</i> (2013)	Yang & Carlsson (2016)
Cheung <i>et al.</i> (2022)	Liebenehm & Waibel (2014)	Yin (2018)
Coller <i>et al.</i> (2012)	Linardi & Tanaka (2013)	Zhang <i>et al.</i> (2024)
Corbett (2016)	Lindner & Rose (2017)	
Courtemanche <i>et al.</i> (2015)	Liu <i>et al.</i> (2014)	

Notes: This table lists all primary studies included in the analysis.

3.2 Data Adjustments

Some studies did not directly report standard errors, hence they had to be computed from other statistics, such as standard deviations or p -values. In other studies, the β estimates were reported in various functional forms, such as $\ln(\beta)$ in Mørkbak *et al.* (2017) or $\gamma = (1 - \beta)/\beta$ in Meissner *et al.* (2022). In such cases, the estimates were then recomputed, and their standard errors recovered using the Delta method (for details, see Appendix B).

In rare instances, studies were found to err in the reporting of the precision measure. The reported magnitude of standard errors or standard deviations was simply out of feasible bounds, relative to the estimate and other reported statistics. All corrections, along with reasoning, can be found in Table A.2 in Appendix A.

A notable distinction between the present bias estimates is whether they were estimated on the individual or aggregate level. Individual level estimation entails calculating present bias for each subject independently, and then reporting summary statistics for the whole sample, whereas aggregate level estimation pools the data on all subjects and uses it to compute one estimate with the best fit for the entire sample. Where individual-level summary statistics are reported, we use the mean estimate as a comparable measure, though in some studies the mean suffers from tremendous size inflation likely due to anomalies in some subjects' decisions or estimation artifacts, such as in Lindner & Rose (2017) or Koch & Nafziger (2019).

Extreme estimates are, however, not exclusive to individual estimation but rarely crop up even amongst aggregate estimates (Ambrus *et al.* 2015; Jones & Mahajan 2015), likely as estimation or statistical artifacts. To prevent these oddities from influencing our analysis, we winsorize our estimates of present bias and their standard errors at the 5% level on both tails. The level of winsorization was chosen so that the results of selective reporting tests are not driven by the extreme values, which was found to be the case for lower levels of winsorization. The results of a robustness check including estimates winsorized at the 2.5% level is described in Section 4.7 in the chapter on selective reporting.

3.3 Additional Variables

Together with present bias estimates and their standard errors, various other variables were collected. Firstly, we recorded estimates of the discount factor δ

or discount rate $\rho = (1 - \delta)/\delta$, along with their standard error. The discounting parameter is always accompanied by the discount period (daily, weekly, monthly, annual, ...). Likewise, estimates of the utility curvature parameter α were collected from studies that reported them.

Secondly, to analyze sources of heterogeneity, we collected information on various characteristics of the estimate and study, such as elicitation and estimation method, study setting, reward domain, payment method and timing of payments, to name a few. The data on these variables was first collected in the form of categorical variables, which were then coded as multiple dummy variables for each category after the collection was completed. Both Imai *et al.* (2021) and Cheung *et al.* (2025) applied a similar procedure with their datasets, which enabled us to conveniently pool and use some of their categorical variables without having to adhere to their specific coding strategies.

Thirdly, publication status, publication year, and various other publication measures, such as number of citations or impact factor of the journal, were also collected. To ensure consistency, publication measures were collected on a single day (January 19, 2026) for all 130 primary studies. The complete list of the 116 collected variables can be found in Table A.3 in Appendix A.

3.4 Initial Analysis

Figure 3.1 shows the histogram of present bias estimates. The unweighted mean of the collected estimates is 0.919 with a 95% confidence interval of [0.912, 0.927], implying moderate present bias on average. The distribution is left-skewed with a median of 0.96, which is closer to the no-bias benchmark of $\beta = 1$. However, we also have to consider that some studies report many estimates, such as Augenblick & Rabin (2019), who report 103 estimates, whereas a significant portion of studies report only a single estimate, such as Akesaka (2019) or Imas *et al.* (2022). Since the simple unweighted mean overvalues studies reporting a larger number of estimates, we also calculate a weighted mean, where the weight is the inverse number of estimates per study. The overall weighted mean equals 0.907 with a corresponding 95% confidence interval of [0.885, 0.930]. The point estimate is slightly lower than its unweighted counterpart, but can still be interpreted as only moderate present bias.

To lump all estimates together would nevertheless be a great generalization. Figure 3.2 shows the distribution of estimates for studies reporting multiple estimates. From the figure alone, there is evident heterogeneity in reported present

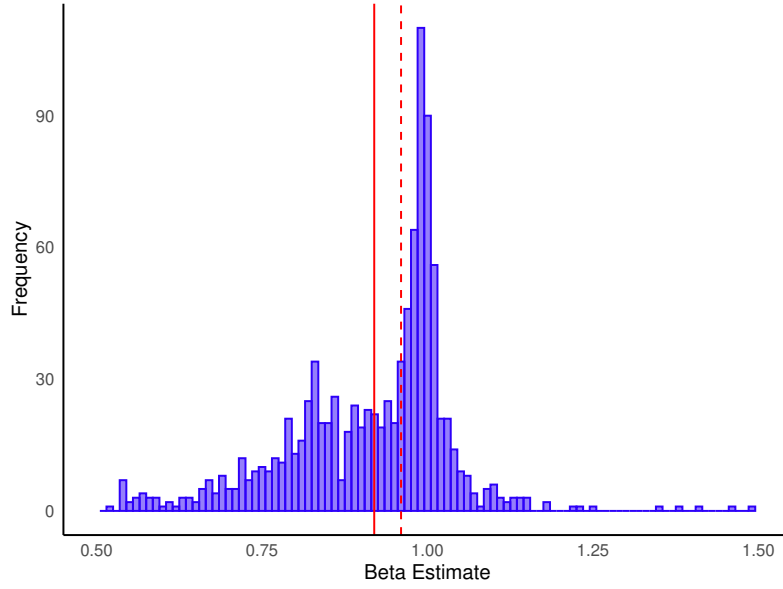


Figure 3.1: Histogram of present bias estimates

Notes: This figure displays the histogram of present-bias parameter (β) estimates. The solid vertical red line denotes the mean estimate, and the dashed red line denotes the median estimate. Extreme values are omitted from the graph, but included in all regressions.

bias estimates, both within and between studies. What follows is a brief investigation into the various subsets of data, with a full analysis of heterogeneity using model averaging methods in Chapter 5.

Table 3.2 presents the unweighted and weighted mean estimates of present bias for selected subgroups, as well as the corresponding 95% confidence intervals. Figure 3.3 displays the distributions of present bias estimates grouped by various characteristics.

At a first glance, there are no substantial differences between aggregation levels, treatment specifications or between real and hypothetical incentives. The last point might come as a surprise, as people would instinctively assume a difference between real and hypothetical decision-making, but has been widely reported on in the literature (Wang *et al.* 2016; Imai & Camerer 2018).

Considering reward domain, there is a large disparity between present bias estimates for money and real effort. Whereas the unweighted mean for money is 0.937, with a 95% confidence interval of [0.929, 0.945], for real effort it is considerably lower at 0.885, with a confidence interval of [0.872, 0.897], which is in line with the idea of domain independence, defined as the low correlation between time preference parameters for different domains, and with studies consistently finding stronger present bias for non-monetary outcomes (Imai

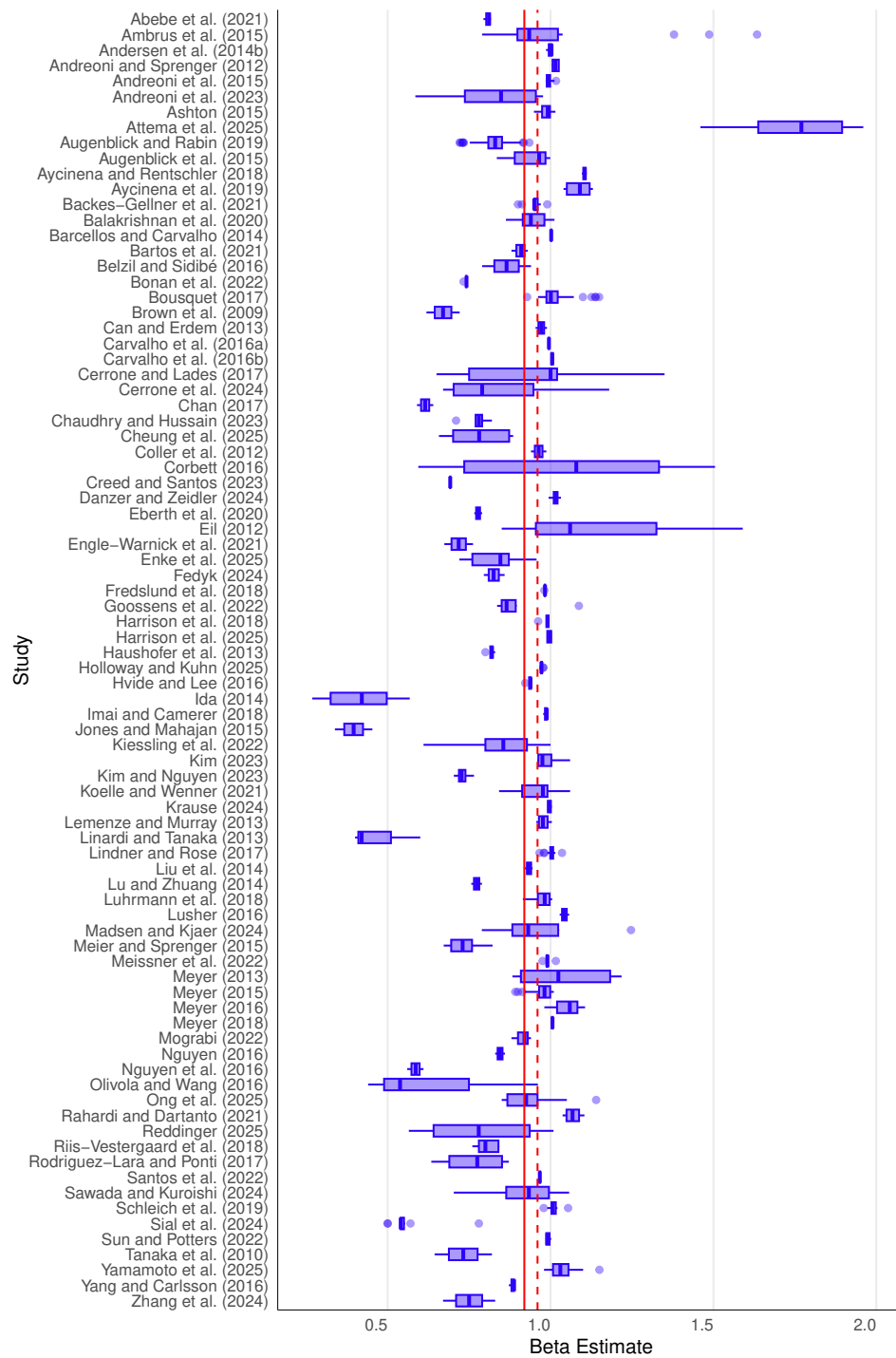


Figure 3.2: Forest plot of present bias estimates

Notes: This figure displays a box plot of reported estimates for each study, in alphabetical order. The solid vertical red line denotes the mean estimate for all studies, and the dashed red line denotes the median estimate. Only studies reporting more than one estimate are included; the full plot can be found in Figure A.2 in Appendix A. Extreme values are omitted from the graph, but included in all regressions.

et al. 2021). The gap narrows when using study-weighting, however, with 0.916 for money and 0.902 for real effort, suggesting that the difference may at least partly be driven by studies reporting a large number of estimates. In spite of that, the corresponding confidence intervals are fairly wide to draw any conclusions. Besides real effort, we also coded non-monetary outcome variables for food (consumables, snacks, beverages), health outcomes, and environmental goods. However, these categories are all based on only a small number of observations drawn from too few primary studies to allow meaningful statistical inference.

Estimates also vary noticeably across study settings. Laboratory and on-line studies report average β above the overall sample mean (0.934 and 0.952, respectively), while field studies report stronger present bias (unweighted mean 0.870). Panel (e) of Figure 3.3 shows a concentration of laboratory estimates around 1, whereas field estimates seem to be nearly evenly distributed amongst lower values.

Table 3.2: Mean statistics for selected subsets

	Unweighted		Weighted		N
	Mean	95% CI	Mean	95% CI	
All Estimates	0.919	[0.912, 0.927]	0.907	[0.885, 0.930]	1050
<i>Estimation Level</i>					
Aggregate	0.919	[0.912, 0.927]	0.914	[0.886, 0.942]	890
Individual	0.920	[0.897, 0.943]	0.897	[0.858, 0.935]	160
<i>Reward Domain</i>					
Money	0.937	[0.929, 0.945]	0.916	[0.893, 0.940]	747
Real effort	0.885	[0.872, 0.897]	0.902	[0.828, 0.976]	276
Food	0.746	[0.674, 0.818]	0.706	[0.608, 0.804]	7
Health	0.734	[0.667, 0.801]	0.733	[0.560, 0.907]	15
Environmental	1.008	[0.913, 1.104]	1.008	[0.794, 1.222]	5
<i>Incentives</i>					
Real	0.919	[0.912, 0.926]	0.910	[0.885, 0.935]	957
Hypothetical	0.926	[0.894, 0.959]	0.897	[0.844, 0.950]	93
<i>Study Type</i>					
Experiment	0.916	[0.909, 0.923]	0.909	[0.886, 0.932]	996
Survey	0.975	[0.931, 1.019]	0.891	[0.799, 0.983]	54
<i>Treatment Type</i>					
Treatment	0.925	[0.910, 0.940]	0.916	[0.850, 0.981]	223

	Unweighted		Weighted		N
	Mean	95% CI	Mean	95% CI	
Neutral Treatment	0.918	[0.910, 0.926]	0.906	[0.882, 0.930]	827
<i>Experimental Setting</i>					
Lab	0.934	[0.926, 0.941]	0.924	[0.889, 0.959]	592
Field	0.864	[0.848, 0.880]	0.873	[0.835, 0.911]	252
Online	0.952	[0.932, 0.973]	0.947	[0.900, 0.995]	174
School/Workplace	0.911	[0.870, 0.953]	0.864	[0.766, 0.961]	32
<i>Subject Pool</i>					
University Students	0.935	[0.927, 0.943]	0.930	[0.896, 0.965]	620
General population	0.897	[0.883, 0.910]	0.892	[0.860, 0.923]	378
Adolescents	0.904	[0.871, 0.936]	0.905	[0.811, 1.000]	46
Clinical	0.906	[0.777, 1.034]	0.870	[0.656, 1.085]	6
<i>Region</i>					
Africa	0.919	[0.894, 0.943]	0.887	[0.815, 0.958]	52
North America	0.878	[0.865, 0.891]	0.898	[0.858, 0.937]	345
Europe	0.963	[0.954, 0.971]	0.949	[0.915, 0.983]	436
Asia	0.884	[0.866, 0.901]	0.867	[0.821, 0.914]	198
<i>Elicitation Method</i>					
MPL	0.915	[0.900, 0.929]	0.915	[0.884, 0.945]	303
Matching	0.856	[0.806, 0.907]	0.853	[0.770, 0.936]	34
CTB	0.952	[0.944, 0.960]	0.942	[0.909, 0.975]	575
Work for Wage	0.826	[0.814, 0.838]	0.790	[0.682, 0.898]	122
<i>Estimation Method</i>					
Switch point	0.967	[0.943, 0.991]	0.911	[0.865, 0.958]	106
ML	0.876	[0.863, 0.889]	0.892	[0.851, 0.933]	312
OLS	0.953	[0.914, 0.992]	0.961	[0.877, 1.044]	28
NLS	0.929	[0.918, 0.941]	0.900	[0.851, 0.950]	313
Tobit	0.950	[0.939, 0.960]	0.950	[0.913, 0.987]	272
<i>Payment Method</i>					
Same method	0.923	[0.915, 0.931]	0.907	[0.883, 0.932]	900
Different method	0.898	[0.881, 0.915]	0.908	[0.847, 0.968]	150
Cash	0.901	[0.887, 0.914]	0.887	[0.848, 0.926]	262
Check	0.873	[0.856, 0.890]	0.985	[0.934, 1.036]	134
Bank	0.963	[0.944, 0.981]	0.955	[0.888, 1.023]	109
Voucher	0.942	[0.898, 0.987]	0.939	[0.854, 1.023]	20
PayPal	0.984	[0.974, 0.994]	0.955	[0.891, 1.019]	214
Postal	0.880	[0.834, 0.926]	0.914	[0.800, 1.029]	34
<i>Earlier Payment Timing</i>					
End of Experiment	0.923	[0.912, 0.933]	0.918	[0.881, 0.956]	402

	Unweighted		Weighted		<i>N</i>
	Mean	95% CI	Mean	95% CI	
End of Day	0.980	[0.972, 0.989]	0.958	[0.922, 0.995]	295
Different Day	0.846	[0.833, 0.859]	0.877	[0.810, 0.944]	191
<i>Publication Status</i>					
Published	0.913	[0.905, 0.922]	0.904	[0.878, 0.930]	681
Not published	0.931	[0.918, 0.943]	0.917	[0.873, 0.962]	369
Published top 5	0.843	[0.830, 0.856]	0.871	[0.778, 0.964]	133
<i>Elicitation Procedure</i>					
FED variation: Yes	0.933	[0.926, 0.940]	0.942	[0.919, 0.964]	941
FED variation: No	0.773	[0.747, 0.799]	0.790	[0.743, 0.837]	52
Nonlinear Utility: Yes	0.929	[0.922, 0.936]	0.924	[0.898, 0.950]	827
Nonlinear Utility: No	0.885	[0.865, 0.904]	0.882	[0.842, 0.922]	223
Certainty: Yes	0.937	[0.930, 0.945]	0.922	[0.894, 0.950]	660
Certainty: No	0.889	[0.875, 0.903]	0.895	[0.860, 0.929]	390
Trans. Costs Addressed: Yes	0.964	[0.957, 0.971]	0.952	[0.924, 0.980]	582
Trans. Costs Addressed: No	0.864	[0.853, 0.876]	0.873	[0.841, 0.904]	468

Notes: This table presents summary statistics across selected subsets of the data. Weighted mean uses the inverse of number of estimates per study as weights. *N* = number of observations. CI = confidence interval. MPL = multiple price list. CTB = convex time budget. FED = front-end delay. The definitions of variables can be found in Table A.3 in Appendix A.

Notably, there seems to be strong evidence for differences between geographical regions. Due to a shortage of estimates, we cannot compare countries or specific regions directly, and we aggregate at the continent level. European studies find a weak present bias, with an unweighted mean of 0.963, and 95% confidence interval of [0.954, 0.971], whereas Asian and North American studies report much stronger bias, with point estimates of 0.884 and 0.878, respectively. Panel (g) of Figure 3.3 shows European estimates concentrated around 1, while Asian estimates seem much more evenly distributed, and North American estimates show a seemingly bimodal distribution.

Lastly, there is systematically weaker present bias in studies that account for confounding factors, and by quite sizable margins. The largest raw difference is associated with transaction costs: studies that address them report an unweighted mean of 0.964, compared with 0.864 in studies that do not. This 10-percentage-point gap is larger than most raw cross-region differences in the sample. Such a pattern suggests that present bias may be relatively small once transaction costs are equalized. Addressing uncertainty increases β estimates

from 0.889 to 0.937 on average, while accounting for nonlinear utility from 0.885 to 0.926. This provides evidence for the much discussed estimation bias (Frederick *et al.* 2002), which is induced by failing to account for any of these factors. Reported parameters are then likely to mirror preferences other than the intertemporal ones.

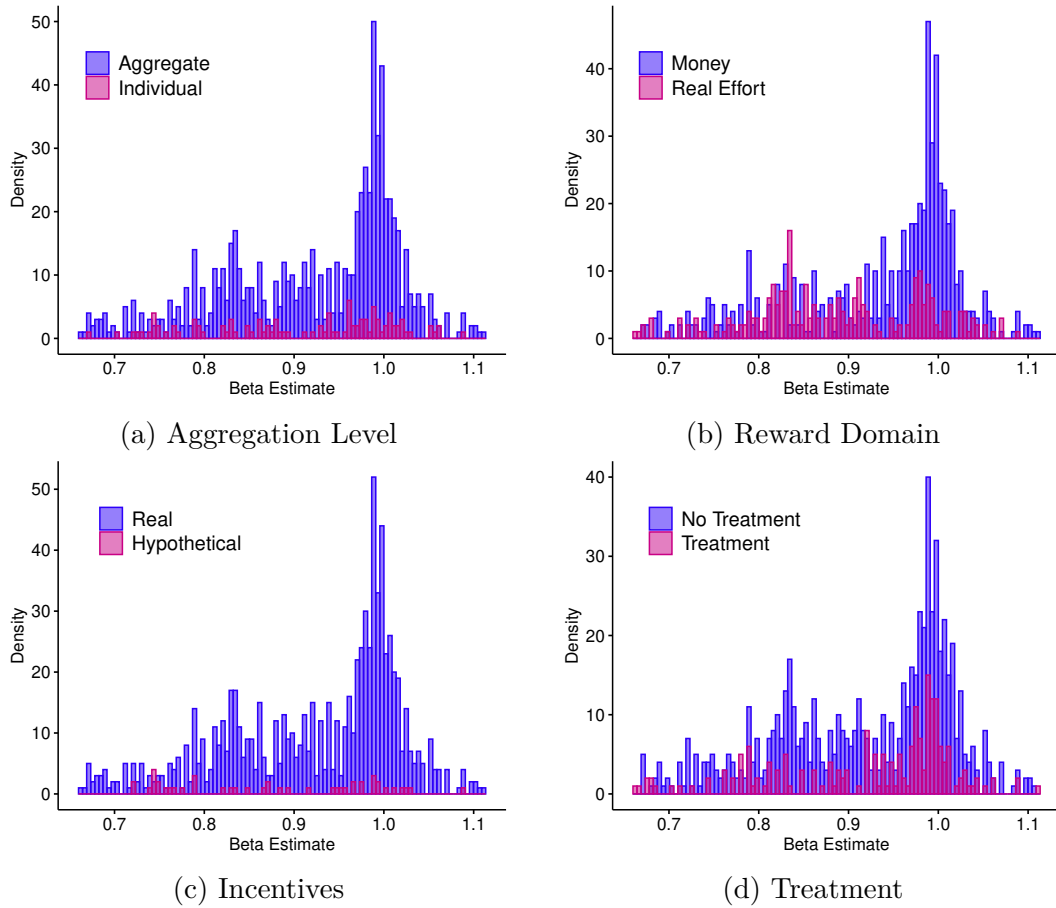


Figure 3.3: Differences across subgroups

Notes: This figure displays histograms and density plots of present bias parameter β estimates across selected subgroups of the data. CTB = convex time budget. MPL = multiple price list. WfW = work for wage. The definitions of variables can be found in Table A.3 in Appendix A

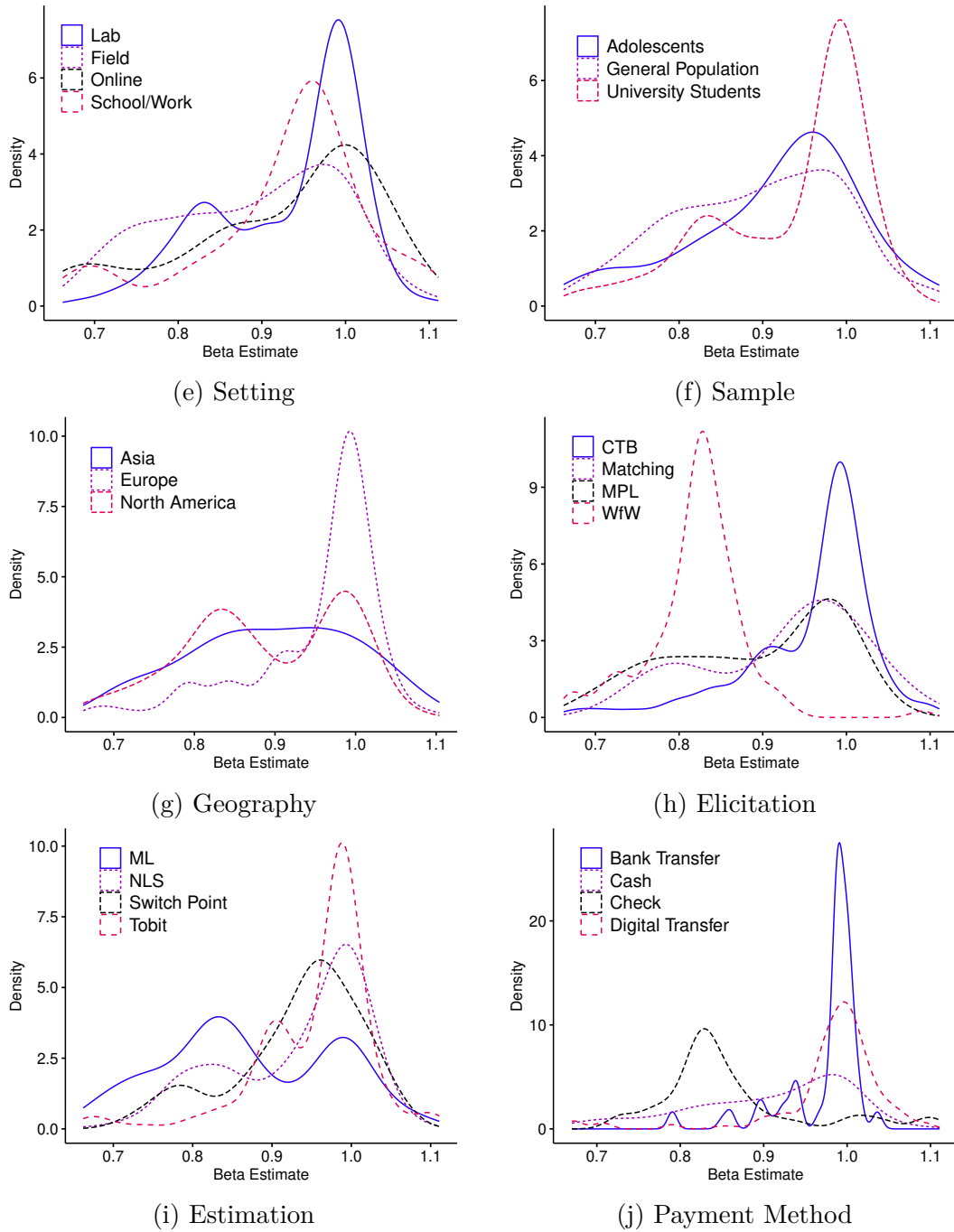


Figure 3.3: Differences across subgroups (continued)

Notes: This figure displays histograms and density plots of present bias parameter β estimates across selected subgroups of the data. CTB = convex time budget. MPL = multiple price list. WfW = work for wage. The definitions of variables can be found in Table A.3 in Appendix A

Chapter 4

Selective Reporting

4.1 Publication Bias and p -hacking

Selective reporting is defined as the tendency of researchers or publishers to report certain, selected, findings. Results that are statistically significant are published more frequently, since editors and reviewers tend to treat them more favorably. Similarly, findings that adhere to the conventional or prevailing theory are favored, or results that are predicted by current understanding are used as a baseline for model selection (Stanley 2005). Negative individual discount rates are, for example, perceived as absurd, yet given a small enough sample and liberal elicitation methods, obtaining negative estimates is not implausible, as documented in Matousek *et al.* (2022). Nevertheless, that is also how one obtains large positive estimates, which are not held to the same level of scrutiny. The systematic omission of certain estimates then manifests as a bias in the literature, commonly referred to as selection or publication bias, and threatens the validity and reliability of modern research for policy and practice (Ioannidis *et al.* 2017).

Testing and correcting for selective reporting is a vital part of any meta-analysis. Kvarven *et al.* (2019) compare results from large-scale pre-registered replications in psychology with 15 meta-analyses on the same topic and find reported meta-analytic effect sizes almost three times as large on average. Recently, Bartoš *et al.* (2024) investigate results from over 68,000 meta-analyses, and find the median probability of the presence of an effect decreased most prominently in economics, from 99.9% to 29.7%, after correcting for publication bias. To ignore selective reporting is therefore inexcusable in modern meta-analyses (Irsova *et al.* 2024).

Several features of the present-bias literature make it a natural candidate for selective reporting. The parameter β carries a strong theoretical prior: values below one correspond to the expected and intuitive outcome of present bias, self-control problems, and procrastination, while values above one correspond to future bias, which is widely regarded as behaviorally implausible. This asymmetry in plausibility creates a corresponding asymmetry in how estimates are treated: findings consistent with β below one confirm the prevailing theory and tend to be viewed as interpretable and interesting, whereas estimates near or above one are liable to be dismissed as noise, reinterpreted, or simply not emphasized.

The convention of treating β as effectively bounded in $[0, 1]$ reinforces this asymmetry, since experimental designs and estimation procedures can be calibrated around this range, shaping the distribution of reported estimates before any editorial stage is reached. On top of these parameter-specific pressures, the usual forces operating across empirical economics apply as well: statistically significant and theoretically aligned results are easier to publish, and the discretion involved in design and estimation leaves room for selective specification. Together, these considerations motivate a careful examination of whether the reported evidence on β is representative of the underlying distribution of estimates, or has been shaped toward the prior.

Selective reporting operates on two distinct, yet intertwined levels. Selection on the publication level, or across-study selection, implies reviewers and editors discriminating studies based on otherwise unbiased results. This type of selection is commonly known as publication bias in the narrow sense, as it operates on the *publication* level. However, researchers themselves may decide against submitting results they anticipate would not pass peer review; hence it is also sometimes referred to as the file-drawer problem (Rosenthal 1979). Methods to quantify and correct for publication bias have been developed since 1979 (Marks-Anglin & Chen 2020), and broadly fall into two families: funnel plot asymmetry tests and methods based on selection models (Jin *et al.* 2015).

Selection on the analysis level, or within-study selection, involves researchers cherry-picking amongst models, data or methodology to provide a conforming or statistically significant result. Commonly referred to as *p*-hacking, within-study selection distorts reported individual point estimates themselves, and can thus circumvent traditional publication bias tests which assume a selection on individually unbiased estimates (Bruns *et al.* 2024; Mathur 2024; Irsova *et al.* 2024).

From the battery of selective reporting tests we employ, funnel-asymmetry tests are best-viewed as a first line of defense against across-study selection, since they primarily detect systematic relationships between reported estimates and their precision across studies. By contrast, selection models, and, in particular, the RTMA method, can also speak to within-study selection, because they are designed to capture distortions arising from selective specification search and p -hacking within papers.

4.2 Funnel Plot

As a starting point, the funnel plot serves as a visual aid for most publication bias tests. It is essentially a scatter plot of present bias estimates ($\hat{\beta}$) on the horizontal axis against their precision ($1/SE$) on the vertical axis. The most precise estimates (ones with the lowest standard error) are at the top of the figure, and are assumed to lie close to the true mean, while more imprecise estimates at the bottom are more diffused. Importantly, in the absence of publication bias the estimates should form an inverted funnel that is symmetrical about the true effect (which, in the absence of publication bias, coincides with the sample mean). If the funnel plot is asymmetrical, it implies a publication bias in favor of either smaller or larger results. Moreover, if the funnel appears wide and hollow, it signifies a preference for statistically significant estimates (Stanley 2005; Stanley & Doucouliagos 2010). The funnel plot for all present bias estimates is presented in Figure 4.1.

At first glance, the constructed funnel plot resembles the desired shape; three features, however, warrant comment. First, the most precise estimates appear around $\beta = 1$, meaning no present bias, which is however considerably different from the sample mean of 0.919, indicating a noticeable discrepancy between the most precise estimates and the average estimate in the literature. Second, the lower left part of the funnel appears denser than the lower right, suggesting that less precise estimates implying stronger present bias are reported more frequently than equally imprecise estimates implying weaker present bias or future bias. Third, although the plot broadly retains the expected shape of a funnel, the distribution appears relatively compact around $\beta = 1$, with limited dispersion to the right, which may point to some truncation or compression of estimates near the benchmark of no present bias. Future bias is, anecdotally, recognized as unlikely by most people, and the convention is then to consider the present bias parameter $\beta \in [0, 1]$. If researchers restrict

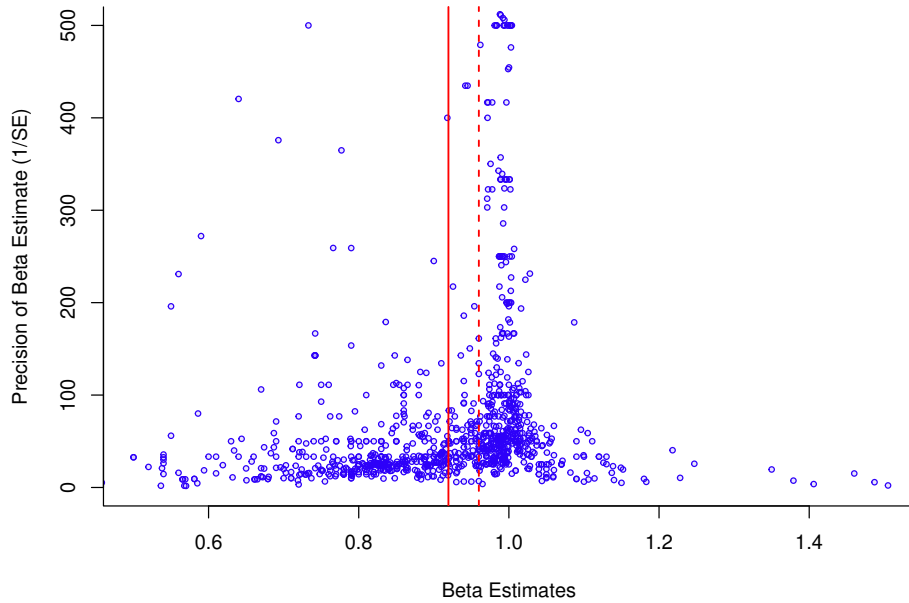


Figure 4.1: Funnel Plot

Notes: This figure displays the funnel plot of present-bias parameter (β) estimates. The solid vertical red line denotes the mean estimate, and the dashed red line denotes the median estimate. Extreme values are omitted from the graph, but included in all regressions.

β to this interval by design, the resulting truncation of the distribution would mimic publication bias visually. I return to this confound in Section 4.6.

The funnel plot is nevertheless only a visual tool, and its asymmetry can have multiple sources. Egger *et al.* (1997) note that one such source is true heterogeneity of the underlying effect. When effect sizes differ systematically across substantively distinct subgroups, pooling them in a single funnel may generate a pattern that resembles publication-bias asymmetry. For example, Bortnikova *et al.* (2024) show that the reported estimates of the beauty premium display markedly different funnel plots for sex workers and the rest of the sample. The interpretation of funnel asymmetry is therefore not straightforward and should not rely on visual inspection alone.

4.3 Linear Tests

Linear regression provides a simple formal examination of selective reporting and a quantification of the extent of publication bias. Based on the funnel

plot, the essential assumption in these tests is the independence of empirical estimates and their standard errors in the absence of publication selection (Egger *et al.* 1997; Stanley & Doucouliagos 2010). The estimated equation can be written algebraically as

$$\hat{\beta}_{ij} = \beta_1 + \gamma \cdot SE(\hat{\beta}_{ij}) + \epsilon_{ij}, \quad (4.1)$$

where $\hat{\beta}_{ij}$ is the i -th estimate of the present bias parameter from the j -th study, $SE(\hat{\beta}_{ij})$ is its corresponding standard error, and ϵ_{ij} is the error term. The slope coefficient γ then quantifies publication bias, while the intercept β_1 represents the true effect beyond bias. This simple regression model for detecting publication bias is also referred to as the Funnel Asymmetry Test (FAT)-Precision Effect Test (PET) procedure (Stanley & Doucouliagos 2012).

Table 4.1: Linear tests for publication bias

	OLS	Study	Precision
Effect Beyond Bias	0.944***	0.918***	0.961***
SE	(0.019)	(0.013)	(0.012)
Bootstrap CI	[0.900, 0.985]	[0.891, 0.944]	[0.932, 0.986]
Publication Bias	-0.734	-0.325	-1.236**
SE	(0.525)	(0.404)	(0.493)
Bootstrap CI	[-1.850, 0.562]	[-1.180, 0.552]	[-2.310, -0.140]
Total observations	1,050	1,050	1,050
	FE	BE	RE
Effect Beyond Bias	0.924***	0.921***	0.913***
SE	(0.005)	(0.015)	(0.010)
Publication Bias	-0.138	-0.431	-0.191
SE	(0.129)	(0.352)	(0.121)
Total observations	1,050	1,050	1,050

Notes: This table displays the results obtained from estimating Equation 4.1. SE = Standard Error, CI = Confidence Interval, OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Precision = Estimates are weighted by the inverse standard error. Study = Estimates are weighted by the inverse number of observations reported per study. Standard errors, clustered at the study level, are included in parentheses. Wild bootstrap confidence intervals at 95% confidence level, bootstrapped over 9,999 iterations, are reported in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

We estimate the equation using various specifications. As baseline, we use standard Ordinary Least Squares (OLS). Next, we use two weighted least

squares specifications with the number of estimates per study (Study) and estimate precision ($1/SE$) (Precision) as weights. The former downweights the dominance of studies reporting many estimates by giving each study the same weight and the latter deals with the inherent heteroscedasticity in the regression equation. To account for unobserved heterogeneity, we employ study-level Fixed Effects (FE), Between Effects (BE), which uses the mean estimate and standard error from each study, and Random Effects (RE) estimators. All standard errors are clustered at the study level, and bootstrap confidence intervals are computed where possible.

The results of all linear tests for all estimates are presented in Table 4.1. Five out of the six linear specifications cannot reject the null of no publication bias at any conventional significance level. Only precision-weighted OLS suggests a negative bias significant at the 5% level and corrected effect of 0.961, higher than the unweighted mean of 0.919 and the five other specifications. Simple OLS however also shows a higher corrected effect of 0.944 relative to the unweighted mean. The rest of the corrected effect sizes are comparable to the mean. The results overall thus provide only weak evidence of selective reporting, driven primarily by the precision-weighted specification. Since linear tests assume a linear estimate–standard error relationship that may not hold here, we turn to nonlinear tests in the next section.

4.4 Nonlinear Tests

The critical assumption of the FAT-PET procedure requires a linear relationship between the estimate and standard error. The assumption is found generally unlikely to hold in practice (Bom & Rachinger 2019). Furthermore, Stanley & Doucouliagos (2014), among others, show that while linear tests are useful for testing for the presence of publication bias and of a true underlying effect, they can provide biased results of the corrected effect when it is non-zero. To address these concerns, we use six different non-linear estimation techniques in total. Results for all nonlinear techniques are displayed in Table 4.2.

The first method is the Weighted Average of Adequately Powered estimates (WAAP) developed by Ioannidis *et al.* (2017). Essentially, the method selects the estimates with statistical power at least 80%—those adequately powered—and computes their inverse-variance weighted average. In the case of the present

Table 4.2: Nonlinear tests for publication bias

	WAAP	Top 10	Stem
Effect Beyond Bias <i>SE</i>	0.950*** (0.004)	0.962*** (0.008)	0.970 (0.025)
Total observations	1,050	1,050	1,050
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias <i>SE</i>	0.964*** (0.003)	0.034*** (0.011)	0.012*** (0.000)
Publication Bias <i>SE</i>	-1.463*** (0.392)	0.132*** (0.005)	0.026 (0.000)
Effect Beyond Bias (β)		0.966	0.988
Total observations	1,050	1,050	1,050

Notes: This table reports results from nonlinear publication bias correction methods. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method applied on non-winsorized dataset. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis method. DPB = Degree of Present Bias $X = 1 - \beta$. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

bias parameter β , we classify an estimate $\hat{\beta}_{ij}$ as adequately powered if it satisfies

$$|\hat{\mu}_W - 1|/2.8 > SE(\hat{\beta}_{ij}),$$

where $\hat{\mu}_W$ is the unrestricted WLS estimate using all estimates. The constant 2.8 is the sum of the two-sided 5% critical value (1.96) and the critical value corresponding to 80% power (0.84). It therefore reflects the standard threshold for an estimate to have at least 80% statistical power to detect a deviation from the null of no present bias $\beta = 1$ (Stanley *et al.* 2017). This procedure yields 465 adequately powered estimates out of the total 1,050. The results show that among nonlinear methods, WAAP produces the smallest correction from the sample mean (0.950, against the unweighted mean of 0.919), but even this is larger than most linear corrections.

Second is the Top 10 method (Stanley *et al.* 2010), which analyzes the 10% of estimates with the highest precision. The idea is to focus only on the top part of the funnel where estimates are assumed to be subject to selection based on statistical significance the least. The Top 10 method results in a corrected effect size of 0.962, suggesting a nontrivial bias relative to the unweighted mean.

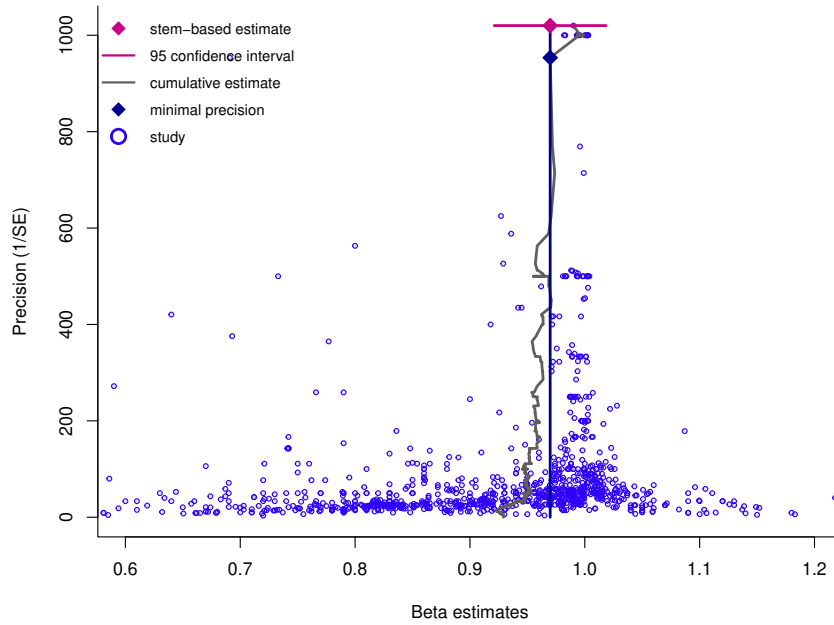


Figure 4.2: Stem-based method

Notes: This figure displays the plot based on the stem-based estimation method by Furukawa (2021). The purple diamond marks the corrected estimate of the present bias parameter β , with the purple line indicating the 95% confidence interval. The grey line shows how the estimate changes when less precise estimates are included. The blue diamond indicates the minimum precision level that all included studies satisfy.

The stem-based method introduced by Furukawa (2021) is a generalization of the Top 10 method. Whereas Top 10 declares a somewhat arbitrary cutoff at 10%, the stem-based bias correction method determines the number of included most precise estimates (the stem of the funnel plot) endogenously—by minimizing the estimated mean squared error, thus optimizing the compromise between efficiency and bias (Imai *et al.* 2021; Matousek *et al.* 2022). The method relies on the validity of the most precise estimates; however, in our case, the lowest standard errors are compressed due to winsorization, which produces multiple estimates with artificially identical precision. This undermines the method’s selection of estimates, and, for that reason, we decide to use the original, non-winsorized dataset. The result is robust to outliers—the point estimate remains unchanged when excluding the estimates that are winsorized, and becomes statistically significant at the 5% level. Figure 4.2 presents the plot complementing the estimation results of the stem-based method. The results yield a corrected estimate of 0.97, in agreement with both prior nonlinear estimates; however the estimate is statistically indistinguishable from 1 on any conventional significance level, meaning it cannot rule out no present bias.

The fourth method is the Endogenous Kink (EK) technique by Bom & Rachinger (2019), which, similarly to the two previous methods, assumes that sufficiently precise studies are unsullied by publication selection. The EK model fits a piece-wise linear regression of estimates on their standard errors, with a kink at the cutoff point below which standard errors are low enough to assume no selection. The regression then consists of a constant segment, with no relationship between estimates and standard errors, and a sloped segment indicating publication bias. The EK method presents a corrected estimate of 0.964, similar to previous results, and negative publication bias significant at any conventional level.

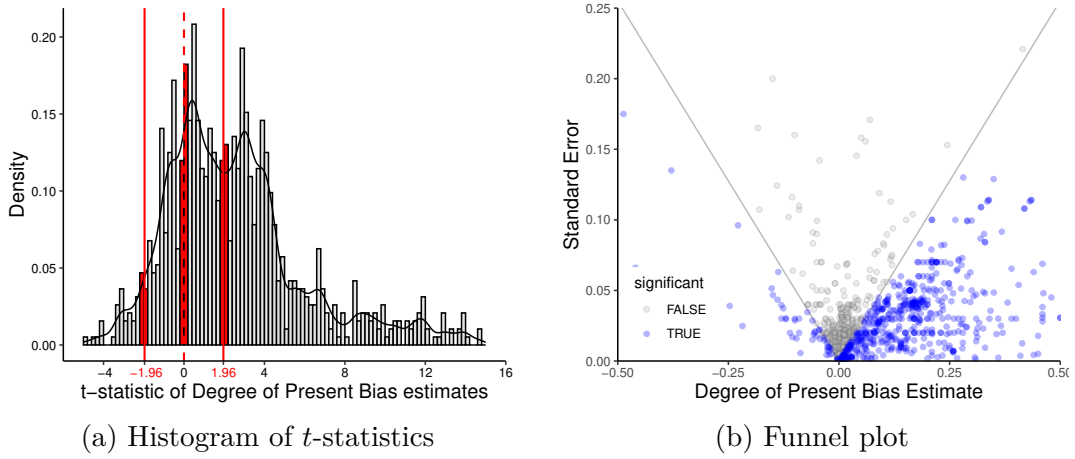


Figure 4.3: Complementary plots for nonlinear methods

Notes: This figure displays complementary plots for nonlinear tests for publication bias. Plot (a) presents the histogram and density of t -statistics of Degree of Present Bias (DPB) estimates, where the dashed red line denotes zero (no present bias) and the solid red lines denote the thresholds for 5% statistical significance ± 1.96 . Bins just below and above thresholds are highlighted. Plot (b) shows the distribution of DPB estimates against their standard errors. The gray lines mark the ± 1.96 thresholds. Significant estimates are colored blue, while insignificant estimate are colored gray.

The last two nonlinear methods are formulated around the null hypothesis of no effect $H_0 : \theta = 0$, where θ is the parameter of interest. Since the null of no effect for the present bias parameter corresponds to $H_0 : \beta = 1$, we follow Imai *et al.* (2021) and use a linear transformation by defining DPB as $X = 1 - \beta$, such that the null hypothesis then becomes $H_0 : X = 0$. The transformed estimates then follow $\hat{X}_{ij} = 1 - \hat{\beta}_{ij}$, and the stronger the present bias, the higher the estimate of DPB. Figure 4.3 shows the distribution of t -statistics recomputed for DPB and the distribution of the estimates and standard errors.

The selection model by Andrews & Kasy (2019) models the conditional

publication probability $p(\cdot)$ for latent estimates based on their z -statistic; it assumes constant publication probabilities for its intervals, for example based on the conventional thresholds of -1.96 and 1.96 signifying statistical significance at the 5% level. The observed estimates are then re-weighted by the inverse of the publication probability recovering the unbiased distribution of effect estimates. Note that publication probability in this case does not necessarily indicate the probability of being published in a journal per se, but the probability that the meta-analyst observes the (latent) result, for example as a working paper or unpublished manuscript (Imai *et al.* 2021). The selection model yields a corrected effect size of 0.034, which, transforming back to β , corresponds to 0.966, very much in line with the other results. The method also suggests a publication bias significant at any conventional level.

The novel Right-Truncated Meta-Analysis (RTMA) method introduced by Mathur (2024) is unique since, compared to previous nonlinear methods, it explicitly tackles p -hacking, or, more broadly, selection within studies. The method hinges on the idea that under various forms of p -hacking, the distribution of published affirmative (significant and positive) estimates is distorted, whereas the distribution of published non-affirmative (negative, or non-significant) estimates remains accurate. Result selection on the researcher level then induces an observed distribution that has been right-truncated. RTMA then imputes the underlying distribution of population effects using Bayesian methods based on the published non-affirmative part of the distribution. RTMA yields the highest point estimate of the nonlinear methods of 0.988 (0.012 as DPB), nonetheless still significantly different from no present bias, though the implied economic effect is tiny. Mathur (2024) suggests examining two diagnostic plots for identifying violations of the assumptions underlying RTMA. The quantile-quantile plot in Figure C.1 in Appendix C unfortunately suggests likely violations of the model's assumptions. We therefore do not treat RTMA as a primary estimator, but report it for completeness and as a check for the direction of bias.

Overall, the nonlinear methods shown in Table 4.2 provide stronger evidence of publication bias than the linear tests: all corrected estimates are at or above 0.95, implying only mild present bias after correction, and both the Endogenous Kink and Andrews and Kasy methods point to statistically significant negative publication bias.

4.5 Allowing for Endogeneity

The preceding publication bias tests all hinge on a common, principal assumption that standard errors are exogenous, which implies that any correlation between them and the corresponding parameter estimates is due to publication bias. However, the correlation can also arise due to unobserved heterogeneity—methodological choices or measurement errors that affect both the estimate and its standard error, which are also prone to p -hacking (Irsova *et al.* 2024; 2025).

Table 4.3: Endogeneity-robust tests for publication bias

	p -uniform* (DPB)	MAIVE
Effect Beyond Bias	0.111***	1.090*
<i>SE</i>		(0.047)
(<i>Anderson-Rubin</i>) <i>CI</i>	[0.081, 0.140]	[0.973, 1.207]
Publication Bias	L = 37	-6.400***
<i>SE</i>	($p = 0.001$)	(1.960)
<i>Anderson-Rubin</i> <i>CI</i>		[-10.580, -3.610]
Effect Beyond Bias (β)	0.889	
First-stage F-statistic		0.844
Total observations	1,050	1,050

Notes: This table reports results of endogeneity-robust publication bias correction methods. p -uniform* = method by van Aert & van Assen (2026), where L represents the publication bias test statistic with the corresponding p -value for the null of no publication bias in parentheses. MAIVE = Meta-Analysis Instrumental Variable Estimator by Irsova *et al.* (2025). DPB = Degree of Present Bias $X = 1 - \beta$. Standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

We utilize two novel methods to counter the endogeneity issue. Firstly, the p -uniform* method by van Aert & van Assen (2026), which is a generalization and improvement of the p -uniform (Van Assen *et al.* 2015) and p -curve (Simonsohn *et al.* 2014) publication bias correction methods. p -uniform* is based on a selection model approach, and depends on the statistical principle that p -values are to be uniformly distributed at the true effect size. The method then involves estimating the true effect via maximum likelihood, so that the implied p -value distribution is consistent with a uniform one (van Aert & van Assen 2026; Marks-Anglin & Chen 2020). p -uniform* yields a corrected point estimate of 0.111, which corresponds to a β of 0.889, implying a stronger present bias than the unweighted mean of 0.919; it is thus the only nonlinear method suggesting that the literature over-reports β (i.e., underestimates present bias).

It also yields an L statistic of 37 and a corresponding p -value of 0.001 for the null of no publication bias.

The second endogeneity-robust method is the Meta-Analysis Instrumental Variable Estimator (MAIVE) introduced by Irsova *et al.* (2025), which uses the inverse of sample size as an instrument for reported variance. Using two-stage least squares, MAIVE uses the following regression:

$$SE(\hat{\beta}_{ij})^2 = \psi_0 + \psi_1(1/N_{ij}) + \nu_{ij},$$

where N_{ij} is the size of the sample used for the estimation of $\hat{\beta}_{ij}$, and ν_{ij} is the error term. Reported standard errors can then be replaced in any meta-analysis technique by the fitted values $\sqrt{\hat{\psi}_0 + \hat{\psi}_1(1/N_{ij})}$. Using an instrumental variable approach remedies the aforementioned concerns—sample size is assumed to not be affected by methodological changes, is not subject to measurement error, and is arguably difficult to be used for p -hacking. We use the PET-PEESE specification (Stanley & Doucouliagos 2014) without inverse variance weights as suggested by the authors. The reported F -statistic from the first-stage regression is merely 0.844—far below the conventional threshold of 10, suggesting a weak instrument and rendering the point estimates unreliable. We therefore also report the computed 95% Anderson-Rubin confidence intervals, which are robust to weak instruments, as primary inference (Irsova *et al.* 2025; Keane & Neal 2024). The computed valid confidence interval for the corrected effect ranges from 0.973 to 1.207, unable to reject no present bias. Even the interval's lower bound is higher than most nonlinear corrected estimates. MAIVE also suggests a significant negative publication bias, with the corresponding Anderson-Rubin interval excluding zero.

The results of both MAIVE and RTMA indicate larger magnitudes of publication bias than simpler methods, which is consistent with possible within-study selection in the literature. Nevertheless, both methods in our case suffer from violations of their specific assumptions, unable to reliably uncover any telltale signs of p -hacking. Any strong inference on this matter would be speculation at this point.

4.6 Caliper Tests

Additionally, we conduct caliper tests introduced by Gerber & Malhotra (2008). Caliper tests are commonly used to compare the number of observations above

and below psychologically important values of the t -statistic (± 1.96) in sufficiently narrow intervals, uncovering any asymmetries in reporting around these significance thresholds.

Table 4.4: Caliper tests for different ranges of present-bias estimates

Caliper test for $\beta \in [0.80, 1.20]$	OLS	Precision
Publication bias	0.383***	-0.640***
SE	(0.126)	(0.124)
Observations	864	864
Caliper test for $\beta \in [0.90, 1.10]$	OLS	Precision
Publication bias	-0.060	-0.195*
SE	(0.117)	(0.104)
Observations	603	603
Caliper test for $\beta \in [0.95, 1.05]$	OLS	Precision
Publication bias	-0.014	0.027
SE	(0.087)	(0.072)
Observations	462	462
Caliper test for $\beta \in [0.55, 0.95]$	OLS	Precision
Publication bias	-1.147***	-0.488
SE	(0.142)	(0.302)
Observations	491	491
Caliper test for $\beta \in [0.60, 0.90]$	OLS	Precision
Publication bias	-0.839***	0.159
SE	(0.130)	(0.238)
Observations	381	381
Caliper test for $\beta \in [0.70, 0.80]$	OLS	Precision
Publication bias	-0.009	0.022
SE	(0.100)	(0.128)
Observations	101	101

Notes: This table reports the results of caliper regressions obtained from estimating Equation 4.1 for different intervals of present-bias estimates β . Only estimates within the indicated bounds are included in each regression. OLS = Ordinary Least Squares. Precision = Estimates are weighted by the inverse standard error. Standard errors, clustered at the study level, are included in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In our case, however, we apply the caliper tests in an alternative manner, which we find more appropriate due to the nature of the parameter in question. Because experimental designs are often calibrated around $\beta = 1$, we test for publication bias within narrow values of β values (rather than of the

t -statistics). Borrowing the approach of Matousek *et al.* (2022), who conduct analogous tests for the estimates of the discount rate, we focus on the baseline linear regression using OLS and precision weights and test for the presence of publication selection bias on two sets of intervals of the present bias estimates.

First, we focus our attention on estimates around one, suggesting only weak present or future bias. Studies' elicitation and estimation methods are likely to be calibrated to precisely estimate values around one, relative to the full range of possible β values. Hence, a correlation between the estimates and standard errors may arise regardless of selection. If the correlation persists for these smaller windows, then the full-sample asymmetries are less likely to be driven by this notion.

In a similar vein, experiments may be specifically designed to elicit only values below one, discriminating against results suggesting future bias. We therefore also conduct the tests for values of β that are safely below one. If the correlation survives here, then it is not fully driven by experiment design allowing only for present, and not future, bias.

The results are presented in Table 4.4. We do not show the corrected point estimates since they are not informative for the restricted samples. The results show the correlation waning as the caliper narrows in both sets of windows. Hence, we cannot rule out that the weak full-sample asymmetry previously described reflects either elicitation methods calibrated around $\beta = 1$, or experimental designs that mechanically truncate β at 1, rather than selective reporting per se. Note that, however, the linear relationship was very weak to begin with: OLS showed no significant publication bias on the whole sample, and using precision weights was the only specification where the standard error's coefficient was significant at the 5% level. The results are then to be interpreted with caution, as the relationship between estimates and standard errors is likely nonlinear in our case.

4.7 Robustness Checks

As robustness checks, we conduct the set of whole-sample publication bias tests for all estimates winsorized at the 2.5% level, and restricting the sample to experimental studies only. The full results can be found in Table C.1 and Table C.2 in Appendix C for 2.5% winsorization and experiments, respectively.

The linear tests for 2.5% winsorization suggest a predominantly *positive* publication bias, significant in three out of six specifications. The corrected

point estimates decrease to around 0.89, compared to the unweighted mean of 0.933. The nonlinear tests, however, provide results that are either quantitatively very similar to, or at least qualitatively consistent with our main results. This discrepancy implies that there exists a small number of outlier estimates above $\beta = 1$ that, when considered, disproportionately influence the linear specifications, affecting the estimated estimate–standard error relationship. The nonlinear methods are robust to such outliers and hence corroborate our main results.

The majority of the data comes from experimental studies (996 out of 1050 estimates). By running the publication bias tests on this subset, we can gauge whether including survey-based studies in our dataset affects the main analysis. The results for experimental studies are qualitatively identical and quantitatively very similar to the main results: mixed evidence of a negative publication bias in linear tests, and predominantly significant publication bias in nonlinear methods, with corrected point estimates mostly in the 0.95–1.0 range. Therefore, we can confirm that our results are also robust to including survey-based studies.

4.8 Selection Within Subgroups

Thus far, we have considered selective reporting only for the literature as a whole; publication selection bias can nevertheless also be correlated with other variables (Stanley & Doucouliagos 2012), meaning, publication bias may not be ubiquitous in the literature, but can be exclusive, or at least more prevalent, for certain subgroups. Doucouliagos & Stanley (2009) conduct a meta-analysis on minimum-wage research, and in their meta-regression analysis find significant correlations between the estimate standard error and two model specification variables. Examining 13,440 hypothesis tests reported in top economics journals, Brodeur *et al.* (2020) show that the severity of publication bias and *p*-hacking substantially differs across identification methods. Similarly, Havránek *et al.* (2024) find considerably less publication selection bias among IV estimates, and estimates from studies in developing countries in their meta-analysis of studies measuring skill substitution.

Motivated by this prior evidence, and by the considerable differences in sample means within certain subsets of the data presented in Table 3.2, which, apart from true heterogeneity might also mirror varying inclinations to selectively report results, we investigate whether publication selection bias is not

concentrated in certain subgroups, where it might be induced by prior expectations, study characteristics, settings or methods. We provide results for the same battery of tests employed for the whole sample for each of the subgroups in Appendix C. This serves as a sort of prelude to the following heterogeneity section, where we use a meta-regression approach identifying the drivers of heterogeneity in present bias estimates.

Reported present bias estimates for real effort (unweighted mean = 0.885) are substantially lower than for monetary rewards (0.937). We analyze whether the diversity in the reported present bias is exaggerated by the preference to adhere to the prevailing theory, past empirical findings, and common intuition: that present bias for real effort, which reflects the degree of procrastination on tasks, is strong, and widely reported in the literature. In their meta-analysis of CTB estimates, Imai *et al.* (2021) find publication bias for real effort, whereas no substantial evidence for selection in the monetary domain. On the other hand, Cheung *et al.* (2025) find moderate publication bias for monetary present bias, but no evidence for selection among the broader category of non-monetary estimates.

Our nonlinear tests for money are consistent with very modest selective reporting, with a corrected effect around 0.95–0.97, and in some models even closer to one, with mixed significance of publication bias. Results for real effort, on the other hand, show evidence consistent with a stronger publication selection bias, with corrected point estimates mostly between 0.95–0.98 in both linear and nonlinear models. While the significance of publication bias is also mixed and some methods imply lower values, the overall tendency to correct estimates closer to one is apparent.

Studies conducted in the field report substantially lower estimates (0.864) than studies conducted in a laboratory (0.934) or online (0.952). The potential noise in field studies induced by confounding factors, an uncontrolled environment relative to laboratory experiments, and the promise of providing greater external validity and policy relevance may provide both leeway and incentive to report stronger effects. Hence, we analyze and compare publication bias between the three settings.

Results for lab are consistent with some degree of selective reporting, with most nonlinear models implying a corrected estimate between 0.98 and 1, suggesting only a very minor effect. The significance of publication bias is mixed across methods. Results for online studies share a similar tendency. For field studies, however, publication bias is ubiquitously significant (except for

MAIVE), with corrected point estimates being quite heterogeneous, but some nonlinear methods agreeing on a coefficient around 0.91–0.93, which is still substantially lower than corrected laboratory results.

Each estimation technique entails making different assumptions and various degrees of researcher discretion. Maximum likelihood, nonlinear least squares, and Tobit are all, to some degree, subject to functional form choices, parameter specifications, or censoring thresholds—decisions made by the researcher. We investigate whether publication selection is associated with methods that allow for more *wiggle room* (Brodeur *et al.* 2020).

Results for subsets estimated by ML, NLS, and Tobit all tell a similar story: mostly significant publication bias, with higher corrected estimates, specifically around 0.93–0.97 for ML (compared to 0.876 unweighted mean), 0.96–1.0 for NLS (0.929), and 0.97–1.0 for Tobit (0.95). Switch point estimates, however, show a predominantly significant *positive* correlation between estimates and standard errors. The unweighted mean of 0.967 gets, in most instances, corrected to considerably lower point estimates, around 0.89–0.91. This is surprising, as the relationship in this particular subset is opposite to what results suggested up to this point. One possible explanation is mechanical, rather than selective, and has to do with the design of the experiment and the reporting of results: switch-point estimates are mostly individually estimated, and then reported as summary statistics, where we take the mean as representative. As was noted in section 4.6, researchers may specifically design experiments so that they precisely estimate values close to, or below one. If then an individual displays future bias, the estimation of the individual’s β parameter may be less precise, generating higher values and greater dispersion. These future-biased outliers would then inflate the mean estimate and standard deviation, creating the observed asymmetry. The proposed mechanism is consistent with the results, but is nevertheless only speculation.

Further subgroup analyses do not uncover any consequential differences in publication selection bias. Results for both published and unpublished studies, as well as for preferred and appendix estimates, follow largely the same patterns as the whole sample. North American estimates seem to be subject to more selection than European or Asian estimates, but the effect is not robust after controlling for other potential confounding variables.

Overall, the subgroup analysis reveals three patterns that motivate the heterogeneity analysis in the next chapter. First, the evidence for selective reporting appears strongest in real-effort and field studies, where several cor-

rection methods tend to shift estimates of β closer to one. Second, switch-point estimates display a markedly different, and in some cases inverted, estimate–standard error relationship, which is more plausibly explained by mechanical features of the design and reporting of these estimates than by selective reporting alone. Third, we find no consistent difference in publication selection between published and unpublished work, nor between preferred and appendix estimates. We therefore next turn to a meta-regression framework that jointly controls for study characteristics and allows the estimate–standard error relationship, and hence the extent of selective reporting, to vary systematically across subgroups.

Chapter 5

Heterogeneity

In the previous sections, we have on several occasions encountered considerable differences between estimates of present bias in various subgroups. In this chapter, we thoroughly analyze the sources of heterogeneity by regressing present bias estimates on collected variables reflecting study design, setting, and publication. To account for model uncertainty we utilize model averaging techniques. Subsequently, we also confirm whether our previous findings about selective reporting hold up when accounting for covariates.

5.1 Explanatory Variables

In total, we collected information on 116 variables; the full list can be found in Table A.3 in Appendix A. We narrowed down the variables which potentially reflect the differences in present bias estimates. These are included in both model averaging techniques, grouping (or excluding if there are no variables to be grouped with) variables which have low variability and omitting variables which have a high degree of collinearity or a variance inflation factor above 10. Missing values for variables with less than 1,050 observations were imputed using the median observation for numerical and dummy variables, and using the mean observation for ratios (the variable Female). Based on the subgroup analyses of selective reporting in the previous chapter, we also include the interaction terms $SE \times \text{Real Effort}$, $SE \times \text{Field}$, and $SE \times \text{Switch Point}$, capturing the distinct estimate–standard error relationships for these subsets. Lastly, several dummies serve as reference categories to avoid the dummy variable trap in each categorical group. For ease of exposition, we group the variables into ten categories. A description and summary statistics

of the 54 included explanatory variables and omitted reference variables can be found in Table 5.1.

Table 5.1: Summary statistics for explanatory variables

Variable Name	Description	Mean	SD
SE	Standard error of present bias estimate	0.033	0.029
SE $\times$ Real Effort	Interaction term between standard error and real effort domain	0.013	0.025
SE $\times$ Field	Interaction term between standard error and field setting	0.007	0.019
SE $\times$ Switch Point	Interaction term between standard error and switch point estimation	0.004	0.019
<i>Estimate Characteristics</i>			
Preferred	= 1 if estimate is presented as main result by authors	0.337	0.473
Appendix	= 1 if estimate is in study appendix	0.472	0.499
Aggregate*	= 1 if estimated on aggregate level	0.848	0.360
Individual	= 1 if estimated on individual level	0.152	0.360
Treatment	= 1 if any treatment was administered	0.212	0.409
<i>Study Characteristics</i>			
Experiment*	= 1 if study was conducted through experiment	0.949	0.221
Survey	= 1 if study was conducted through survey	0.051	0.221
Longitudinal	= 1 if study was longitudinal	0.438	0.496
<i>Sample Characteristics</i>			
Sample Size	Number of participants in sample	373.541	906.152
University Students	= 1 if sample is university students	0.590	0.492
General Population*	= 1 if sample is general adult population	0.360	0.480
Adolescents	= 1 if sample is adolescents	0.044	0.205
Subsample	= 1 if estimated on a subsample	0.523	0.500
Female	Percentage of female participants	0.492	0.234
Age Mean	Mean age of participants	32.550	11.072
<i>Study Setting</i>			
Lab*	= 1 if elicitation was conducted in a lab	0.564	0.496
Field	= 1 if elicitation was conducted in the field	0.240	0.427
Online	= 1 if elicitation was conducted online	0.166	0.372
School/Work	= 1 if elicitation was conducted at participants' school or workplace	0.031	0.172
<i>Geography</i>			
Africa	= 1 if study was conducted in Africa	0.050	0.217

Variable Name	Description	Mean	SD
North America	= 1 if study was conducted in North America	0.329	0.470
Europe*	= 1 if study was conducted in Europe	0.415	0.493
Asia	= 1 if study was conducted in Asia	0.189	0.391
Continent Other	= 1 if study was conducted in Australia or South America	0.017	0.130
<i>Rewards</i>			
Real	= 1 if study incentives were real	0.911	0.284
Money*	= 1 if reward was monetary	0.711	0.453
Real Effort	= 1 if outcome was a real effort task	0.263	0.440
Reward Other	= 1 if other reward domain (food, health, environmental)	0.026	0.158
<i>Delivery Method and Timing</i>			
Soon Cash*	= 1 if earlier payment was paid in cash	0.250	0.433
Soon Check	= 1 if earlier payment was paid by check	0.128	0.334
Soon Bank	= 1 if earlier payment was paid by bank transfer	0.104	0.305
Soon PayPal	= 1 if earlier payment was paid by any digital money transfer	0.204	0.403
Delivery Other	= 1 if earlier payment was delivered by other method	0.051	0.221
Payment Same	= 1 if earlier and delayed payment method were same	0.857	0.350
End*	= 1 if immediate outcome was delivered at the end of the experiment	0.383	0.486
Day	= 1 if immediate outcome was delivered before the end of the day	0.281	0.450
Different Day	= 1 if immediate outcome was delivered at a different day	0.182	0.386
<i>Elicitation</i>			
CTB*	= 1 if elicited through CTB protocol	0.548	0.498
MPL*	= 1 if elicited using a multiple price list	0.289	0.453
Work for Wage	= 1 if elicited using a work-for-wage task	0.116	0.321
Elicitation Other	= 1 if elicited using a task not classified as MPL, CTB, or Wage (includes Matching)	0.048	0.213
Num Sooner	Number of sooner option dates t	2.096	0.554
Num Delay	Number of different delays k	3.127	4.457
FED	= 1 if a varying Front-End Delay was introduced	0.948	0.223
Deal Uncertainty	= 1 if study deals with reward uncertainty	0.629	0.483
Deal Transaction Cost	= 1 if study deals with transaction costs	0.554	0.497

Variable Name	Description	Mean	SD
<i>Estimation</i>			
Switch Point	= 1 if estimated directly from indifference points	0.101	0.301
ML*	= 1 if estimated using Maximum Likelihood	0.297	0.457
NLS	= 1 if estimated using Nonlinear Least Squares	0.298	0.458
Tobit	= 1 if estimated using a Tobit estimator	0.259	0.438
Estimation Other	= 1 if estimated using OLS or different method	0.042	0.200
Utility Control	= 1 if study accounts for non-linear utility	0.788	0.409
CRRA	= 1 if CRRA utility function assumed	0.385	0.487
Fixed	= 1 if background consumption is fixed	0.171	0.377
Estimated	= 1 if background consumption is estimated	0.065	0.246
Joint Estimation	= 1 if β and δ are estimated jointly	0.259	0.438
<i>Publication Characteristics</i>			
Economics*	= 1 if study discipline is economics	0.929	0.258
Discipline Other	= 1 if neuroscience or psychology discipline	0.033	0.180
Published	= 1 if study is published	0.649	0.478
Pub Year	Year in which study was published	2019.554	3.745
Log Citations per Year	Natural logarithm of one plus the average number of citations per year from Google Scholar since first version	1.680	1.160
Impact Factor	IDEAS/RePEc recursive discounted impact factor for journals	1.028	0.908

Notes: This table reports summary statistics and descriptions for variables used in model averaging techniques. SD = standard deviation. * = reference category for the corresponding set of dummy variables. Missing value for variables with less than 1050 observations were imputed using the median observation for numerical and dummy variables, and using the mean observation for ratios.

Estimate Characteristics

We collect information on the role of the estimate in the paper: whether it is a preferred estimate by the authors—an estimate presented as the main result of the paper, frequently included in the abstract or introduction. About half of the reported estimates are reported in study appendices, usually as robustness checks or using different estimation specifications. Comparing preferred and appendix values to the reference category of non-preferred main-paper estimates,

we can gauge whether authors systematically emphasize or suppress certain estimates. We also differentiate between estimations on the aggregate and individual level. Aggregate-level estimates (reference category) are computed from pooling the observations on all participants, whereas individual-level estimates are computed for each participant individually and then reported as a summary statistic—in our case we collect the mean. Some studies also involve variation in treatments. These can account to anything from poverty priming (Bartoš *et al.* 2021), hunger or cognitive fatigue (Ashton 2015) to deciding for another participant (Kölle & Wenner 2023) or priming Chinese and Taiwanese subjects with Confucianism (Liu *et al.* 2014). We follow Imai *et al.* (2021) and distinguish between a control/neutral treatment and any treatment variation.

Study Characteristics

Furthermore, we record information on the general structure of the study. An overwhelming majority of studies are experimental, and only a handful are survey-based. An example survey study is provided by Akesaka (2019), who uses a Japan Household Panel Survey conducted by Osaka University to examine whether individuals' time preferences were affected by the damage from the Great East Japan Earthquake in 2011. The survey features various behavioral-economic questions, enabling the estimation of time preference parameters. We also distinguish longitudinal studies, which typically require eliciting time preference parameters from the same participants at different points in time. Akesaka (2019) is also an example of a longitudinal study, since the data was collected on the same households in 4 different years.

Sample Characteristics

The mean sample size in our dataset is 374, but the range is quite vast; the minimum number of subjects is 4 for a certain subsample in Bousquet (2017), whereas the maximum is 13,436 from a large-scale multi-country experiment conducted by Schleich *et al.* (2019). About half of the present bias estimates are computed from data on a subsample. Subsamples are very study-specific, but commonly involve treatment specifications, excluding attritors or outliers, or demographic groups. We control for the demographic differences by accounting for the ratio of female participants and mean participant age.

The subject pool is of high importance as external validity is at stake if the effect is inferred from a specific subset of the population (Roe & Just

2009). Over half of the present bias estimates come from university students, a convenient and, subsequently, overrepresented subject pool. We compare the university student sample to the neutral reference category including the general adult population and a small number of clinical population estimates, which had too few observations to be considered individually. Finally, we can also compare to a younger population—adolescents, typically high school or middle school students.

Study Setting

Both laboratory and field experiments have been scrutinized for different validity concerns. Laboratory studies typically offer greater replicability and internal validity, that is, a stronger ability to isolate causal relationships, but they may suffer from limited external validity. Field studies, by contrast, are often seen as more externally relevant, yet they are also more exposed to uncontrolled variation and confounding factors (Roe & Just 2009). To capture heterogeneity across study settings, we include indicators for field, online, and school/workplace studies, with laboratory studies serving as the reference category. In addition, because the subgroup analysis in the previous chapter suggested that the estimate–standard error relationship may differ specifically in field settings, we also include the interaction term $SE \times \text{Field}$, allowing the magnitude of publication-bias to vary between field and other studies.

Geography

Heterogeneity in the present-bias parameter may also reflect broad geographic differences across study settings. We control for geography-based heterogeneity by including indicators for Africa, North America, Asia, and grouping South America and Australia as a residual category due to a lack of observations. Europe then serves as the reference category. While the aggregation may seem too coarse, there exists evidence suggesting that differences may be relatively modest within regions, but more pronounced across broader cultural groups. For example, Schleich *et al.* (2019) in their large-scale survey find little variation across eight European countries, while Cheung *et al.* (2025) in their meta-analysis find substantial evidence for stronger present bias within Asian studies.

Rewards

The concern with hypothetical incentives in intertemporal choice experiments is well documented: without real stakes, subjects may misreport preferences. However, a growing body of evidence suggests that the difference between real and hypothetical rewards in time preference experiments is negligible. Brañas-Garza *et al.* (2023) conclude that results for paid and hypothetical preferences are *mostly the same*, and Cheung *et al.* (2025) in their meta-analysis find no effect of real versus hypothetical incentives.

The leading theory and supporting empirical evidence points to domain independence—the low correlation between time preference parameters for different domains (Matousek *et al.* 2022). Augenblick *et al.* (2015) conduct a real effort experiment with a complementary monetary study and find a minor present bias for monetary rewards but a considerably stronger present bias for real effort tasks. Both prior meta-analyses conclude reward domain as a major source of heterogeneity (Imai *et al.* 2021; Cheung *et al.* 2025).

Our dataset distinguishes between several reward-related characteristics. First, we record whether incentives are real or hypothetical. Second, we code the reward domain as monetary, real effort, food, health, or environmental. Because food, health, and environmental outcomes are represented by too few observations to be analyzed separately, we do not model them as distinct domains in the heterogeneity analysis. Instead, monetary rewards serve as the reference category, real effort is treated separately, and the rest are combined into a residual category.

Delivery Method and Timing

A vital aspect of executing a time preference study is the method of delivery and timing of outcomes. A payment in cash and a payment in voucher, for example, may represent different utility to the subject. Hence, we code variables for different monetary payment methods. These variables then mainly pertain to studies with real, paid rewards and are therefore most relevant for monetary incentive schemes, although they may also apply in other designs such as work-for-wage tasks. We collect information on payment methods for both the earlier (soon) and delayed (late) payments; however, due to collinearity concerns, we include only the earlier payment options in the analysis. We also include a dummy variable identifying whether the earlier and delayed payments were conducted using the same method.

In the estimation, payment by cash is included in the reference category as the most neutral and most frequent option (for examples, see Lührmann *et al.* (2018) or Meyer (2015)). The reference category therefore captures cash payments, a small number of real-incentive studies for which the delivery method is not reported, and observations without real monetary incentives. However, since we also include the dummy variable for real incentives, coefficients on the delivery methods are primarily identified from the variation within real-incentive studies, and can be interpreted relative to cash payments.

Apart from cash payments, we individually consider payments by check (for example, Augenblick & Rabin (2019)), by bank transfer (Lindner & Rose 2017), and by digital money transfer, such as PayPal (Yamamoto *et al.* 2025). Voucher and postal payments are included in the residual category.

Furthermore, we include indicators for the timing of delivery of the present outcome (at $t = 0$). The outcome is in all cases real, but can consist of a monetary payment, completion of a real-effort task or other non-monetary outcome. Having the present outcome be delivered at the end of the study compared to the present outcome being delivered at a different day may affect the perceived immediacy and salience of the reward. We differentiate between delivery being at the end of the experiment (i.e. immediately accessible), delivery later on the same day, and delivery at a future date. The reference category includes delivery at the end of the experiment and a few real-incentive studies for which the timing was not reported.

Elicitation

Perhaps the largest focus in the literature as to the heterogeneity of estimated time preference parameters is put on the elicitation methodology. How researchers collect information on the subjects' intertemporal decision-making can vastly alter the results, often without reflecting true heterogeneity of time preferences, but instead reflecting the influence of other confounding factors (Frederick *et al.* 2002; Cheung 2016; Harrison & Lau 2005; Lipman & Attema 2024).

We code variables for the four tasks most frequently used in quasi-hyperbolic discounting studies: multiple price list, the matching task, the CTB protocol, and the work-for-wage task (Augenblick & Rabin 2019). In the final heterogeneity specification, however, only the work-for-wage task and the residual category including the matching task enter separately, whereas CTB and MPL

are absorbed into the reference category. This pooling is a pragmatic choice rather than a claim of methodological equivalence: both variables display a variance inflation factor above ten, which can reduce the reliability and precision of the estimated coefficients. Several features that differentiate CTB from MPL (most importantly the treatment of utility curvature and related design choices) are nevertheless controlled for separately elsewhere in the specification.

If the elicitation task features choices between binary outcomes, following Imai *et al.* (2021), we also collect the number of sooner option dates t and the number of different delays k .

The FED is a delay to the early payment at time t ; it serves two purposes: (i) if applied uniformly to all choices, it minimizes the imposed transaction costs (Harrison & Lau 2005), and (ii) if applied only to a subset of the choices, it allows for a separate estimation of the discount factor δ based on the time frames shifted into the future ($t > 0$), and subsequently for the estimation of the present bias parameter β considering the choices made including the present ($t = 0$). Studies identifying β based on the variation in the front-end delay have been shown to estimate only a weak present bias (Cohen *et al.* 2020). In our specification, the dummy variable for FED then reflects the latter—a delay imposed on *any* earlier payment (see Courtemanche *et al.* (2015) or Bonan *et al.* (2022) as examples of studies eliciting β without FED variation).

The common methodological challenge lies in ensuring the reward at the earlier and later dates are equivalent in the subjects' perspective, save for the effect of the time delay. Two factors have been widely cited to introduce confounding effects: transaction cost differences and uncertainty in receiving the delayed payment. Transaction costs are frictions that decrease the perceived utility of a reward in the eyes of the subject. If, for example, the rewards are handed out via cash, then there could be additional effort required to pick up the delayed payment at the laboratory relative to the earlier payment delivered at the end of the experiment. Uncertainty then, likewise, decreases the potential utility of the delayed payment. Subjects who may not fully trust the researchers view the delayed reward with a certain risk factor, discounting its utility more heavily than what the intertemporal discounting function implies. We code dummy variables equal to one if the respective issue has been addressed by the authors of the study. Transaction costs are usually equalized by using instant electronic payments, or by splitting a fixed participation reward between the two dates. Certainty is generally instilled using a statement or guarantee issued by an official, credible institution—a bank or university, for

example.

Estimation

Systematic heterogeneity may also arise from differences in estimation methodology. Each method represents varying levels of researcher discretion and requires different functional-form or parameter choices. Cheung *et al.* (2025), as an example in the time-preferences literature, find estimation by OLS to be a major predictor of the discount factor δ . The three most frequently used methods in our dataset are maximum likelihood estimation, nonlinear least squares, and Tobit estimation, with each amounting to just above a quarter of the collected estimates. Less frequently studies also infer the time preference parameters directly from switching (indifference) points and also by standard OLS. In the analysis, we use ML as the reference category and pool OLS with other methods, which did could not be assigned to any of the main variables, as the residual category.

Crucially, we also control for whether the study accounts for nonlinear forms of the utility function. Nonlinear utility has been argued to be a prominent confounding factor in eliciting time preferences, providing downward-biased estimates of β if unaccounted for (Frederick *et al.* 2002; Cheung 2016). There are two main methods for measuring the instantaneous utility function: equating preferences under risk and over time by assigning a complementary risk preference task and eliciting the utility curvature parameter from intertemporal substitution. A more detailed description of the two methods is provided in Chapter 2. In the heterogeneity analysis, however, we do not distinguish between the methods, and retain only the main indicator for utility control, focusing on the overall impact rather than the specific approaches.

We also collect information on the specification of background consumption, whether it was fixed or estimated (see Kiessling *et al.* (2022) as an example of both), and we also include a dummy variable indicating whether the time preference parameters β and δ were estimated jointly.

Publication Characteristics

Lastly, we collect information on the studies' publication status and various publication measures. An overwhelming majority of the time preference studies are from the field of economics or business, which becomes the reference category along with a small number of studies from adjacent social sciences;

less frequent are psychology and neuroscience papers. For lack of observations, we merge the latter two together. Per meta-analytical practice, we include logarithms of the average number of citations per year on Google Scholar, accounting for zero citations by adding one: $\ln(1 + \text{avg. cit. per year})$. Moreover, we include the IDEAS/RePEc recursive discounted impact factor for journals for published studies, and a dummy variable indicating whether the study has been published in one of the top five economics journals (American Economic Review, Econometrica, Journal of Political Economy, Quarterly Journal of Economics, and Review of Economic Studies).

5.2 Bayesian Model Averaging

The general form of the meta-regression analysis model can be expressed as

$$\hat{\beta}_{ij} = \alpha_k + \mathbf{X}_{ijk}\lambda_k + \epsilon_{ij},$$

where $\mathbf{X}_{ijk}$ is a subset of explanatory variables selected from the set of all potential variables $\{\mathbf{X}_{ij}\}$, which, in our case, includes all collected explanatory variables, estimate standard errors, and the three variable–standard error interaction terms. The problem with estimating the above equation given many potential covariates comes down to model uncertainty: which explanatory variables $\mathbf{X}_{ijk} \in \{\mathbf{X}_{ij}\}$ should be included in the model? Model averaging circumvents this issue by estimating the regression equation for all possible combinations of $\{\mathbf{X}_{ij}\}$, assigning each model a weight based on its fit. The weights are then used to compute a weighted average of the coefficient estimates λ (Havránek 2019).

In the Bayesian framework, the coefficient estimates are weighted using principles derived from Bayes' theorem. Suppose a set of K possible models $M = \{M_1, M_2, \dots, M_K\}$, where model $M_k \in M$ contains the parameters $\theta_k \in \Theta_k$. The goal is to estimate the effect size λ , whose posterior distribution given the data D is defined as

$$p(\lambda|D) = \sum_{k=1}^K p(\lambda|M_k, D)P(M_k|D).$$

The above is essentially a weighted average of the posterior distributions $p(\lambda|M_k, D)$ under each of the models $M_1, \dots, M_K$ with weights given by the model's posterior probability $P(M_k|D)$. The Posterior Model Probability (PMP) assigned to

model M_k is defined as

$$P(M_k|D) = \frac{p(D|M_k)P(M_k)}{\sum_{i=1}^K p(D|M_i)P(M_i)} = \frac{p(D|M_k)P(M_k)}{p(D)},$$

where

$$p(D|M_k) = \int_{\Theta_k} p(D|\theta_k, M_k)p(\theta_k|M_k)d\theta_k$$

is the marginal likelihood of model M_k . The prior for the model parameters $p(\theta_k|M_k)$ reflects the belief regarding the values of θ_k under M_k , and $P(M_k)$ is the prior probability that M_k is the true model. Both priors are chosen by the researcher before considering the data (Steel 2020; Hoeting *et al.* 1999; Feldkircher & Zeugner 2022).

However, one cannot go on and estimate all possible model combinations with even a moderately high number of potential explanatory variables. The matrix $\mathbf{X}_{ij}$ in our case contains a total of 54 variables, which amounts to 2^{54} possible models—roughly 18 quadrillion, which is simply not computationally feasible. The solution is to approximate using the Markov chain Monte Carlo algorithm (Zeugner & Feldkircher 2015), which primarily considers models with high PMPs, markedly reducing the runtime.

As our baseline, we use the routinely employed priors suggested by Eicher *et al.* (2011): the unit information prior (UIP) as a specification of Zellner’s g-prior, which is a weakly informative prior easily overridden by the data, and the dilution prior (George 2010) as the model prior, which penalizes models with higher collinearity. In Appendix D, we conduct robustness checks using different sets of priors: UIP g-prior with uniform model prior, BRIC g-prior with random model prior, and Hannan–Quinn g-prior with random model prior.

5.3 Frequentist Model Averaging

Additionally, we employ the Mallows Model Averaging (MMA) method due to Hansen (2007), which is a Frequentist Model Averaging (FMA) alternative. Being a frequentist method, it does not hinge on specifying prior beliefs about parameter values or models. In MMA, the weights w_k are computed by minimizing the Mallows criterion:

$$\min_{w_k} C_p(\mathbf{\Omega}) = \mathbf{\Omega}' \hat{\epsilon}' \hat{\epsilon} \mathbf{\Omega} + 2\hat{\sigma}^2 L' \mathbf{\Omega},$$

subject to

$$\sum_{k=1}^K w_k = 1, \quad 0 \leq w_k \leq 1, \forall k = 1, 2, \dots, K,$$

where $\mathbf{\Omega} = (w_1, w_2, \dots, w_K)'$ is the $(K \times 1)$ vector of weights, $\hat{\epsilon} = (\hat{\epsilon}_1, \hat{\epsilon}_2, \dots, \hat{\epsilon}_K)$ is the $(n \times K)$ matrix of all residual vectors, $\hat{\sigma}^2 = (n - K)^{-1} \hat{\epsilon}'_K \hat{\epsilon}_K$ is the estimator of the error variance σ^2 for the full model, and $L = (k_1, k_2, \dots, k_K)'$ is the vector of the number of regressors included in each of the models (Amini & Parmeter 2012).

To reduce the computational burden of brute-forcing the estimation of all the 2^{54} models, we orthogonalize the regressors according to Amini & Parmeter (2012), based on the work of Magnus *et al.* (2010). This allows us to estimate only a set of $K = 54$ nested models, where we linearly add regressors to the model space. The coefficients are then recovered by transforming to the original basis. For inference, we calculate cluster-robust standard errors using the R code provided in the supplementary material of this thesis, and also as part of the automatic replication tools for meta-analysis on GitHub.

5.4 Results

The results of BMA are visualized in Figure 5.1. Each column denotes an individual model; models are displayed according to descending PMPs on the horizontal axis. The explanatory variables are sorted by Posterior Inclusion Probability (PIP) in descending order. PIP is the sum of posterior model probabilities for the models in which the variable is included. PIP values are then akin to measures of statistical significance: the higher the PIP, the greater the importance. For a given model, variables in blue (darker in grayscale) have a positive coefficient, variables in red (lighter in grayscale) have a negative coefficient, and variables with no color are not included in the model.

The numerical results of both BMA and FMA are presented in Table 5.2. Excluding the intercept, 18 variables have a PIP above 95%. These largely overlap with the 19 variables with a p -value < 0.1 . The model averaging techniques have thus identified a considerable amount of impactful drivers. The remainder of this section provides an account of selected consequential effects.

The results suggest little evidence of a general association between standard errors and present-bias estimates. However, this average pattern masks substantial subgroup differences: all three standard-error interaction terms have a PIP above 99% and two have p -values below 0.05, indicating that the estimate–

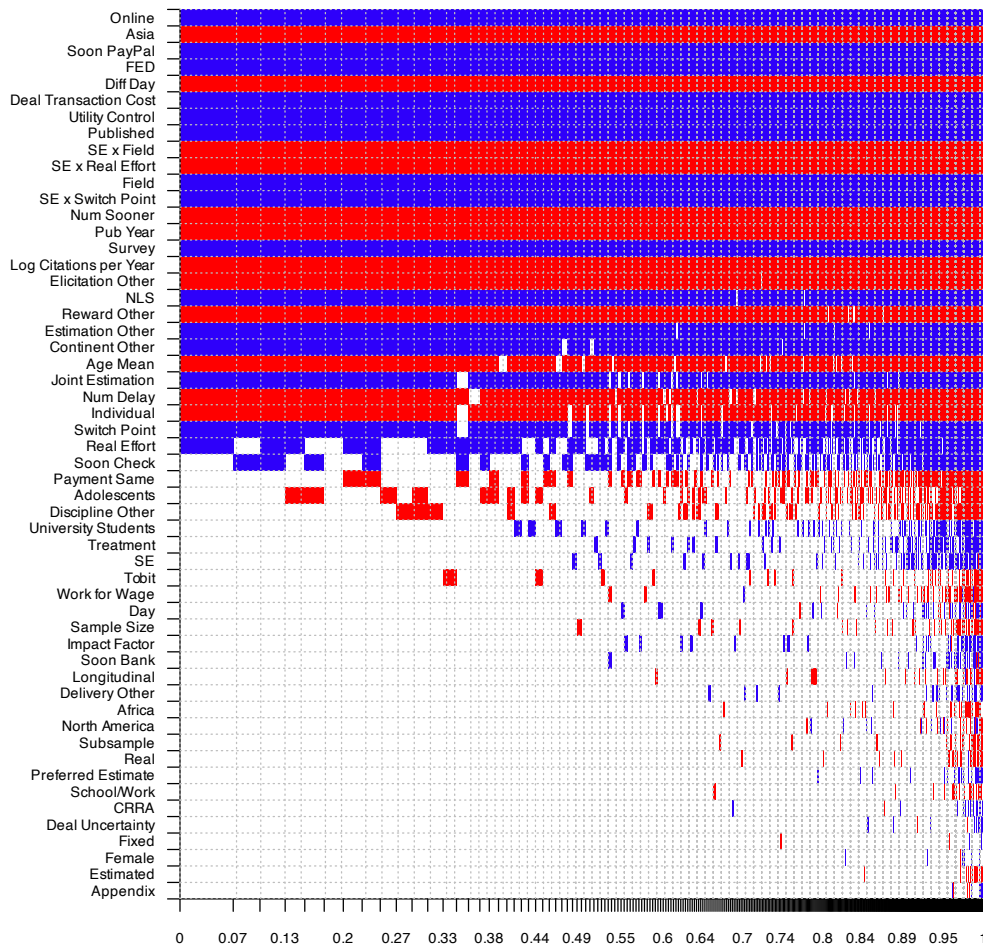


Figure 5.1: Bayesian model averaging results

Notes: This figure displays the results of Bayesian model averaging using the unit information g -prior (Eicher *et al.* 2011) and the dilution model prior (George 2010). Each column denotes an individual model. Variables are sorted by posterior inclusion probability in descending order. The horizontal axis denotes cumulative posterior model probabilities. For a given model, variables in blue (darker in grayscale) have a positive coefficient, variables in red (lighter in grayscale) have a negative coefficient, and variables with no color are not included in the model. For graphical rendering purposes, the computation for this plot uses 20,000 iterations, whereas the numerical results presented in Table 5.2 use 3,000,000 iterations; the results are qualitatively identical. For a detailed description of explanatory variables, see Table 5.1.

standard error relationship varies systematically across these subsets. Field studies seem to harbor the strongest evidence consistent with selective reporting in favor of smaller estimates of β (stronger present bias), with the same tendency also being observed for real effort estimates, lagging behind only slightly in the magnitude of the effect. Switch point estimates, on the other hand, display a positive relationship between estimates and their standard errors.

Study setting appears to have a substantial impact: after publication selection bias correction, field studies seem to be associated with larger estimates of β (BMA: post. mean = +0.050; PIP = 1.00) relative to those conducted in the lab—in fact the exact opposite of what the preliminary exploratory analysis suggested. On average field studies report lower values for β than laboratory or online studies (the unweighted mean β is lower in field studies). This raw difference, however, conflates the study setting with differences in study design and estimate precision. Lower β estimates are also associated with several design features that are plausibly more common outside the laboratory, such as unaddressed transaction costs, the absence of a front-end delay, no control for nonlinear utility, or immediate payments delivered on a different day. Once these factors are controlled for, the main effect of field setting turns positive. At the same time, the interaction term $SE \times \text{Field}$ is strongly negative, implying the conditional Field effect declines with imprecision. Hence, for sufficiently precise and otherwise comparable estimates, field studies tend to report higher β estimates (weaker present bias) than laboratory studies, but among less precise field estimates the effect reverses. The lower unconditional mean for field studies is therefore consistent with selective reporting (or noise) among imprecise field studies. Online studies are confirmed to be associated with higher estimates of β as well (BMA: post. mean = +0.083; PIP = 1.00), compared to laboratory studies.

Table 5.2: Results of model averaging techniques

Response variable:	Bayesian model averaging			Frequentist model averaging		
Present bias estimate $\hat{\beta}$	Post Mean	Post SD	PIP	Coef.	SE	<i>p</i> -value
Constant	8.7609	NA	1.0000	7.6790	4.5668	0.093
SE	0.0220	0.0890	0.0809	0.4186	0.4151	0.313
SE $\times$ Field	-1.6184	0.1944	1.0000	-1.9371	0.4982	0.000
SE $\times$ Switch Point	1.2524	0.2576	0.9996	0.6828	0.5004	0.172
SE $\times$ Real Effort	-1.2069	0.2830	1.0000	-1.7129	0.8369	0.041
<i>Estimate Characteristics</i>						
Preferred Estimate	0.0001	0.0013	0.0280	0.0068	0.0076	0.371
Appendix	0.0000	0.0009	0.0187	-0.0000	0.0000	—
Individual	-0.0240	0.0151	0.7860	-0.0322	0.0186	0.083
Treatment	0.0014	0.0045	0.1173	0.0064	0.0211	0.762
<i>Study Characteristics</i>						
Survey	0.0614	0.0190	0.9742	0.0806	0.0346	0.020

	Post Mean	Post SD	PIP	Coef.	SE	<i>p</i> -value
Longitudinal	-0.0003	0.0023	0.0368	-0.0149	0.0165	0.367
<i>Sample Characteristics</i>						
Sample Size	-0.0000	0.0000	0.0552	-0.0000	0.0000	—
University Students	0.0013	0.0051	0.0853	0.0081	0.0221	0.714
Adolescents	-0.0082	0.0149	0.2788	-0.0257	0.0222	0.247
Subsample	-0.0001	0.0012	0.0289	-0.0048	0.0055	0.383
Female	0.0003	0.0035	0.0239	-0.0134	0.0238	0.573
Age Mean	-0.0019	0.0009	0.8839	-0.0018	0.0011	0.102
<i>Study Setting</i>						
Field	0.0502	0.0115	0.9993	0.0569	0.0282	0.044
Online	0.0830	0.0109	1.0000	0.0767	0.0256	0.003
School/Work	-0.0005	0.0044	0.0335	-0.0047	0.0361	0.896
<i>Geography</i>						
Africa	-0.0002	0.0025	0.0231	-0.0165	0.0315	0.600
North America	-0.0001	0.0021	0.0311	-0.0094	0.0203	0.643
Asia	-0.0408	0.0092	0.9978	-0.0494	0.0179	0.006
Continent Other	0.0793	0.0301	0.9440	0.0743	0.0372	0.046
<i>Rewards</i>						
Real	-0.0001	0.0021	0.0201	-0.0179	0.0309	0.562
Real Effort	0.0214	0.0216	0.5690	0.0605	0.0396	0.127
Reward Other	-0.0641	0.0218	0.9685	-0.0433	0.0319	0.175
<i>Payment Method and Timing</i>						
Soon Check	0.0174	0.0219	0.4615	0.0645	0.0340	0.058
Soon Bank	0.0010	0.0052	0.0597	0.0257	0.0259	0.321
Soon PayPal	0.0415	0.0098	0.9997	0.0778	0.0298	0.009
Payment Same	-0.0087	0.0133	0.3542	-0.0366	0.0239	0.126
Day	0.0005	0.0032	0.0437	-0.0010	0.0220	0.964
Diff Day	-0.0757	0.0118	1.0000	-0.0791	0.0265	0.003
Delivery Other	0.0004	0.0037	0.0325	0.0350	0.0358	0.328
<i>Elicitation</i>						
Work for Wage	-0.0023	0.0101	0.0744	-0.0630	0.0407	0.122
Elicitation Other	-0.0560	0.0198	0.9557	-0.0708	0.0326	0.030
Num Sooner	-0.0289	0.0078	0.9913	-0.0329	0.0131	0.012
Num Delay	-0.0020	0.0010	0.8748	-0.0023	0.0010	0.021
FED	0.1799	0.0141	1.0000	0.1771	0.0342	0.000
Deal Uncertainty	0.0002	0.0016	0.0267	-0.0005	0.0148	0.973
Deal Transaction Cost	0.0412	0.0082	1.0000	0.0312	0.0205	0.128
<i>Estimation</i>						
Switch Point	0.0376	0.0257	0.7506	0.0528	0.0334	0.114

	Post Mean	Post SD	PIP	Coef.	SE	<i>p</i> -value
NLS	0.0237	0.0102	0.9132	0.0278	0.0152	0.067
Tobit	-0.0012	0.0045	0.0935	-0.0055	0.0104	0.597
Estimation Other	0.0402	0.0194	0.8827	0.0451	0.0172	0.009
Utility Control	0.0518	0.0105	0.9998	0.0576	0.0232	0.013
CRRA	0.0000	0.0014	0.0208	-0.0078	0.0156	0.617
Fixed	-0.0001	0.0014	0.0240	-0.0098	0.0081	0.226
Estimated	-0.0001	0.0016	0.0194	-0.0072	0.0084	0.391
Joint Estimation	0.0227	0.0125	0.8406	0.0171	0.0175	0.328
<i>Publication Characteristics</i>						
Published	0.0677	0.0106	1.0000	0.0680	0.0195	0.000
Pub Year	-0.0040	0.0012	0.9770	-0.0034	0.0023	0.139
Log Citations per Year	-0.0158	0.0047	0.9900	-0.0147	0.0097	0.130
Impact Factor	0.0007	0.0031	0.0685	0.0084	0.0140	0.549
Discipline Other	-0.0080	0.0178	0.2072	-0.0352	0.0353	0.319

Notes: This table presents the results of Bayesian model averaging using the unit information g-prior (Eicher *et al.* 2011) and the dilution model prior (George 2010); and results from frequentist model averaging, where weights are computed by minimizing the Mallows criterion C_p (Hansen 2007). Post mean = posterior mean. Post SD = posterior standard deviation. PIP = posterior inclusion probability. SE = standard error. The posterior mean in BMA and coefficient in FMA denote the marginal effect of the study characteristic on the estimated present bias parameter β . Standard errors are clustered at the study level. PIPs > 0.95 and *p*-values < 0.1 are bolded. For a detailed description of explanatory variables, see Table 5.1.

A key finding that directly revises the conclusions of Imai *et al.* (2021) is that, once we control for study characteristics and allow the estimate–standard error relationship to vary across subgroups, we find limited evidence of a direct difference between monetary and real-effort present-bias estimates. In Table 5.2, BMA reports a positive Real Effort coefficient of +0.021 with a PIP of 0.57, suggesting that the raw money–effort gap is at least partly accounted for by other study characteristics rather than by reward domain alone. The positive sign indicates a tendency toward weaker present bias for real effort, although the evidence is not sufficiently strong to draw firm conclusions.

The results do, however, show a relevant difference for the other reward domains (food, health, environmental), where present bias is estimated to be considerably stronger (BMA: post. mean = -0.064; PIP = 0.97). This suggests that the domain-independence hypothesis is not cleanly violated when considering the broader category of non-monetary outcomes. Theoretically, this effect is quite intuitive: money is fungible and storable, while the considered

food, health and environmental outcomes are often perishable or irreversible. This effect is, however, only weakly identified since the variable has only 27 observations.

Regarding the switch-point result (BMA: $SE \times \text{Switch Point post. mean} = +1.252$; $PIP = 1.00$), the positive estimate–standard error relationship is consistent with the mechanical explanation proposed in the previous chapter: future-biased outliers inflate the sample mean and standard deviation of individually estimated parameters due to study design being calibrated for β near or below one, generating a correlation even in the absence of selective reporting.

Methodological choices further stand out as impactful drivers of estimates present bias. The most economically consequential finding is that three methodological controls: variation in the front-end delay (FED), transaction-cost equalization, and nonlinear utility curvature, together account for a shift in β of approximately 27.3 percentage points, all having a PIP of virtually one. The FED coefficient alone contributes +0.180 (FMA: +0.177, $p < 0.001$); transaction cost control +0.041; and the utility-curvature control adds a further +0.052. In contrast, the unweighted mean of β is only 8.1 percentage points below the no-bias benchmark of $\beta = 1$; the methodological controls thus move the estimated β by more than three times the magnitude of the estimated behavioral effect itself.

The interpretation of this pattern is that a substantial portion of the present bias reported by the literature is, in fact, a measurement artifact. The theoretical mechanisms have been discussed since Frederick *et al.* (2002) and Harrison & Lau (2005), who argue that the alleged evidence for forms of hyperbolic discounting *can be easily explained as an artifact of experimental procedures*. The participant choosing between a sooner and later payment evaluates not a pure intertemporal tradeoff, but instead a sort of compound lottery, considering the level of trust in the researcher, the transaction costs, and the preference to smooth payoffs across different points in time (Cheung 2016). Each of these confounds, if not controlled, biases the estimated β downward, toward stronger present bias.

A fourth methodological choice—the delivery day of the sooner payment, operates in a more subtle way. The dummy variable indicating that the immediate reward was, in fact, delivered on a different day has a coefficient of -0.076 ($PIP = 1.00$). From a theoretical perspective the sign of the effect is puzzling: if $t = 0$ is itself postponed, both rewards become future, which should push β towards one. If β instead declines, it suggests that different-day delivery

creates distrust in the researcher or introduces asymmetric transaction costs, which more than offset the benefit of the universal front-end delay, or that the estimation procedure may confound β with the discount factor δ . Cheung *et al.* (2025) also find this effect to have an estimated negative sign, and fourth highest PIP from their explanatory variables. At any rate, the practical implication for study design is clear: same-day delivery of the sooner payment should be preferred together with institutional guarantees ensuring credibility of delayed payments and equalizing transaction costs using other approaches (see, for example, Balakrishnan *et al.* (2020); Andreoni *et al.* (2015) or Meyer (2015)).

Two further results warrant interpretation. First, studies conducted in Asia find stronger present bias (BMA: post. mean = -0.041; PIP = 1.00) compared to European studies, even after controlling for methodological covariates and selection. This echoes the finding of Cheung *et al.* (2025) that geographical heterogeneity of present bias survives in the literature. This result is consistent with two non-exclusive mechanisms: (i) the Asian subset is mainly composed of studies conducted in developing economies (Pakistan, Bangladesh, Vietnam, Indonesia), where intertemporal preferences may be confounded with liquidity constraints (Carvalho *et al.* 2016a), and (ii) cultural factors, such as those examined by Liu *et al.* (2014), which may moderate elicited preferences. Distinguishing between these explanations is beyond the present data, but the result of geographical heterogeneity is robust. Furthermore, the merged Australia and South America variable has an estimated positive effect implying a weaker present bias; this effect is, however, very weakly identified, since the residual category only has 18 observations.

Second, published studies are associated with higher β estimates (BMA: post. mean = +0.068; PIP = 1.00) relative to unpublished works (unpublished manuscripts, working papers, theses, ...), holding all other variables constant. This finding runs contrary to the conventional expectation that publication selection favors more extreme effects (Ioannidis *et al.* 2017) and instead suggests editorial conservatism: journals favoring findings that align with the canonical exponential discounting, rather than large deviations from it, which may be attributed to measurement error. More recent and highly cited studies, however, report lower β (stronger present bias), with coefficients of -0.004 (PIP = 0.98) and -0.016 (PIP = 0.99) for publication year and log citations per year, respectively. Together, these results provide a justification for our inclusive literature search; meta-analyses restricted to the published literature would likely

overestimate β and thereby underestimate present bias.

In Appendix D, we provide robustness checks using different sets of BMA priors mentioned previously. Figure D.5 then displays the posterior inclusion probabilities for variables across different prior specifications. Our main results hold in all specifications.

Chapter 6

Best-Practice Estimate

As our last piece of contribution, we aim to derive the best-practice estimate of present bias using the results of the model averaging methods in Chapter 5. Table 6.1 reports the fitted values from the main Bayesian Model Averaging (BMA) specification conditional on a set of methodologically defensible design choices that we treat as best-practice.

Table 6.1: Best-practice estimates across scenarios

	Present Bias β	95% Conf. int.	
Overall best practice	0.969	0.945	0.993
Money	0.965	0.939	0.991
Real Effort	0.986	0.953	1.020
Reward Other	0.901	0.857	0.945
Lab	0.943	0.917	0.969
Field	0.993	0.962	1.025
Online	1.026	0.996	1.057
School/Work	0.943	0.897	0.988
Europe	0.975	0.951	1.000
Asia	0.935	0.905	0.964
North America	0.975	0.947	1.004
Africa	0.975	0.937	1.013

Notes: This table reports best-practice estimates using fitted values from the main BMA estimation in Table 5.2. Confidence intervals for implied β parameters are approximate and are computed based on a frequentist specification. For context-specific estimates, we set the respective dummy variable to one and the rest of the corresponding set of dummy variables to zero.

The one caveat of this procedure is the degree of subjectivity in picking the ideal study design. While we follow the broad consensus in the literature

regarding certain study-design features, we echo Imai *et al.* (2021) in that we do not think our meta-analysis is capable of determining the best way to identify present bias. We therefore impose restrictions on variables that are known to confound the effect and specific choices that seem to have undesirable effects, but do not incline to any particular identification method.

First, we address the well-documented methodological confounds. We set Deal Transaction Cost and Deal Uncertainty to one, as the resulting estimate of present bias should not be conflated with other preferences arising from the delayed payment being perceived as more risky or requiring more effort to acquire. Further, we set the front-end delay (FED) variable to one, which reflects the prevailing identification strategy in the literature (90% of the collected estimates), in which the measured present bias relies on variation in the timing of sooner rewards. Crucially, FED variation helps to disentangle present bias β from the discount factor δ , with the heterogeneity analysis suggesting that estimation without FED variation seems to generate considerably lower β values. Perhaps most consequentially on a theoretical basis, we set Utility Control to one: assuming a linear form of the utility function when, in fact, it is concave ascribes a part of the curvature to time preferences, creating a downward bias in the estimated β (Frederick *et al.* 2002; Cheung 2016). In addition, we set Joint Estimation to one, since sequential estimation (first recovering δ and subsequently estimating β conditional on the recovered δ) may induce additional measurement error across the two estimation stages. Finally, we set Diff Day to zero; we therefore restrict ourselves to reward delivery during the day of the experimental session.

Second, regarding experimental design, we avoid ranking the standard elicitation and estimation methods, except for methods that are associated with bias in our heterogeneity analysis. Each estimator (ML, NLS, OLS, Tobit) is appropriate under different study contexts; we therefore set the values for the corresponding variables equal to their sample means. Analogously, we cannot objectively rank elicitation methods, leaving them at their sample means as well. We do, however, prefer aggregate-level estimation, as the sample mean of individual-level estimates is sensitive to outliers outside the $[0, 1]$ interval. For the same reason, we set Switch Point to zero, as it generates the spurious positive correlation between estimates and standard errors we identified in Section 4.8. Furthermore, we focus our attention on incentivized experimental studies as the methodological standard and natural reference point, setting Survey to zero and Real to one.

Third, for publication characteristics we impose a ‘high profile’ study at the frontier of the literature, setting Publication to one, Publication Year to 2025 (the most recent study in the sample), and Log citations per Year and the Journal Impact Factor to their 95th percentiles (3.51 and 2.62, respectively).

Lastly, we set SE and the three standard error–interaction variables to zero, nullifying publication selection bias. All variables not explicitly listed are set to their sample means. We then vary the contextual variables in Table 6.1, obtaining implied estimates of present bias for different economic environments.

The overall best-practice estimate is 0.969, consistent with the corrected point estimates of most nonlinear publication-bias correction methods, and implies a minor present bias: to make an individual indifferent between a consumption today and tomorrow, tomorrow’s consumption has to be approximately 3.2% higher due to present bias relative to standard exponential discounting. The estimate for Money (0.965) is roughly comparable to the overall estimate; real effort, on the other hand, is calculated to have a higher β of 0.986, implying a weaker present bias. Nevertheless, the approximate confidence interval is wide and the BMA coefficient for Real Effort does not have a high PIP relative to other variables; as such, the best-practice point estimate is imprecisely measured, and should be interpreted with caution. The results are similar to the corrected point estimates of the subgroup nonlinear publication-bias tests, ranging from 0.95–1.00.

Field and online settings have point estimates near the benchmark of no present bias $\beta = 1$ (0.993 and 1.026, respectively), while lab and school/workplace settings are computed to have stronger present bias (0.943 for both). Asia has a best-practice estimate of 0.935, while the rest of the continents all have a much weaker present bias of $\beta = 0.975$.

To provide an illustrative example of the magnitude of the implied present bias, consider an individual with quasi-hyperbolic time preferences, with the parameters $\beta = 0.969$ and $\delta_{\text{annual}} = 0.75$, which is consistent with results of the annual discount rate in both Matousek *et al.* (2022) and Cheung *et al.* (2025). Assuming linear utility for the sake of simplicity, if the individual chooses between consuming 100 units today and c units in one year, c would have to be equal to $100/\beta\delta_{\text{annual}} \approx 137.6$ units for the individual to be indifferent between the two options. An exponential discounter, without present bias, would only require $c = 100/\delta_{\text{annual}} \approx 133$ units.

For a short-run illustration, the implied daily discount factor is computed as $\delta_{\text{daily}} = \delta_{\text{annual}}^{1/365} \approx 0.9992$. If the quasi-hyperbolic discounter then considers

the choice between 100 units today, and c units tomorrow, c would have to equal $100/\beta\delta_{\text{daily}} \approx 103.3$, whereas the exponential discounter would require approximately $c \approx 100.1$ units. This implies that present bias increases the required compensation for delaying present consumption by about 3% relative to exponential discounting.

Chapter 7

Conclusion

This thesis presents a meta-analysis of empirical estimates of present bias in quasi-hyperbolic discounting. We investigate whether the literature supports a strong and robust behavioral effect, or whether part of the reported evidence is an artifact of selective reporting and study design. Using the largest dataset to date, consisting of 1,050 β estimates from 130 primary studies, we show that the raw literature reports a moderate present bias overall, but that this conclusion becomes much weaker once selective reporting is accounted for.

Linear publication-bias tests report a weak estimate–standard error relationship. Most nonlinear correction methods, however, revise β upward from the raw mean of 0.919 to roughly 0.95–0.99, implying a mild present bias. This result does not imply that present bias is absent, but it suggests that the available literature tends to exaggerate its magnitude. Caliper tests conducted around β values cannot rule out that the observed weak linear estimate–standard-error asymmetry reflects experimental designs calibrated around $\beta = 1$ or methods that mechanically truncate β at this boundary. Subgroup analyses of publication selection bias further imply that selective reporting is not uniform; evidence consistent with publication bias is concentrated in field and real-effort studies, while switch-point estimates display an opposite pattern that is more plausibly mechanical than selective.

The heterogeneity analysis conducted using 54 explanatory variables employs Bayesian and frequentist model-averaging methods and, overall, the results suggest that study design and methodology matter at least as much as behavior. Front-end delay variation, equalization of transaction costs, and control for nonlinear utility all shift the reported β upward, towards weaker present bias, and delivering the ‘immediate’ reward on a different day shifts it

downward, towards stronger present bias. Together, these results suggest that a nontrivial part of what is often interpreted as present bias may in fact reflect confounding factors in measurement rather than true intertemporal preferences. This is consistent with much of the long-standing concerns in the literature: in experimental intertemporal choice tasks, subjects compare convenience, trust, and the curvature of utility over rewards together with pure intertemporal discounting (Frederick *et al.* 2002; Harrison & Lau 2005; Andreoni *et al.* 2015; Cheung 2016).

Crucially, once we account for the subgroup-specific selection patterns, the commonly stressed gap between monetary and real-effort estimates narrows considerably. On the other hand, some heterogeneity remains even after correction, especially for Asian studies and other non-monetary outcomes (food, health, and environmental goods), which both exhibit lower β (stronger present bias). The overall implication is that the literature is not shaped by one universal present bias, but instead by a more complex combination of study-design choices and subgroup-specific selection (or mechanical) patterns as well as genuine heterogeneity (Imai *et al.* 2021; Brodeur *et al.* 2020; Havránek *et al.* 2024).

These findings have practical implications for both research and policy. For researchers, the results point to a minimum methodological baseline for credibility: control for nonlinear forms of the utility function, equalize transaction costs appropriately, use a variation in the front-end delay, and report transparent results. For policy, the results suggest caution in calibrating interventions or welfare analyses to raw published estimates of present bias, since these likely overstate the size of the true underlying behavioral parameter (Stanley 2005; Ioannidis *et al.* 2017).

Several limitations, nevertheless, still remain. The non-monetary reward domains, except real effort, are all only thinly represented, making it impossible to reliably study present bias for these areas individually. Further, the violations of the assumptions for the RTMA and MAIVE methods make it impossible to provide a robust detection of *p*-hacking. Lastly, our analysis is restricted only to quasi-hyperbolic estimates rather than the full set of dynamic-inconsistency models, which may reflect different underlying behavioral mechanisms; extending the analysis to alternative frameworks therefore remains an important avenue for future research.

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Appendix A

Additional Information on Dataset Construction

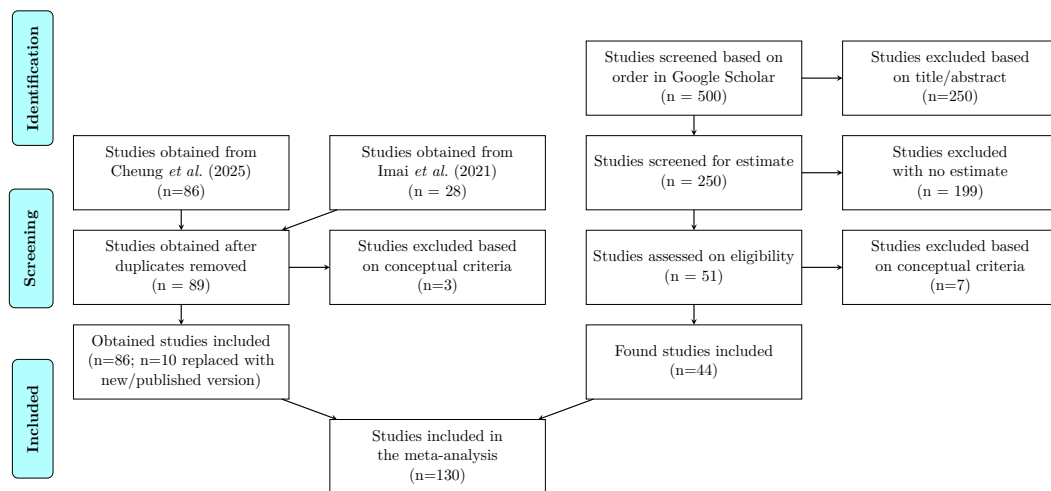


Figure A.1: PRISMA flow diagram

Table A.1: List of excluded studies

Abaluck <i>et al.</i> (2018)	Fang & Silverman (2009)	Paserman (2008)
Bai <i>et al.</i> (2021)	Gautier & van Vuuren (2020)	Wang & Sloan (2018)
DellaVigna <i>et al.</i> (2017)	Golder (2025)	
Fang & Wang (2015)	Martinez <i>et al.</i> (2023)	

Notes: This table lists the studies excluded from the analysis based on the conceptual criterion described in the inclusion criteria in Chapter 3.

Table A.2: Audit of fixed entries

Study	Observation ID	Original Value	Correction
Backes-Gellner <i>et al.</i> (2021)	243–248	SEs ≈ 0.2	coded as SD
<i>Reasoning:</i> Study reports SEs and SDs of the same size in different parts of the paper; magnitude-wise, they are most likely SDs.			
Bonan <i>et al.</i> (2022)	287–292	SDs ≈ 0.007	coded as SE
<i>Reasoning:</i> Extremely tight SDs, reported with other variables, likely SE.			
Horn & Kiss (2020)	526	SE = 0.185	coded as SD
<i>Reasoning:</i> Impossibly large SE given summary statistics (percentiles, min, max) provided; very likely SD.			
<i>Notes:</i> This table presents all corrections made to reported precision measures, along with their justification.			

Table A.3: All collected variables

Variable Name	Description	Mean	SD
<i>Identification</i>			
Observation ID	Identification number for estimates β	-	-
Study ID	Identification number for study	-	-
Study Name	Author(s) and year of publication	-	-
Title	Title of study	-	-
Number of Estimates	Number of estimates reported by study		
Source	Estimate source: Imai, Cheung or Own	-	-
Study Source	Source of Study: Imai, Cheung or Own	-	-
<i>Estimates</i>			
Beta Estimate ^w	Estimate of present bias β	0.919	0.117
Beta SE ^w	Standard error of present bias estimate	0.033	0.029
Beta SD	Standard deviation of present bias estimate	1.19e+3	3.81e+4
Discount Rate	Estimate of discount rate ρ	0.612	0.635
Discount Rate SE	Standard error of discount rate estimate	0.160	0.299
Discount Rate SD	Standard deviation of discount rate estimate	0.590	0.676
Delta	Estimate of discount factor δ	0.923	0.159
Delta SE	Standard error of discount factor estimate	0.023	0.049
Delta SD	Standard deviation of discount factor estimate	0.243	0.231
Utility Curvature	Estimate of utility curvature α	-4.53e+5	8.06e+6
Utility Curvature SE	Standard error of utility curvature estimate	1.91e+5	2.52e+6

Variable Name	Description	Mean	SD
Utility Curvature SD	Standard deviation of utility curvature estimate	1.19	2.84
<i>Estimate Characteristics</i>			
Discount Duration	The unit of time considered for the discounting parameters	-	-
Preferred	= 1 if estimate is presented as main result by authors	0.337	0.473
Appendix	= 1 if estimate is in study appendix	0.472	0.499
Aggregate	= 1 if estimated on aggregate level	0.848	0.360
Individual	= 1 if estimated on individual level	0.152	0.360
Treatment	= 1 if any treatment was administered	0.212	0.409
Treatment Description	Description of administered treatment	-	-
Years Collection	The years the data was collected	-	-
<i>Sample Characteristics</i>			
Sample Size	Number of participants in sample	373.541	906.152
University Students	= 1 if sample are university students	0.590	0.492
General Population	= 1 if sample is general adult population	0.360	0.480
Adolescents	= 1 if sample are adolescents	0.044	0.205
Clinical Population	= 1 if sample is any clinical population	0.006	0.075
Subsample	= 1 if estimated on a subsample	0.523	0.500
Subsample Description	Description of the subsample	-	-
Female	Percentage of female participants	0.492	0.234
Number Male	Number of male participants	546	1270
Number Female	Number of female participants	692	1340
Age Mean	Mean age of participants	32.550	11.072
Age Median	Median age of participants	29.710	5.242
Age SD	Standard deviation of participant age	7.493	4.175
Age Min	Minimum participant age	17.911	2.838
Age Max	Maximum participant age	53.384	26.334
<i>Study Setting</i>			
Lab	= 1 if elicitation was conducted in a laboratory	0.564	0.496
Field	= 1 if elicitation was conducted in the field	0.240	0.427
School/Work	= 1 if elicitation was conducted at school or workplace	0.031	0.172
Online	= 1 if elicitation was conducted online	0.166	0.372
<i>Geography</i>			
Geography	Country or countries where study was conducted	-	-
Africa	= 1 if study was conducted in Africa	0.050	0.217

Variable Name	Description	Mean	SD
North America	= 1 if study was conducted in North America	0.329	0.470
South America	= 1 if study was conducted in South America	0.015	0.123
Europe	= 1 if study was conducted in Europe	0.415	0.493
Asia	= 1 if study was conducted in Asia	0.189	0.391
Australia	= 1 if study was conducted in Australia	0.002	0.044
<i>Study Characteristics</i>			
Experiment	= 1 if study was done through experiment	0.949	0.221
Survey	= 1 if study was done through survey	0.051	0.221
Longitudinal	= 1 if study was longitudinal	0.438	0.496
<i>Reward domain</i>			
Real	= 1 if study incentives were real	0.911	0.284
Money	= 1 if reward was monetary	0.711	0.453
Real Effort	= 1 if outcome was a real effort task	0.263	0.440
Food	= 1 if reward was a food/beverage	0.007	0.081
Health	= 1 if outcome was health related	0.014	0.119
Environmental	= 1 if reward was an environmental good	0.005	0.069
<i>Payment Method</i>			
Soon Cash	= 1 if earlier reward was paid in cash	0.250	0.433
Soon Check	= 1 if earlier reward was paid by check	0.128	0.334
Soon Bank	= 1 if earlier reward was paid by bank transfer	0.104	0.305
Soon Voucher	= 1 if earlier reward was paid in voucher	0.019	0.137
Soon PayPal	= 1 if earlier reward was paid by any digital money transfer	0.204	0.403
Soon Postal	= 1 if earlier reward was paid by post	0.032	0.177
Late Cash	= 1 if delayed reward was paid in cash	0.107	0.309
Late Check	= 1 if delayed reward was paid by check	0.143	0.350
Late Bank	= 1 if delayed reward was paid by bank transfer	0.137	0.344
Late Voucher	= 1 if delayed reward was paid in voucher	0.019	0.137
Late PayPal	= 1 if delayed reward was paid by any digital money transfer	0.267	0.442
Late Postal	= 1 if delayed reward was paid by post	0.056	0.230
Payment Same	= 1 if earlier and delayed payment method were same	0.857	0.350
<i>Immediate Delivery Timing</i>			
End	= 1 if immediate payment was delivered at the end of the experiment	0.383	0.486

Variable Name	Description	Mean	SD
Day	= 1 if immediate payment was delivered before the end of the day	0.281	0.450
Different Day	= 1 if immediate payment was delivered at a different day	0.182	0.386
<i>Elicitation Method</i>			
MPL	= 1 if elicited through MPL	0.289	0.453
Matching	= 1 if elicited using a Matching task	0.032	0.177
CTB	= 1 if elicited through CTB protocol	0.548	0.498
Work for Wage	= 1 if elicited using a work-for-wage task	0.116	0.321
Elicit Other (excl.)	= 1 if task not classified as MPL, Matching, CTB, or Work for Wage	0.015	0.123
<i>Elicitation Details</i>			
Num Sooner	Number of sooner option dates t	2.096	0.554
Num Delay	Number of different delays k	3.127	4.457
Num Frames	Number of pairs (t, k) used	5.594	6.148
FED	= 1 if a varying Front-End Delay was introduced	0.948	0.223
Deal Uncertainty	= 1 if study deal with reward uncertainty	0.629	0.483
Deal Transaction Cost	= 1 if study deals with transaction costs	0.554	0.497
<i>Utility Control Details</i>			
Utility Control	= 1 if study accounts for non-linear utility	0.788	0.409
Risk Preference	= 1 if a complementary risk preference was used	0.146	0.353
Intertemporal Sub	= 1 if controlled using intertemporal substitution (includes CTB)	0.642	0.480
CRRA	= 1 if CRRA utility function assumed	0.385	0.487
Bg Cons	= 1 if background consumption is considered	-	-
Fixed	= 1 if background consumption is fixed	0.171	0.377
Estimated	= 1 if background consumption is estimated	0.065	0.246
Bg Cons Sooner	estimated background consumption at sooner date	9.85e+7	1.01e+9
Bg Cons Later	estimated background consumption at later date	9.85e+7	1.01e+ 9
Joint Estimation	= 1 if β and δ are estimated jointly	0.259	0.438
<i>Estimation Method</i>			
Switch Point	= 1 if estimated directly from indifference points	0.101	0.301
ML	= 1 if estimated using Maximum Likelihood	0.297	0.457

Variable Name	Description	Mean	SD
OLS	= 1 if estimated using Ordinary Least Squares	0.027	0.161
NLS	= 1 if estimated using Nonlinear Least Squares	0.298	0.458
Tobit	= 1 if estimated using a Tobit estimator	0.259	0.438
Estimate Other (excl.)	= 1 if estimation method not classified as Switch Point, ML, OLS, NLS, or Tobit	0.016	0.126
<i>Publication Characteristics</i>			
Economics	= 1 if study field is Economics	0.929	0.258
Neuroscience	= 1 if study field is Neuroscience	0.005	0.069
Psychology	= 1 if study field is Psychology	0.029	0.167
Published	= 1 if study is published	0.649	0.478
Top Five	= 1 if study published Top Five	0.127	0.333
First Year	Year in which study was first available	2018.373	3.503
Publication Year	Year in which study was published	2019.554	3.745
Publication Month	Month in which study was published	5.730	2.899
Number of Citations	Total number of citations of study	85.604	163.404
Log Citations per Year	Natural logarithm of one plus the average number of citations per year from Google Scholar since first version	1.680	1.160
Impact Factor	IDEAS/RePEc recursive discounted impact factor for journals	1.028	0.908

Notes: This table reports summary statistics for all variables collected for the meta-analysis. SD = standard deviation. W = winsorized at the 5% level.

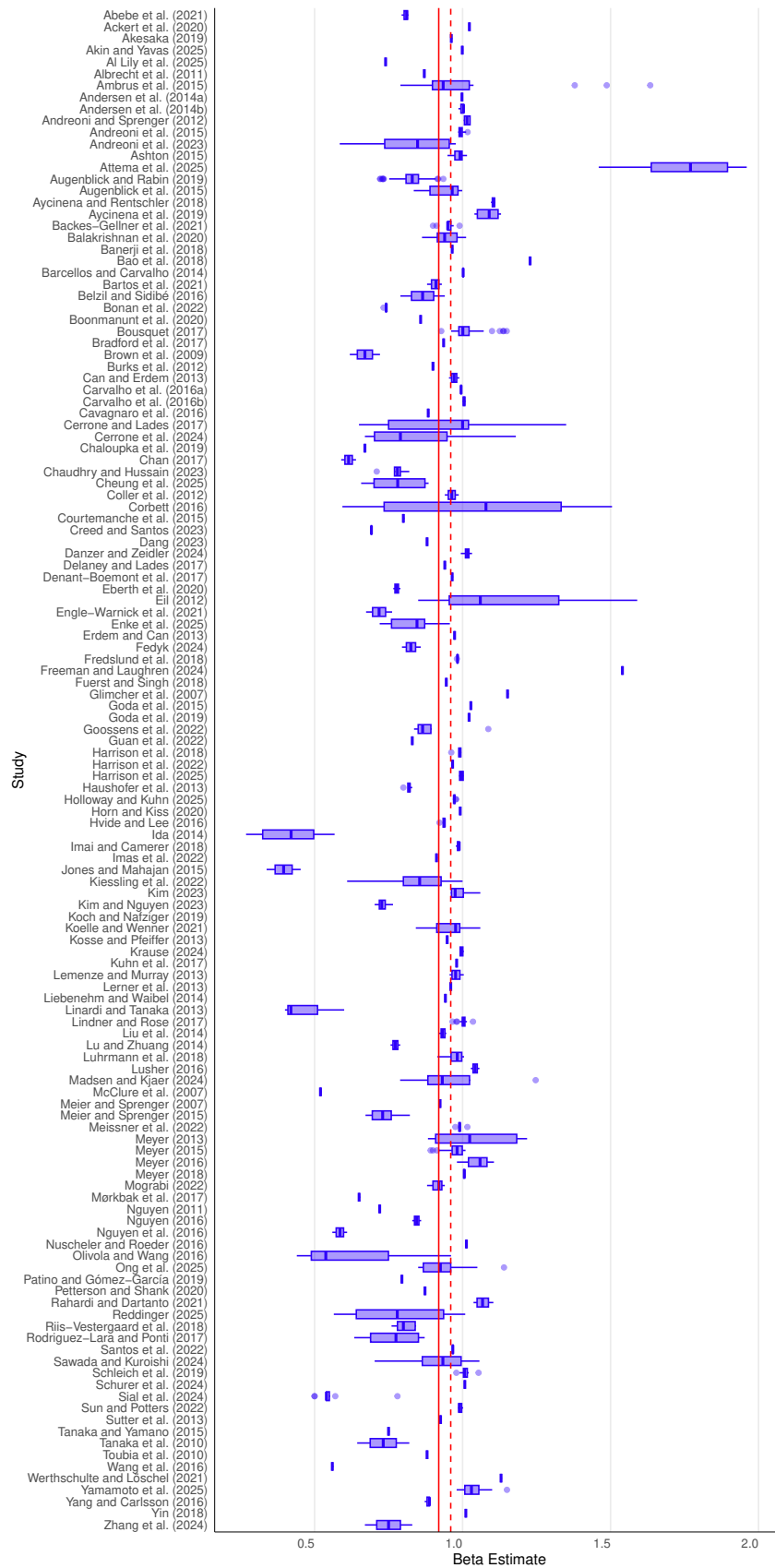


Figure A.2: Forest plot of present bias estimates

Notes: This figure displays a box plot of reported estimates for all studies, in alphabetical order. The solid vertical red line denotes the mean estimate for all studies, and the dashed red line denotes the median estimate. Extreme values are omitted from the graph, but included in all regressions.

Appendix B

The Delta Method

The Delta method is used for calculating the variance of Y , a function of a random variable X . The Taylor series expansion of function $f(\cdot)$ about a value a is

$$f(x) = f(a) + f'(a)(x - a) + f''(a)\frac{(x - a)^2}{2!} + \dots,$$

where we can approximate as

$$f(x) \approx f(a) + f'(a)(x - a).$$

Letting $a = \mu_x$, a Taylor series expansion of $y = f(x)$ about μ_x gives

$$y = f(x) \approx f(\mu_x) + f'(\mu_x)(x - \mu_x).$$

Taking the variance of both sides then yields:

$$\text{Var}(Y) = \text{Var}(f(X)) \approx [f'(\mu_x)]^2 \text{Var}(X). \quad (\text{B.1})$$

Then, as an example, if the present bias estimate is reported in the form of the parameter $\gamma = (1 - \beta)/\beta$, as in Meissner *et al.* (2022), we have

$$\beta = f(\gamma) = \frac{1}{1 + \gamma},$$

and we can recover the standard error of estimate $\hat{\beta}$ by applying the Delta method, where

$$f'(\gamma) = -\frac{1}{(1 + \gamma)^2}.$$

Evaluating the derivative at the estimate $\hat{\gamma}$ gives

$$f'(\hat{\gamma}) = -\frac{1}{(1 + \hat{\gamma})^2},$$

and thus, from Equation B.1, we get

$$\text{Var}(\hat{\beta}) \approx [f'(\hat{\gamma})]^2 \text{Var}(\hat{\gamma}) = \left[-\frac{1}{(1 + \hat{\gamma})^2} \right]^2 \text{Var}(\hat{\gamma}) = \frac{\text{Var}(\hat{\gamma})}{(1 + \hat{\gamma})^4}.$$

Finally, taking square roots yields the desired standard error as

$$SE(\hat{\beta}) = \frac{SE(\hat{\gamma})}{(1 + \hat{\gamma})^2}.$$

Appendix C

Additional Information to Selective Reporting Tests

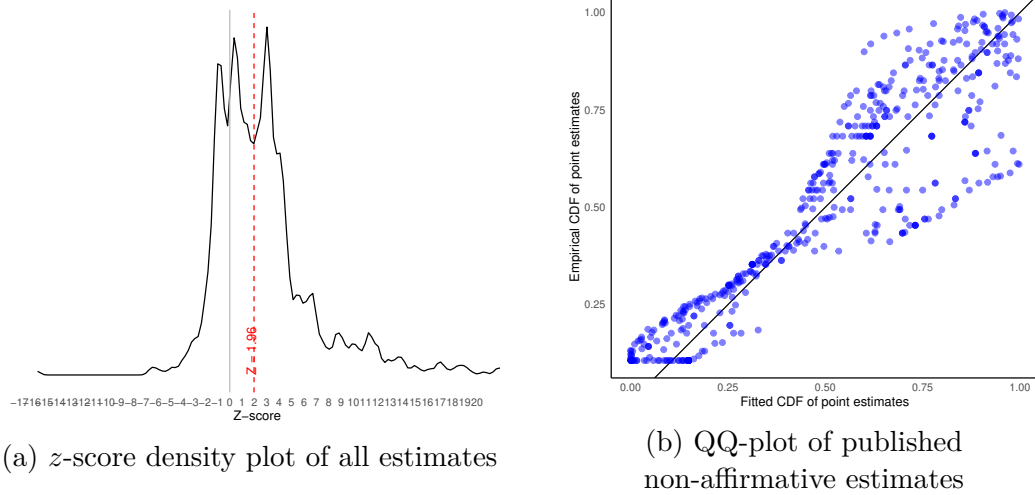


Figure C.1: Diagnostic plots for the RTMA method

Notes: This figure displays the diagnostic plots for the RTMA method by Mathur (2024). Plot (a) is a density plot of observed z -scores for DPB ($X = 1 - \beta$) estimates. The dashed red line represents the threshold for statistical significance at the 5% level. Plot (b) displays the quantile-quantile plot of the observed cumulative distribution function of published non-affirmative estimates and the fitted cumulative distribution function using RTMA estimates. If the estimates do not adhere to the 45-degree line, the method may not fit appropriately.

Results of Publication Bias Tests for Robustness Checks and Subgroups

Table C.1: Publication bias tests for 2.5% winsorization

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.909***	0.890***	0.953***
<i>SE</i>	(0.026)	(0.018)	(0.014)
<i>Bootstrap CI</i>	[0.851, 0.965]	[0.853, 0.927]	[0.920, 0.981]
Publication Bias	0.661	0.835	-0.555
<i>SE</i>	(0.930)	(0.640)	(0.754)
<i>Bootstrap CI</i>	[-1.250, 3.060]	[-0.618, 2.370]	[-2.040, 1.220]
Total observations	1050	1050	1050
	FE	BE	RE
Effect Beyond Bias	0.893***	0.894***	0.882***
<i>SE</i>	(0.006)	(0.020)	(0.015)
Publication Bias	1.095***	0.702*	1.033***
<i>SE</i>	(0.136)	(0.361)	(0.127)
Total observations	1050	1050	1050
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.953***	0.960***	0.970
<i>SE</i>	(0.004)	(0.009)	(0.025)
Total observations	1050	1050	1050
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.964***	-0.002***	0.165***
<i>SE</i>	(0.003)	(0.001)	(0.001)
Publication Bias	-1.431***	0.003*	0.135**
<i>SE</i>	(0.449)	(0.002)	(0.001)
Effect Beyond Bias (β)		1.002	0.835
Total observations	1050	1050	1050
Endogeneity-robust tests	p -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.090***		1.010
<i>SE</i>			(0.066)
<i>Anderson-Rubin CI</i>	[0.048, 0.129]		[0.845, 1.176]
Publication Bias	L = 15.1		-2.250
<i>SE</i>	(p < 0.001)		(1.980)
<i>Anderson-Rubin CI</i>			[-7.160, 1.970]
Effect Beyond Bias (β)	0.910		
First-stage F-statistic			1.560
Total observations	1050		1050

Notes: This table displays results for publication bias tests for all estimates winsorized at the 2.5% level. Unweighted mean: 0.933. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.2: Publication bias tests for experimental studies

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.952***	0.922***	0.968***
<i>SE</i>	(0.017)	(0.014)	(0.012)
<i>Bootstrap CI</i>	[0.913, 0.990]	[0.895, 0.950]	[0.939, 0.992]
Publication Bias	-1.119***	-0.426	-1.608***
<i>SE</i>	(0.376)	(0.411)	(0.450)
<i>Bootstrap CI</i>	[-1.914, -0.251]	[-1.324, 0.466]	[-2.544, -0.628]
Total observations	996	996	996
	FE	BE	RE
Effect Beyond Bias	0.922***	0.926***	0.916***
<i>SE</i>	(0.005)	(0.016)	(0.010)
Publication Bias	-0.163	-0.553	-0.238*
<i>SE</i>	(0.133)	(0.370)	(0.125)
Total observations	996	996	996
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.957***	0.968***	1.002
<i>SE</i>	(0.004)	(0.008)	(0.053)
Total observations	996	996	996
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.970***	0.047***	0.016***
<i>SE</i>	(0.003)	(0.010)	(0.000)
Publication Bias	-1.775***	0.127***	0.027
<i>SE</i>	(0.376)	(0.004)	(0.000)
Effect Beyond Bias (β)		0.953	0.984
Total observations	996	996	996
Endogeneity-robust tests	p -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.113***		1.330***
<i>SE</i>			(0.101)
<i>Anderson-Rubin CI</i>	[0.082, 0.142]		[1.080, 1.580]
Publication Bias	L = 36.9		-18.300***
<i>SE</i>	$(p < 0.001)$		(4.920)
<i>Anderson-Rubin CI</i>			[-30.600, -11.600]
Effect Beyond Bias (β)	0.887		
First-stage F-statistic			0.154
Total observations	996		996

Notes: This table displays results for publication bias tests for experimental studies. Un-weighted mean for subgroup: 0.916. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.3: Publication bias tests for monetary rewards

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.945***	0.927***	0.958***
<i>SE</i>	(0.020)	(0.014)	(0.013)
<i>Bootstrap CI</i>	[0.898, 0.988]	[0.900, 0.954]	[0.929, 0.985]
Publication Bias	-0.276	-0.406	-0.777
<i>SE</i>	(0.671)	(0.487)	(0.615)
<i>Bootstrap CI</i>	[-1.950, 1.340]	[-1.463, 0.756]	[-2.198, 0.563]
Total observations	747	747	747
	FE	BE	RE
Effect Beyond Bias	0.925***	0.932***	0.910***
<i>SE</i>	(0.005)	(0.015)	(0.011)
Publication Bias	0.450***	-0.580	0.287**
<i>SE</i>	(0.155)	(0.391)	(0.145)
Total observations	747	747	747
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.954***	0.955***	0.969
<i>SE</i>	(0.004)	(0.010)	(0.026)
Total observations	747	747	747
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.963**	-0.002*	0.009***
<i>SE</i>	(0.004)	(0.001)	(0.000)
Publication Bias	-1.157**	0.005**	0.024
<i>SE</i>	(0.564)	(0.002)	(0.000)
Effect Beyond Bias (β)		1.002	0.991
Total observations	747	747	747
Endogeneity-robust tests	p -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.109		1.100
<i>SE</i>			(0.055)
<i>Anderson-Rubin CI</i>	[0.079, 0.137]		[0.962, 1.235]
Publication Bias	L = 36		-6.700**
<i>SE</i>	$(p < 0.001)$		(2.810)
<i>Anderson-Rubin CI</i>			[-11.000, -2.400]
Effect Beyond Bias (β)	0.891		
First-stage F-statistic			0.635
Total observations	747		747

Notes: This table displays results for all publication bias tests for monetary rewards. Unweighted mean for subgroup: 0.937. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.4: Publication bias tests for real effort

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.952	0.844**	0.986
<i>SE</i>	(0.059)	(0.061)	(0.040)
<i>Bootstrap CI</i>	[0.788, 1.051]	[0.694, 0.989]	[0.811, 1.040]
Publication Bias	-1.375	0.806	-2.073***
<i>SE</i>	(0.862)	(1.088)	(0.635)
<i>Bootstrap CI</i>	[-3.000, 1.700]	[-2.650, 3.160]	[-3.287, 0.295]
Total observations	276	276	276
	FE	BE	RE
Effect Beyond Bias	0.953***	0.798**	0.976
<i>SE</i>	(0.012)	(0.083)	(0.031)
Publication Bias	-1.400***	1.485	-1.276***
<i>SE</i>	(0.237)	(1.131)	(0.234)
Total observations	276	276	276
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.948***	0.991**	0.979
<i>SE</i>	(0.007)	(0.004)	(0.046)
Total observations	276	276	276
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	1.017*	-3.808	0.019***
<i>SE</i>	(0.009)	-	(0.000)
Publication Bias	-2.897***	0.000	0.016
<i>SE</i>	(0.264)	-	(0.000)
Effect Beyond Bias (β)		4.808	0.981
Total observations	276	276	276
Endogeneity-robust tests	p -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.060		1.030
<i>SE</i>			(0.079)
<i>Anderson-Rubin CI</i>	[-0.047, 0.143]		[0.836, 1.198]
Publication Bias	L = 1.39		-5.520**
<i>SE</i>	(p = 0.239)		(2.510)
<i>Anderson-Rubin CI</i>			[-8.395, -0.735]
Effect Beyond Bias (β)	0.940		
First-stage F-statistic			0.646
Total observations	276		276

Notes: This table displays results for all publication bias tests for real effort. Unweighted mean for subgroup: 0.885. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.5: Publication bias tests for lab studies

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.979	0.932***	0.999
<i>SE</i>	(0.020)	(0.019)	(0.010)
<i>Bootstrap CI</i>	[0.925, 1.020]	[0.894, 0.971]	[0.970, 1.020]
Publication Bias	-1.524**	-0.243	-2.175***
<i>SE</i>	(0.750)	(0.614)	(0.717)
<i>Bootstrap CI</i>	[-3.364, 0.375]	[-2.534, 0.997]	[-3.919, -0.511]
Total observations	592	592	592
	FE	BE	RE
Effect Beyond Bias	0.932***	0.939**	0.923***
<i>SE</i>	(0.005)	(0.027)	(0.013)
Publication Bias	0.066	-0.486	0.016
<i>SE</i>	(0.138)	(0.678)	(0.138)
Total observations	592	592	592
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.983***	0.987***	0.998
<i>SE</i>	(0.004)	(0.004)	(0.025)
Total observations	592	592	592
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.992***	0.041***	0.014***
<i>SE</i>	(0.002)	(0.010)	(0.000)
Publication Bias	-1.759***	0.104***	0.022
<i>SE</i>	(0.210)	(0.006)	(0.000)
Effect Beyond Bias (β)		0.959	0.986
Total observations	592	592	592
Endogeneity-robust tests	p -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.097		1.150*
<i>SE</i>			(0.088)
<i>Anderson-Rubin CI</i>	[0.048, 0.145]		[0.935, 1.373]
Publication Bias	L = 13		-11.000**
<i>SE</i>	(p < 0.001)		(5.360)
<i>Anderson-Rubin CI</i>			[-19.684, -0.096]
Effect Beyond Bias (β)	0.903		
First-stage F-statistic			0.346
Total observations	592		592

Notes: This table displays results for all publication bias tests for lab studies. Unweighted mean for subgroup: 0.934. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.6: Publication bias tests for field studies

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.920***	0.911***	0.923**
<i>SE</i>	(0.024)	(0.022)	(0.035)
<i>Bootstrap CI</i>	[0.870, 0.980]	[0.867, 0.956]	[0.810, 0.990]
Publication Bias	-1.825***	-1.451***	-1.919*
<i>SE</i>	(0.325)	(0.343)	(1.007)
<i>Bootstrap CI</i>	[-2.580, -1.050]	[-2.126, -0.432]	[-4.292, 0.643]
Total observations	252	252	252
	FE	BE	RE
Effect Beyond Bias	0.903***	0.913***	0.907***
<i>SE</i>	(0.011)	(0.023)	(0.019)
Publication Bias	-1.268***	-1.533**	-1.347***
<i>SE</i>	(0.331)	(0.598)	(0.287)
Total observations	252	252	252
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.884***	0.928***	0.919
<i>SE</i>	(0.009)	(0.023)	(0.063)
Total observations	252	252	252
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.930***	0.083***	0.048***
<i>SE</i>	(0.010)	(0.021)	(0.002)
Publication Bias	-2.573*	0.144***	0.051***
<i>SE</i>	(1.338)	(0.007)	(0.001)
Effect Beyond Bias (β)		0.917	0.952
Total observations	252	252	252
Endogeneity-robust tests	<i>p</i> -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.140		0.841***
<i>SE</i>			(0.052)
<i>Anderson-Rubin CI</i>	[0.086, 0.185]		[0.728, 0.970]
Publication Bias	L = 16.7		-1.300
<i>SE</i>	(p < 0.001)		(2.890)
<i>Anderson-Rubin CI</i>			[-4.340, 3.920]
Effect Beyond Bias (β)	0.860		
First-stage F-statistic			1.720
Total observations	252		252

Notes: This table displays results for all publication bias tests for field studies. Unweighted mean for subgroup: 0.864. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.7: Publication bias tests for online studies

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.950*	0.953**	0.968**
<i>SE</i>	(0.027)	(0.019)	(0.015)
<i>Bootstrap CI</i>	[0.884, 1.006]	[0.912, 0.991]	[0.928, 0.997]
Publication Bias	0.054	-0.125	-0.314
<i>SE</i>	(0.804)	(0.630)	(0.690)
<i>Bootstrap CI</i>	[-2.050, 1.970]	[-1.870, 1.260]	[-1.850, 1.300]
Total observations	174	174	174
	FE	BE	RE
Effect Beyond Bias	0.940***	0.957	0.944**
<i>SE</i>	(0.019)	(0.030)	(0.025)
Publication Bias	0.250	-0.231	0.125
<i>SE</i>	(0.347)	(0.533)	(0.292)
Total observations	174	174	174
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.954***	0.987***	1.002
<i>SE</i>	(0.009)	(0.003)	(0.021)
Total observations	174	174	174
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.985***	-0.054***	-0.013***
<i>SE</i>	(0.004)	(0.015)	(0.000)
Publication Bias	-1.816***	0.038***	0.028
<i>SE</i>	(0.605)	(0.009)	(0.000)
Effect Beyond Bias (β)		1.054	1.013
Total observations	174	174	174
Endogeneity-robust tests	p -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.063		1.160
<i>SE</i>			(0.102)
<i>Anderson-Rubin CI</i>	[0.012, 0.107]		[0.906, 1.388]
Publication Bias	L = 5.54		-6.550**
<i>SE</i>	(p = 0.019)		(3.100)
<i>Anderson-Rubin CI</i>			[-11.760, -1.340]
Effect Beyond Bias (β)	1.054		
First-stage F-statistic			1.060
Total observations	174		174

Notes: This table displays results for all publication bias tests for online studies. Unweighted mean for subgroup: 0.952. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.8: Publication bias tests for maximum likelihood estimates

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.915***	0.919***	0.961*
<i>SE</i>	(0.025)	(0.022)	(0.022)
<i>Bootstrap CI</i>	[0.860, 0.975]	[0.874, 0.963]	[0.888, 0.998]
Publication Bias	-0.998**	-0.538	-2.181***
<i>SE</i>	(0.481)	(0.628)	(0.694)
<i>Bootstrap CI</i>	[-2.180, 0.058]	[-1.920, 1.030]	[-3.580, -0.345]
Total observations	312	312	312
	FE	BE	RE
Effect Beyond Bias	0.891***	0.919***	0.914***
<i>SE</i>	(0.009)	(0.027)	(0.018)
Publication Bias	-0.382*	-0.548	-0.414**
<i>SE</i>	(0.204)	(0.550)	(0.192)
Total observations	312	312	312
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.932***	0.964**	0.965
<i>SE</i>	(0.010)	(0.016)	(0.026)
Total observations	312	312	312
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.976***	0.040**	0.003***
<i>SE</i>	(0.006)	(0.019)	(0.000)
Publication Bias	-3.377***	0.140***	0.005
<i>SE</i>	(0.798)	(0.009)	(0.000)
Effect Beyond Bias (β)		0.960	0.997
Total observations	312	312	312
Endogeneity-robust tests	p -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.102		0.884
<i>SE</i>			(0.110)
<i>Anderson-Rubin CI</i>	[0.046, 0.149]		[0.611, 1.158]
Publication Bias	L = 10.6		-0.329
<i>SE</i>	(p = 0.001)		(4.680)
<i>Anderson-Rubin CI</i>			[-3.380, 1.500]
Effect Beyond Bias (β)	0.892		
First-stage F-statistic			1.470
Total observations	312		312

Notes: This table displays results for all publication bias tests for maximum likelihood estimates. Unweighted mean for subgroup: 0.876. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.9: Publication bias tests for nonlinear least squares estimates

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.968	0.945***	0.986
<i>SE</i>	(0.030)	(0.021)	(0.014)
<i>Bootstrap CI</i>	[0.887, 1.027]	[0.901, 0.990]	[0.935, 1.009]
Publication Bias	-1.413**	-1.096*	-2.057***
<i>SE</i>	(0.616)	(0.621)	(0.530)
<i>Bootstrap CI</i>	[-2.664, 0.422]	[-2.880, 0.340]	[-3.104, -0.446]
Total observations	313	313	313
	FE	BE	RE
Effect Beyond Bias	0.893***	0.957*	0.892***
<i>SE</i>	(0.009)	(0.024)	(0.018)
Publication Bias	1.322***	-1.477**	0.552**
<i>SE</i>	(0.303)	(0.553)	(0.272)
Total observations	313	313	313
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.964***	0.994***	0.978***
<i>SE</i>	(0.007)	(0.002)	(0.006)
Total observations	313	313	313
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.997	-0.002*	0.000***
<i>SE</i>	(0.004)	(0.001)	(0.000)
Publication Bias	-2.879***	0.001	0.010
<i>SE</i>	(0.402)	(0.002)	(0.000)
Effect Beyond Bias (β)		1.002	1.000
Total observations	313	313	313
Endogeneity-robust tests	<i>p</i> -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.099		1.050**
<i>SE</i>			(0.025)
<i>Anderson-Rubin CI</i>	[0.047, 0.146]		[0.985, 1.103]
Publication Bias	L = 11.1		-4.030**
<i>SE</i>	(p < 0.001)		(1.770)
<i>Anderson-Rubin CI</i>			[-5.638, 0.339]
Effect Beyond Bias (β)	0.901		
First-stage F-statistic			2.410
Total observations	313		313

Notes: This table displays results for all publication bias tests for nonlinear least squares estimates. Unweighted mean for subgroup: 0.929. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.10: Publication bias tests for Tobit estimates

Linear tests	OLS	Study	Precision
Effect Beyond Bias	1.011	0.977	0.986
<i>SE</i>	(0.017)	(0.017)	(0.012)
<i>Bootstrap CI</i>	[0.965, 1.043]	[0.938, 1.012]	[0.956, 1.009]
Publication Bias	-2.089***	-1.316	-1.243**
<i>SE</i>	(0.464)	(0.860)	(0.512)
<i>Bootstrap CI</i>	[-2.792, -0.479]	[-2.790, 1.440]	[-2.287, 0.285]
Total observations	272	272	272
	FE	BE	RE
Effect Beyond Bias	1.026***	0.973	1.003
<i>SE</i>	(0.007)	(0.020)	(0.013)
Publication Bias	-2.611***	-1.127	-2.456***
<i>SE</i>	(0.213)	(0.694)	(0.207)
Total observations	272	272	272
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.972***	0.975***	1.000
<i>SE</i>	(0.007)	(0.007)	(0.022)
Total observations	272	272	272
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.976***	-0.004**	0.043***
<i>SE</i>	(0.003)	(0.002)	(0.001)
Publication Bias	-0.593*	0.006	0.041**
<i>SE</i>	(0.355)	(0.005)	(0.000)
Effect Beyond Bias (β)		1.004	0.957
Total observations	272	272	272
Endogeneity-robust tests	<i>p</i> -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.073		0.954
<i>SE</i>			(0.031)
<i>Anderson-Rubin CI</i>	[0.033, 0.113]		[0.890, 1.020]
Publication Bias	L = 11.5		-0.031
<i>SE</i>	(p < 0.001)		(1.940)
<i>Anderson-Rubin CI</i>			[-2.800, 1.730]
Effect Beyond Bias (β)	0.927		
First-stage F-statistic			1.210
Total observations	272		272

Notes: This table displays results for all publication bias tests for Tobit estimates. Unweighted mean for subgroup: 0.950. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.11: Publication bias tests for switch point estimates

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.892***	0.891***	0.900***
<i>SE</i>	(0.023)	(0.025)	(0.030)
<i>Bootstrap CI</i>	[0.839, 0.943]	[0.837, 0.944]	[0.825, 0.965]
Publication Bias	1.721***	1.062	1.550**
<i>SE</i>	(0.393)	(0.866)	(0.642)
<i>Bootstrap CI</i>	[-0.580, 2.470]	[-3.960, 2.460]	[-2.150, 4.410]
Total observations	106	106	106
	FE	BE	RE
Effect Beyond Bias	0.911***	0.892***	0.886***
<i>SE</i>	(0.012)	(0.031)	(0.022)
Publication Bias	1.302***	1.027	1.287***
<i>SE</i>	(0.247)	(1.075)	(0.245)
Total observations	106	106	106
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.918***	0.890***	0.846**
<i>SE</i>	(0.012)	(0.038)	(0.077)
Total observations	106	106	106
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.891***	-0.056	-0.008***
<i>SE</i>	(0.014)	(0.053)	(0.000)
Publication Bias	2.202	0.122***	0.023
<i>SE</i>	(1.575)	(0.022)	(0.000)
Effect Beyond Bias (β)		1.056	1.008
Total observations	106	106	106
Endogeneity-robust tests	<i>p</i> -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.108		2.410
<i>SE</i>			(1.21)
<i>Anderson-Rubin CI</i>	[0.032, 0.166]		[-0.559, 5.385]
Publication Bias	L = 6.74		-49.200
<i>SE</i>	(p = 0.009)		(40.600)
<i>Anderson-Rubin CI</i>			[-138.300, 29.500]
Effect Beyond Bias (β)	0.892		
First-stage F-statistic			0.004
Total observations	106		106

Notes: This table displays results for all publication bias tests for switch point estimates. Unweighted mean for subgroup: 0.967. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.12: Publication bias tests for published studies

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.937***	0.910***	0.958***
<i>SE</i>	(0.024)	(0.016)	(0.015)
<i>Bootstrap CI</i>	[0.883, 0.987]	[0.876, 0.943]	[0.922, 0.989]
Publication Bias	-0.726	-0.196	-1.377*
<i>SE</i>	(0.854)	(0.526)	(0.709)
<i>Bootstrap CI</i>	[-2.430, 1.490]	[-1.380, 1.020]	[-2.923, 0.346]
Total observations	681	681	681
	FE	BE	RE
Effect Beyond Bias	0.948***	0.909***	0.933***
<i>SE</i>	(0.005)	(0.018)	(0.012)
Publication Bias	-1.056***	-0.169	-0.959***
<i>SE</i>	(0.135)	(0.428)	(0.130)
Total observations	681	681	681
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.949***	0.954***	0.985
<i>SE</i>	(0.005)	(0.011)	(0.026)
Total observations	681	681	681
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.963***	0.025	0.018***
<i>SE</i>	(0.004)	(0.015)	(0.000)
Publication Bias	-1.727***	0.133***	0.027
<i>SE</i>	(0.490)	(0.006)	(0.000)
Effect Beyond Bias (β)		0.975	0.982
Total observations	681	681	681
Endogeneity-robust tests	p -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.104		0.132
<i>SE</i>			(0.558)
<i>Anderson-Rubin CI</i>	[0.066, 0.139]		[-1.130, 1.460]
Publication Bias	L = 20.9		18.400
<i>SE</i>	(p < 0.001)		(13.100)
<i>Anderson-Rubin CI</i>			[-3.710, 48.350]
Effect Beyond Bias (β)	0.896		
First-stage F-statistic			0.066
Total observations	681		681

Notes: This table displays results for all publication bias tests for published studies. Unweighted mean for subgroup: 0.913. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.13: Publication bias tests for unpublished studies

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.957	0.938***	0.966
<i>SE</i>	(0.035)	(0.023)	(0.021)
<i>Bootstrap CI</i>	[0.859, 1.021]	[0.890, 0.986]	[0.894, 1.003]
Publication Bias	-0.766	-0.641	-1.008*
<i>SE</i>	(0.586)	(0.561)	(0.551)
<i>Bootstrap CI</i>	[-2.084, 0.947]	[-1.902, 0.766]	[-2.103, 0.322]
Total observations	369	369	369
	FE	BE	RE
Effect Beyond Bias	0.907***	0.955*	0.903***
<i>SE</i>	(0.009)	(0.026)	(0.019)
Publication Bias	0.694***	-1.168*	0.417*
<i>SE</i>	(0.233)	(0.598)	(0.217)
Total observations	369	369	369
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.968***	0.963**	1.002
<i>SE</i>	(0.006)	(0.015)	(0.069)
Total observations	369	369	369
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.966***	-0.008	0.003***
<i>SE</i>	(0.006)	(0.005)	(0.000)
Publication Bias	-1.011	0.011**	0.023
<i>SE</i>	(0.653)	(0.005)	(0.000)
Effect Beyond Bias (β)		1.008	0.997
Total observations	369	369	369
Endogeneity-robust tests	<i>p</i> -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.125		1.080
<i>SE</i>			(0.056)
<i>Anderson-Rubin CI</i>	[0.071, 0.178]		[0.942, 1.205]
Publication Bias	L = 16.6		-4.460*
<i>SE</i>	(p < 0.001)		(2.410)
<i>Anderson-Rubin CI</i>			[-7.271, 0.852]
Effect Beyond Bias (β)	0.875		
First-stage F-statistic			2.290
Total observations	369		369

Notes: This table displays results for all publication bias tests for unpublished studies. Unweighted mean for subgroup: 0.931. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.14: Publication bias tests for preferred estimates

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.928***	0.918***	0.940***
<i>SE</i>	(0.017)	(0.014)	(0.015)
<i>Bootstrap CI</i>	[0.890, 0.967]	[0.890, 0.945]	[0.908, 0.971]
Publication Bias	-0.061	-0.577	-0.374
<i>SE</i>	(0.683)	(0.458)	(0.596)
<i>Bootstrap CI</i>	[-1.790, 1.620]	[-1.567, 0.449]	[-1.730, 0.940]
Total observations	354	354	354
	FE	BE	RE
Effect Beyond Bias	0.931***	0.920***	0.910***
<i>SE</i>	(0.010)	(0.015)	(0.012)
Publication Bias	-0.121	-0.666*	-0.264
<i>SE</i>	(0.223)	(0.371)	(0.190)
Total observations	354	354	354
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.941***	0.940***	0.948
<i>SE</i>	(0.007)	(0.018)	(0.040)
Total observations	354	354	354
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.940***	0.034*	0.013***
<i>SE</i>	(0.007)	(0.018)	(0.000)
Publication Bias	-0.376	0.133***	0.037
<i>SE</i>	(0.810)	(0.008)	(0.000)
Effect Beyond Bias (β)		0.966	0.987
Total observations	354	354	354
Endogeneity-robust tests	p -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.127		1.300
<i>SE</i>			(0.714)
<i>Anderson-Rubin CI</i>	[0.098, 0.155]		[-0.475, 3.082]
Publication Bias	L = 47.6		-14.300
<i>SE</i>	$(p < 0.001)$		(26.900)
<i>Anderson-Rubin CI</i>			[-67.100, 31.400]
Effect Beyond Bias (β)	0.873		
First-stage F-statistic			0.009
Total observations	354		354

Notes: This table displays results for all publication bias tests for preferred estimates. Unweighted mean for subgroup: 0.926. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.15: Publication bias tests for appendix estimates

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.948**	0.927***	0.963**
<i>SE</i>	(0.023)	(0.024)	(0.018)
<i>Bootstrap CI</i>	[0.892, 1.000]	[0.877, 0.976]	[0.916, 0.995]
Publication Bias	-1.289**	-0.648	-1.717**
<i>SE</i>	(0.566)	(0.467)	(0.676)
<i>Bootstrap CI</i>	[-2.432, 0.386]	[-1.772, 0.375]	[-3.152, 0.017]
Total observations	496	496	496
	FE	BE	RE
Effect Beyond Bias	0.911***	0.936***	0.918***
<i>SE</i>	(0.007)	(0.024)	(0.015)
Publication Bias	-0.157	-0.971	-0.267
<i>SE</i>	(0.191)	(0.631)	(0.182)
Total observations	496	496	496
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.943***	0.959***	0.953
<i>SE</i>	(0.006)	(0.012)	(0.033)
Total observations	496	496	496
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.966***	-0.006	0.017***
<i>SE</i>	(0.005)	(0.019)	(0.000)
Publication Bias	-1.904***	0.141***	0.018
<i>SE</i>	(0.502)	(0.009)	(0.000)
Effect Beyond Bias (β)		1.006	0.983
Total observations	496	496	496
Endogeneity-robust tests	<i>p</i> -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.124		0.889**
<i>SE</i>			(0.053)
<i>Anderson-Rubin CI</i>	[0.077, 0.169]		[0.761, 1.017]
Publication Bias	L = 18.8		0.800
<i>SE</i>	(p < 0.001)		(2.850)
<i>Anderson-Rubin CI</i>			[-1.550, 2.480]
Effect Beyond Bias (β)	0.876		
First-stage F-statistic			1.050
Total observations	496		496

Notes: This table displays results for all publication bias tests for appendix estimates. Unweighted mean for subgroup: 0.906. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.16: Publication bias tests for Asia estimates

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.897***	0.880***	0.907***
<i>SE</i>	(0.030)	(0.028)	(0.028)
<i>Bootstrap CI</i>	[0.830, 0.970]	[0.821, 0.938]	[0.822, 0.966]
Publication Bias	-0.351	-0.304	-0.622
<i>SE</i>	(0.885)	(0.824)	(0.889)
<i>Bootstrap CI</i>	[-2.350, 2.090]	[-2.010, 2.180]	[-2.620, 1.470]
Total observations	198	198	198
	FE	BE	RE
Effect Beyond Bias	0.904***	0.880***	0.888***
<i>SE</i>	(0.012)	(0.034)	(0.025)
Publication Bias	-0.560*	-0.290	-0.519*
<i>SE</i>	(0.304)	(0.725)	(0.280)
Total observations	198	198	198
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.883***	0.906***	0.878***
<i>SE</i>	(0.009)	(0.024)	(0.012)
Total observations	198	198	198
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.909***	0.006	0.000***
<i>SE</i>	(0.009)	(0.042)	(0.000)
Publication Bias	-0.764	0.159***	0.007
<i>SE</i>	(0.897)	(0.015)	(0.000)
Effect Beyond Bias (β)		0.994	1.000
Total observations	198	198	198
Endogeneity-robust tests	p -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.098		0.816*
<i>SE</i>			(0.104)
<i>Anderson-Rubin CI</i>	[0.001, 0.163]		[0.565, 1.072]
Publication Bias	L = 3.87		3.380
<i>SE</i>	(p = 0.049)		(5.290)
<i>Anderson-Rubin CI</i>			[-8.260, 15.030]
Effect Beyond Bias (β)			0.905
First-stage F-statistic			0.647
Total observations	198		198

Notes: This table displays results for all publication bias tests for Asia estimates. Unweighted mean for subgroup: 0.884. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.17: Publication bias tests for Europe estimates

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.975	0.958**	0.991
<i>SE</i>	(0.017)	(0.017)	(0.007)
<i>Bootstrap CI</i>	[0.927, 1.010]	[0.924, 0.993]	[0.971, 1.004]
Publication Bias	-0.395	-0.214	-0.946*
<i>SE</i>	(0.629)	(0.670)	(0.494)
<i>Bootstrap CI</i>	[-2.030, 1.210]	[-2.220, 1.270]	[-2.154, 0.352]
Total observations	436	436	436
	FE	BE	RE
Effect Beyond Bias	0.937***	0.963*	0.932***
<i>SE</i>	(0.006)	(0.021)	(0.014)
Publication Bias	0.853***	-0.372	0.631***
<i>SE</i>	(0.185)	(0.453)	(0.171)
Total observations	436	436	436
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.987***	0.994***	0.990
<i>SE</i>	(0.004)	(0.001)	(0.012)
Total observations	436	436	436
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.996**	-0.002*	0.006***
<i>SE</i>	(0.002)	(0.001)	(0.000)
Publication Bias	-1.321***	0.004***	0.018
<i>SE</i>	(0.243)	(0.001)	(0.000)
Effect Beyond Bias (β)		1.002	0.994
Total observations	436	436	436
Endogeneity-robust tests	p -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.058		1.010
<i>SE</i>			(0.035)
<i>Anderson-Rubin CI</i>	[0.022, 0.093]		[0.920, 1.090]
Publication Bias	L = 8.87		-1.880
<i>SE</i>	(p = 0.003)		(1.630)
<i>Anderson-Rubin CI</i>			[-5.900, 1.330]
Effect Beyond Bias (β)			0.990
First-stage F-statistic			0.729
Total observations	436		436

Notes: This table displays results for all publication bias tests for Europe estimates. Unweighted mean for subgroup: 0.963. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table C.18: Publication bias tests for North America estimates

Linear tests	OLS	Study	Precision
Effect Beyond Bias	0.922***	0.898***	0.948**
<i>SE</i>	(0.028)	(0.023)	(0.023)
<i>Bootstrap CI</i>	[0.854, 0.984]	[0.852, 0.944]	[0.888, 0.991]
Publication Bias	-1.184**	0.087	-1.870***
<i>SE</i>	(0.513)	(0.609)	(0.601)
<i>Bootstrap CI</i>	[-2.187, 0.238]	[-1.370, 1.460]	[-3.082, -0.319]
Total observations	345	345	345
	FE	BE	RE
Effect Beyond Bias	0.911***	0.899***	0.924***
<i>SE</i>	(0.009)	(0.027)	(0.018)
Publication Bias	-0.888***	0.058	-0.758***
<i>SE</i>	(0.230)	(0.619)	(0.216)
Total observations	345	345	345
Nonlinear tests	WAAP	Top 10	Stem
Effect Beyond Bias	0.935***	0.936***	1.002
<i>SE</i>	(0.008)	(0.018)	(0.059)
Total observations	345	345	345
	EK	AK (DPB)	RTMA (DPB)
Effect Beyond Bias	0.948***	0.094***	0.032***
<i>SE</i>	(0.007)	(0.013)	(0.000)
Publication Bias	-1.859**	0.123***	0.029**
<i>SE</i>	(0.767)	(0.005)	(0.000)
Effect Beyond Bias (β)		0.906	0.968
Total observations	345	345	345
Endogeneity-robust tests	p -uniform* (DPB)		MAIVE
Effect Beyond Bias	0.140		1.230
<i>SE</i>			(0.155)
<i>Anderson-Rubin CI</i>	[0.086, 0.188]		[0.843, 1.577]
Publication Bias	L = 18.7		-13.700*
<i>SE</i>	$(p < 0.001)$		(6.160)
<i>Anderson-Rubin CI</i>			[-20.940, -3.290]
Effect Beyond Bias (β)	0.860		
First-stage F-statistic			0.226
Total observations	345		345

Notes: This table displays results for all publication bias tests for North America estimates. Unweighted mean for subgroup: 0.878. OLS = Ordinary Least Squares, FE = Fixed Effects, BE = Between Effects, RE = Random Effects. Study = weighted by inverse number of estimates per study. Precision = weighted by inverse standard error. WAAP = Weighted Average of Adequately Powered estimates. Top 10 = Top 10% most precise estimates. STEM = stem-based method. EK = Endogenous kink model. AK = Andrews and Kasy selection model. RTMA = Right-Truncated Meta-Analysis. MAIVE = Meta-Analysis Instrumental Variable Estimator. DPB = Degree of Present Bias ($X = 1 - \beta$). Clustered standard errors are reported in parentheses where applicable. Bootstrap and Anderson-Rubin confidence intervals are shown in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix D

BMA Robustness Checks

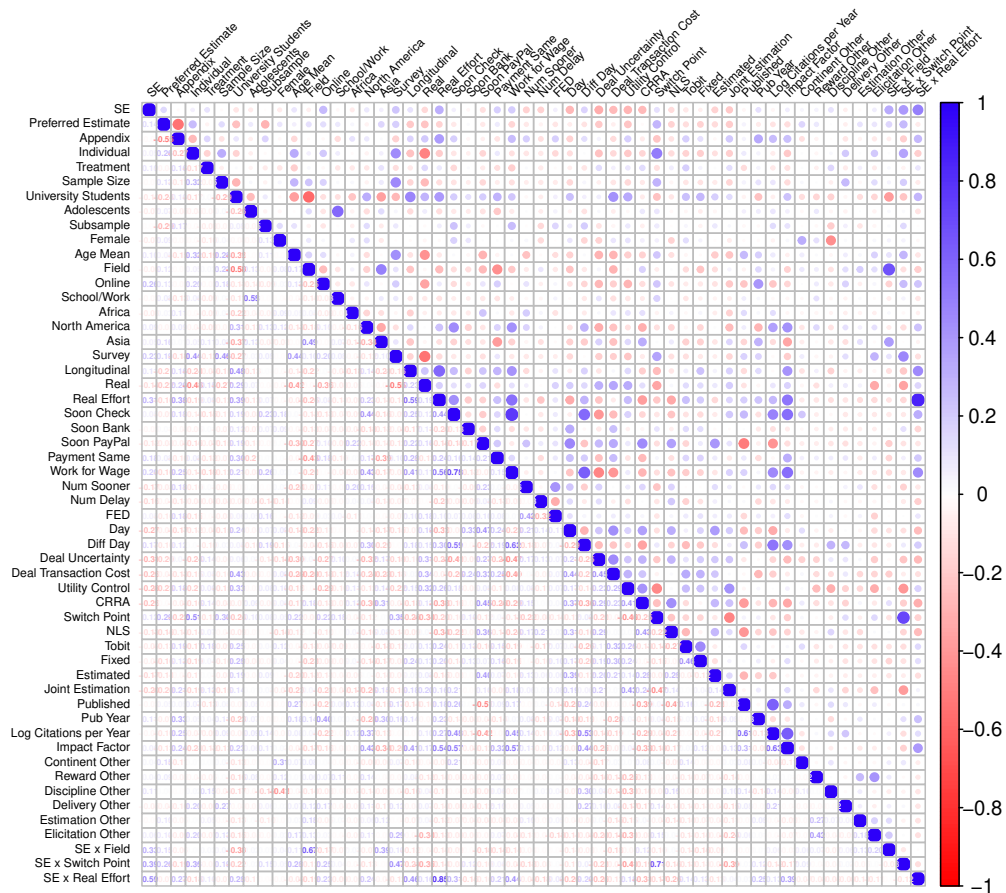


Figure D.1: Correlation matrix

Notes: This figure shows the correlation matrix of the variables used in the model averaging techniques. Blue color denotes a positive correlation; red color denotes a negative correlation. For a detailed description of explanatory variables, see Table 5.1.

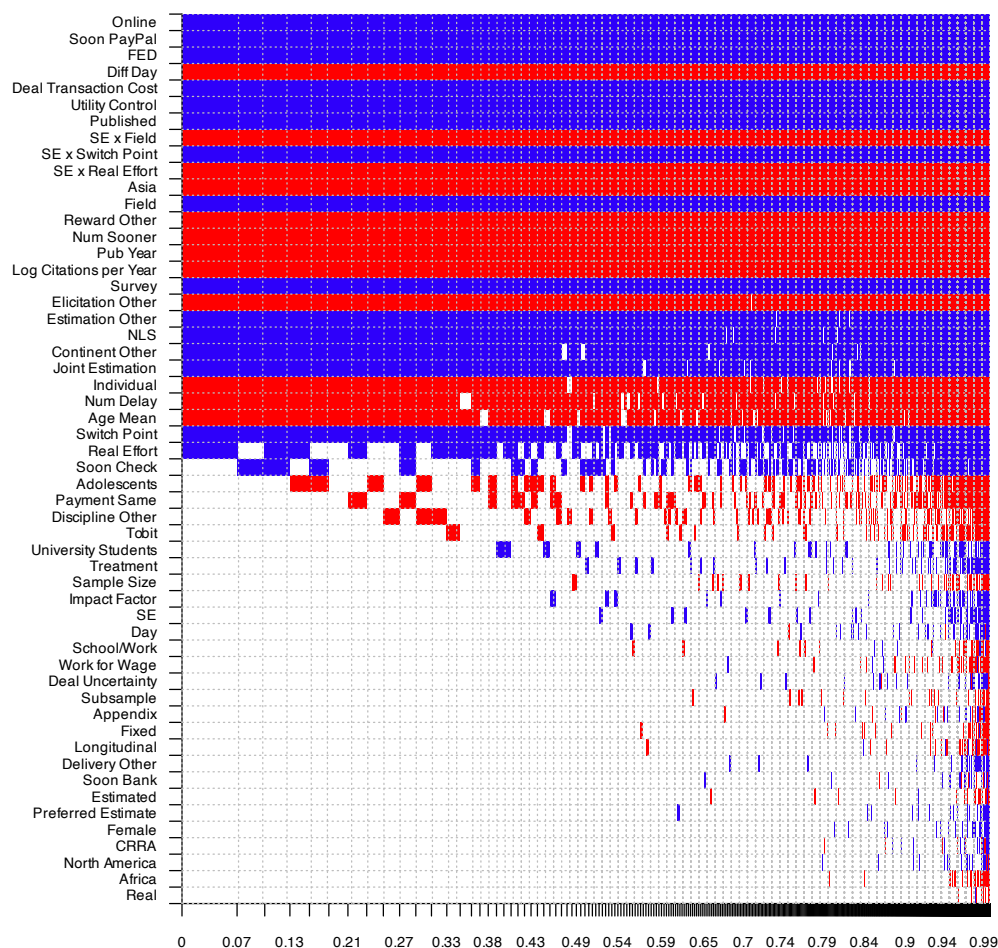


Figure D.2: BMA results (UIP, uniform)

Notes: This figure displays the results of Bayesian model averaging using the unit information g-prior and the uniform model prior. Each column denotes an individual model. Variables are sorted by posterior inclusion probability in descending order. The horizontal axis denotes cumulative posterior model probabilities. Variables in blue (darker in grayscale) depict an estimated positive sign. Variables in red (lighter in grayscale) depict an estimated negative sign. Variables with no color are not included in the given model. For graphical rendering purposes, the computation for this plot uses 20,000 iterations, whereas considered numerical results use 3,000,000 iterations; the results are qualitatively identical. For a detailed description of explanatory variables, see Table 5.1.

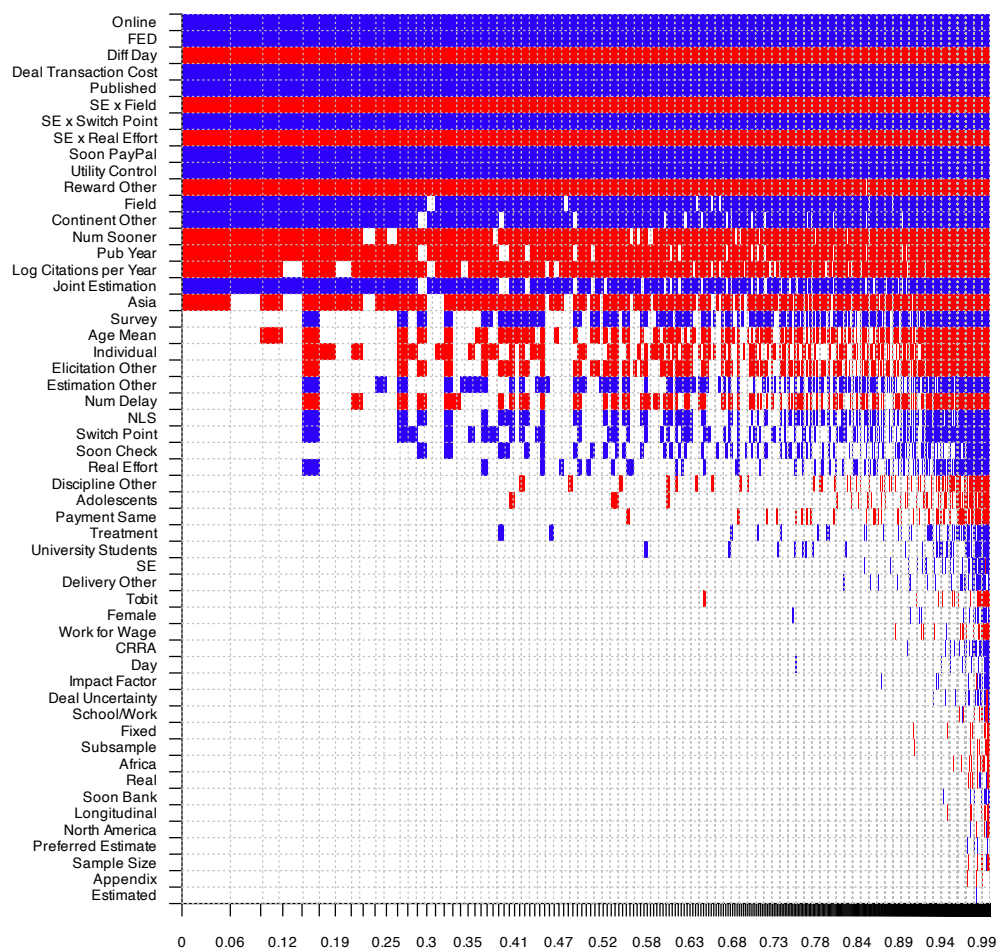


Figure D.3: BMA results (BRIC, random)

Notes: This figure displays the results of Bayesian model averaging using the BRIC g-prior and the random model prior. Each column denotes an individual model. Variables are sorted by posterior inclusion probability in descending order. The horizontal axis denotes cumulative posterior model probabilities. Variables in blue (darker in grayscale) depict an estimated positive sign. Variables in red (lighter in grayscale) depict an estimated negative sign. Variables with no color are not included in the given model. For graphical rendering purposes, the computation for this plot uses 20,000 iterations, whereas considered numerical results use 3,000,000 iterations; the results are qualitatively identical. For a detailed description of explanatory variables, see Table 5.1.

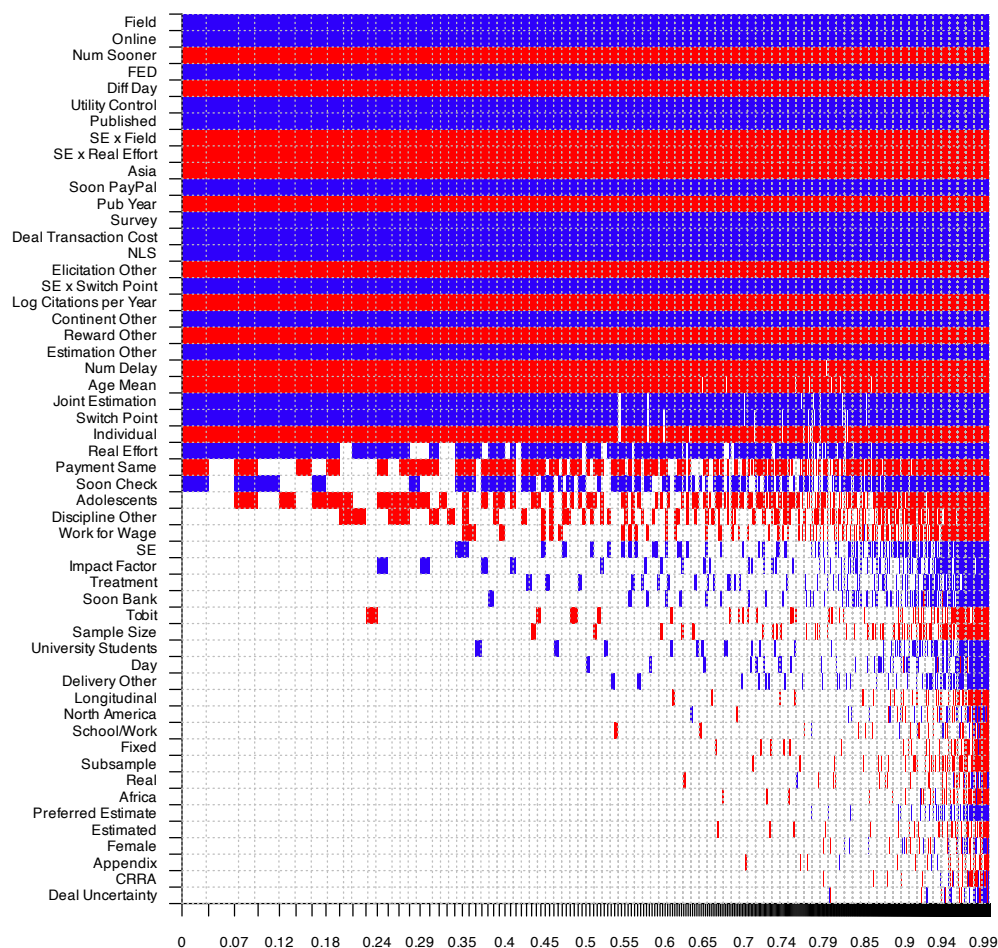


Figure D.4: BMA results (HQ, random)

Notes: This figure displays the results of Bayesian model averaging using the Hannan-Quinn g-prior and the random model prior. Each column denotes an individual model. Variables are sorted by posterior inclusion probability in descending order. The horizontal axis denotes cumulative posterior model probabilities. Variables in blue (darker in grayscale) depict an estimated positive sign. Variables in red (lighter in grayscale) depict an estimated negative sign. Variables with no color are not included in the given model. For graphical rendering purposes, the computation for this plot uses 20,000 iterations, whereas considered numerical results use 3,000,000 iterations; the results are qualitatively identical. For a detailed description of explanatory variables, see Table 5.1.

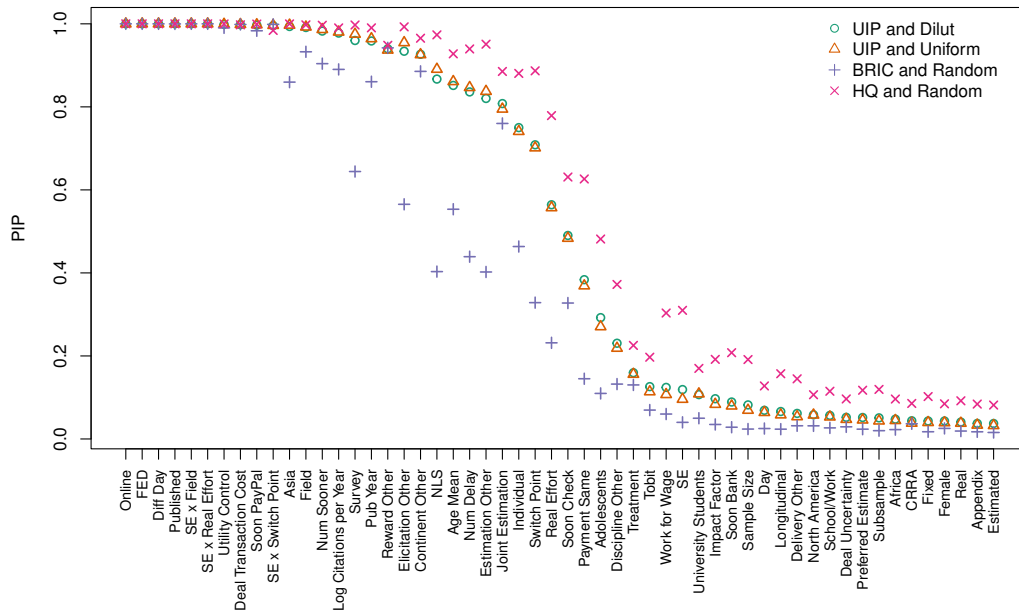


Figure D.5: Posterior inclusion probabilities under different priors

Notes: This figure displays the posterior inclusion probabilities for explanatory variables under different BMA priors. UIP = unit information g-prior. Dilut = dilution model prior. HQ = Hannan–Quinn g-prior. For a detailed description of explanatory variables, see Table 5.1.